

Pre-requisite: Nil

Division IV:

Human physiology: biological axons and neurons, neuromuscular and synaptic junctions; sensory systems - hearing, taste, smell and visual receptors. Dr. Souptick Chanda

Texts:

1. J. L. Tymoczko, J. M. Berg and L. Stryer, Biochemistry, 5th Ed, W. H. Freeman & Co, 2002.
2. D. L. Nelson and M. M. Cox, Lehninger Principles of Biochemistry, Macmillan Worth, 2000.

References:

1. N. Hopkins, J. W. Roberts, J. A. Steitz, J. Watson and A. M. Weiner, Molecular Biology of the Gene, 4th Ed, Benjamin Cummings, 1987.
2. C. R. Cantor and P. R. Schimmel, Biophysical Chemistry (Parts I, II and III), W.H. Freeman & Co., 1980.
3. **C. C. Chatterjee, Human Physiology, Vol 1 & 2, 11th Ed, Medical Allied Agency, 1987.**

<http://shiloi.iitg.ernet.in/~biotech/BT%20Syllabus/BT%20101.htm>

Text book of Medical physiology, Guyton and Hall

Grading Pattern

- Mid semester: 50 marks
- ~~Quiz before mid semester: 10 marks~~
- Final semester : 50 marks
- ~~Quiz after mid semester: 10 marks~~

➤ No re-examination

➤ Attendance guidelines shall be strictly followed

Human physiology:

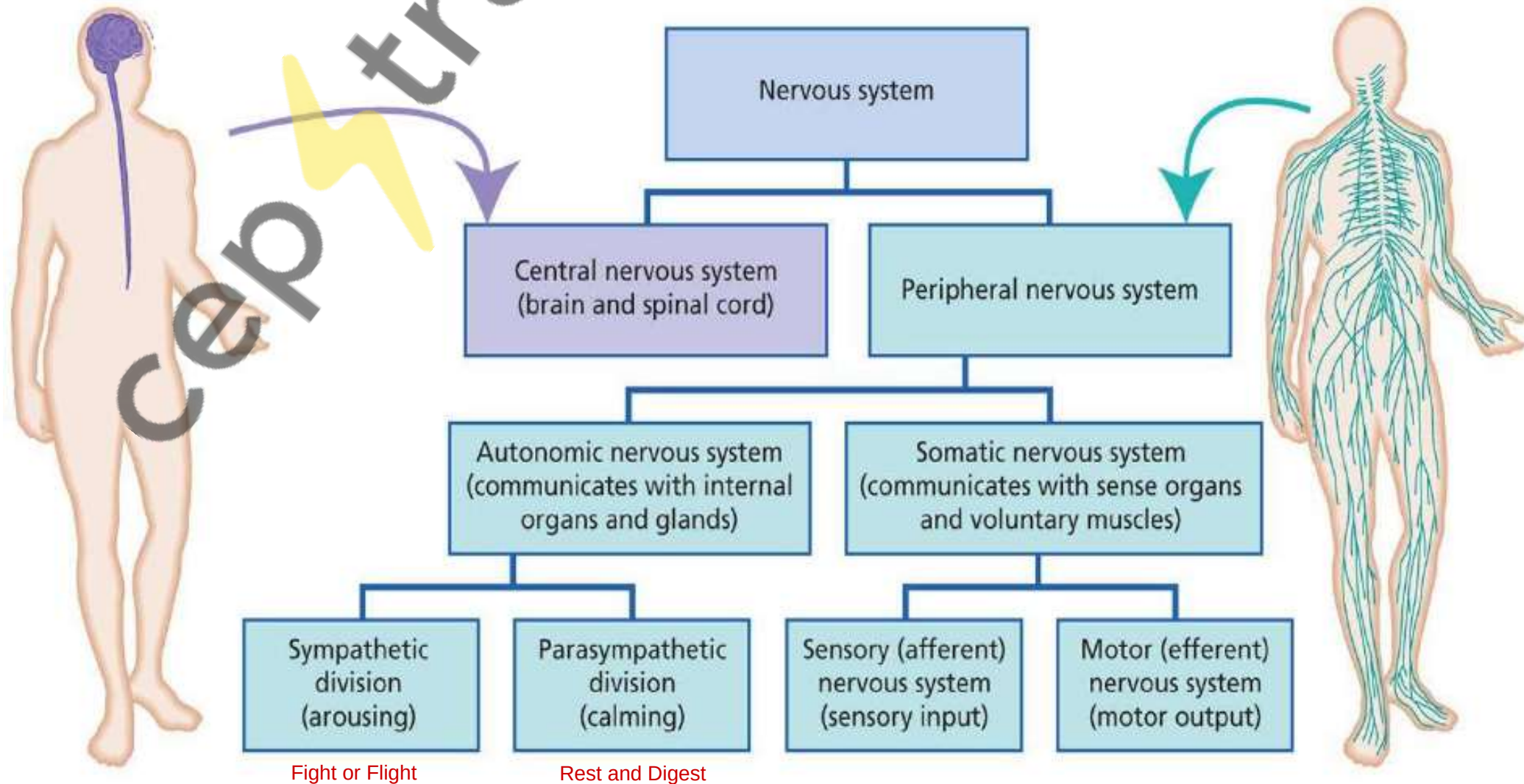
Nervous system, biological axons and neurons, neuromuscular and synaptic junctions;

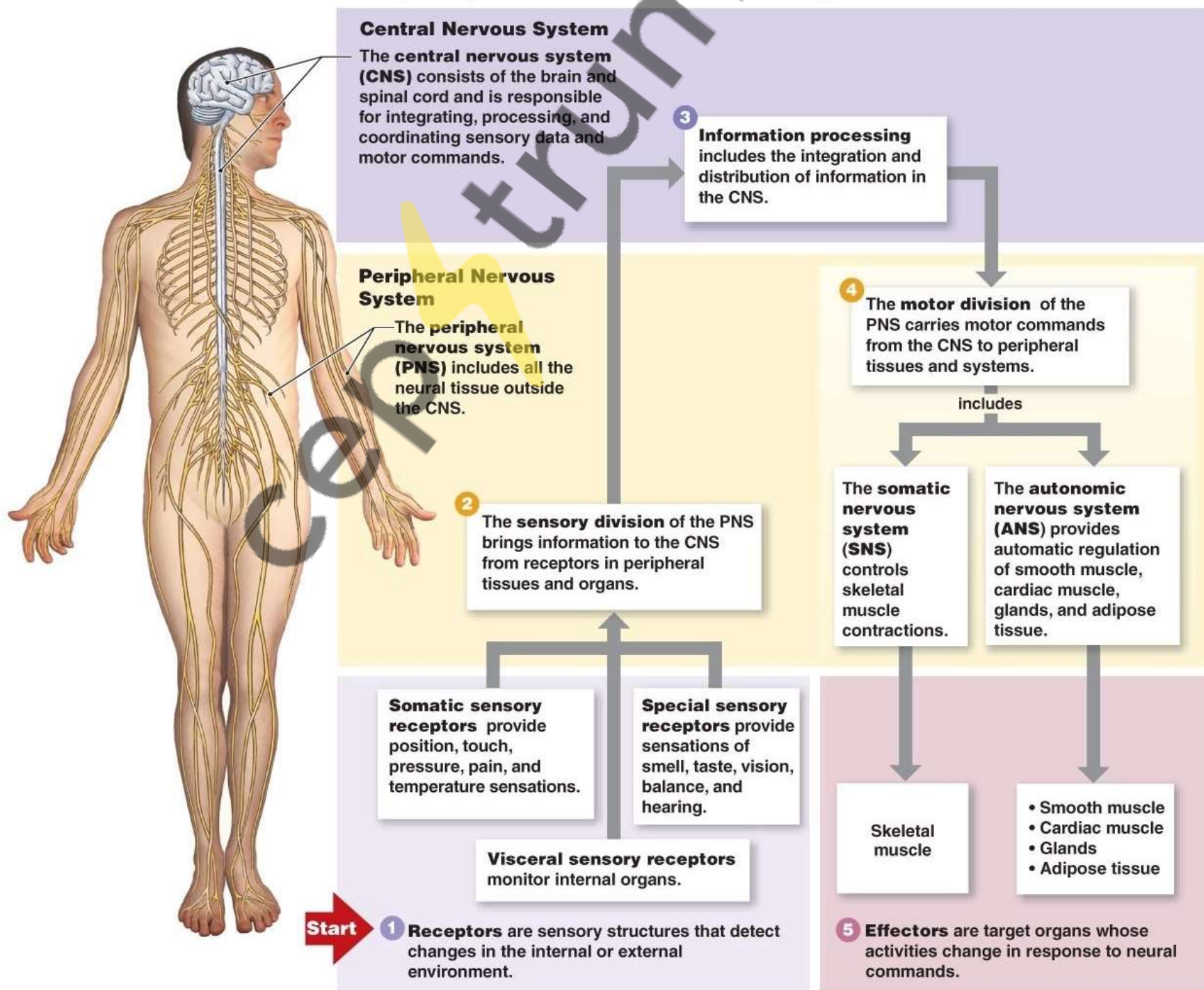
Sensory systems - hearing, taste, smell and visual receptors;

Introduction to ANN.

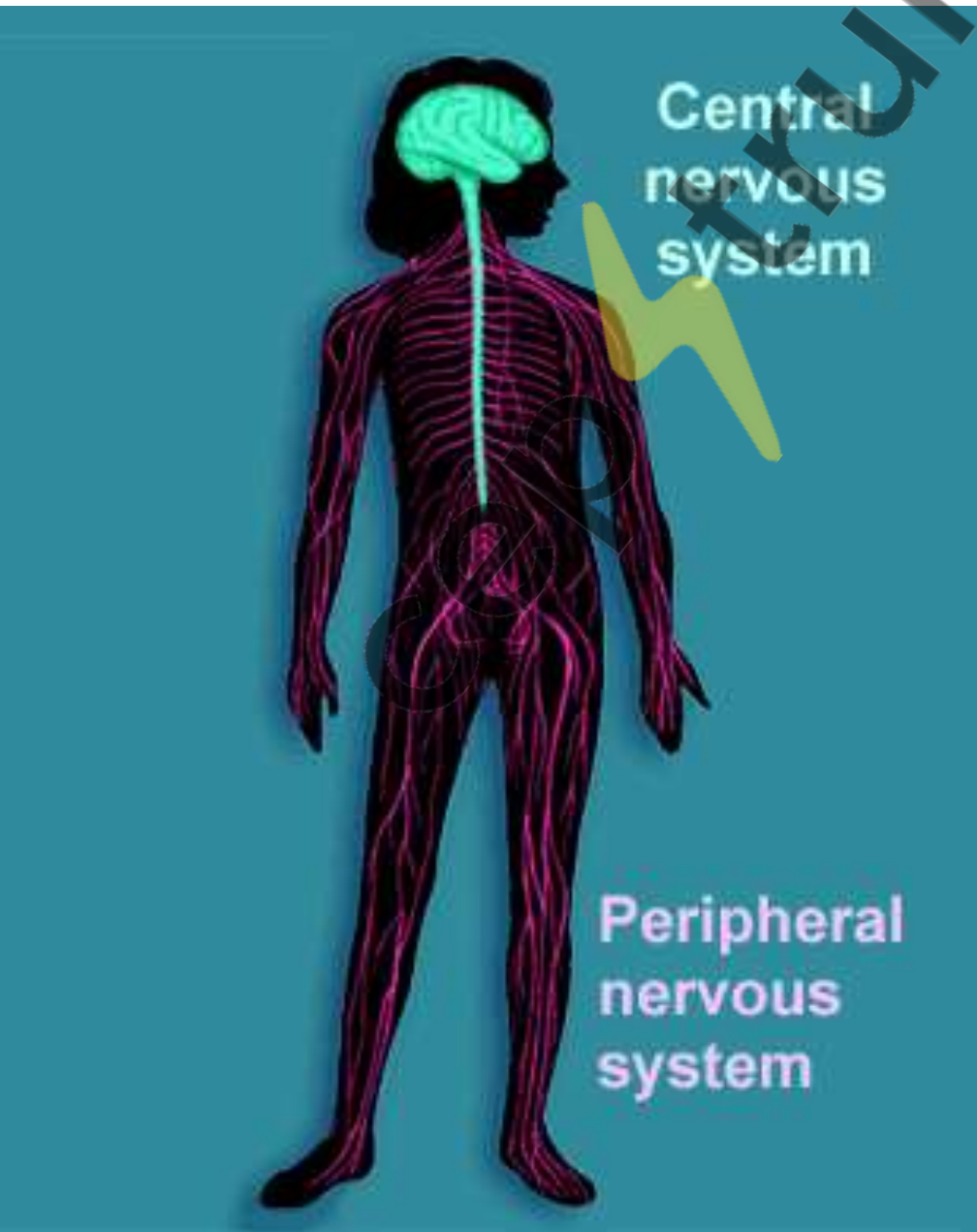
Dr. S. Chanda,
BSBE, IIT Guwahati

The Nervous System

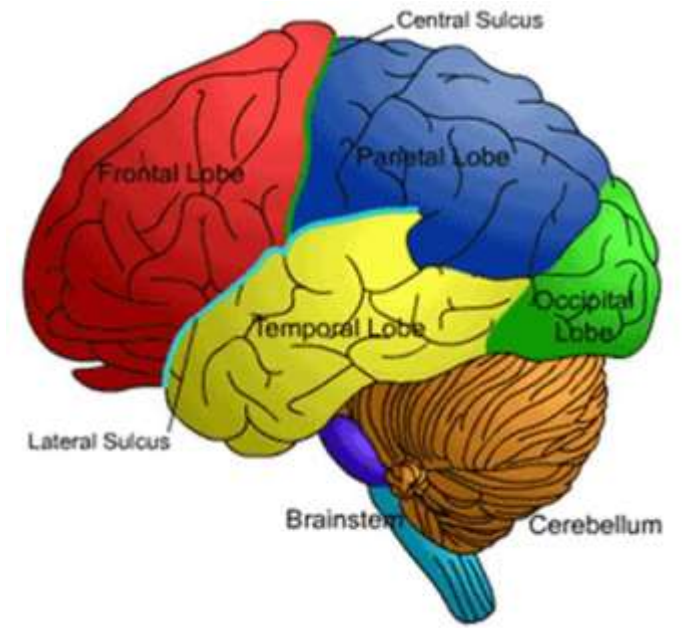
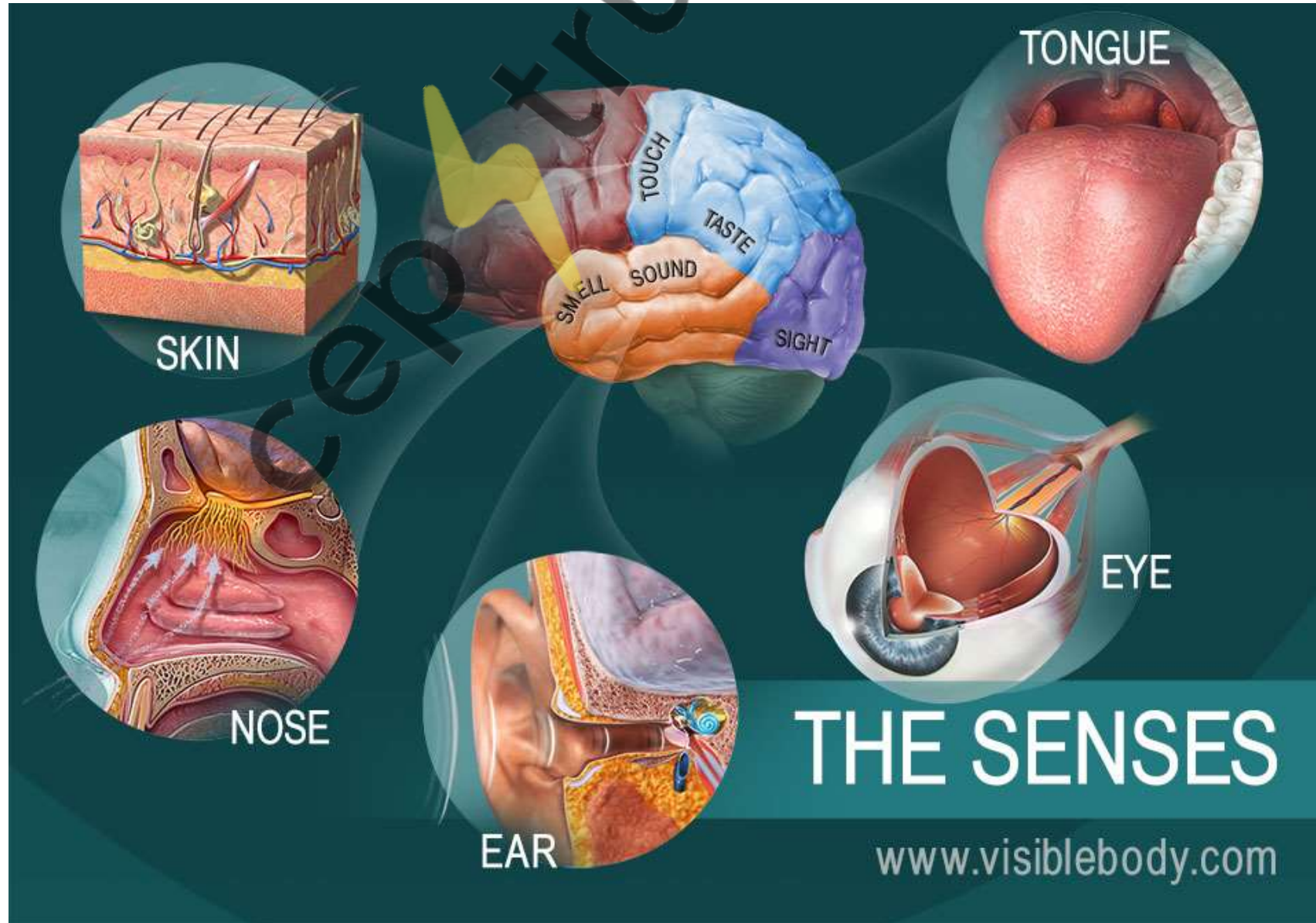




The major components and functions of the central nervous system (CNS)



The Special Senses



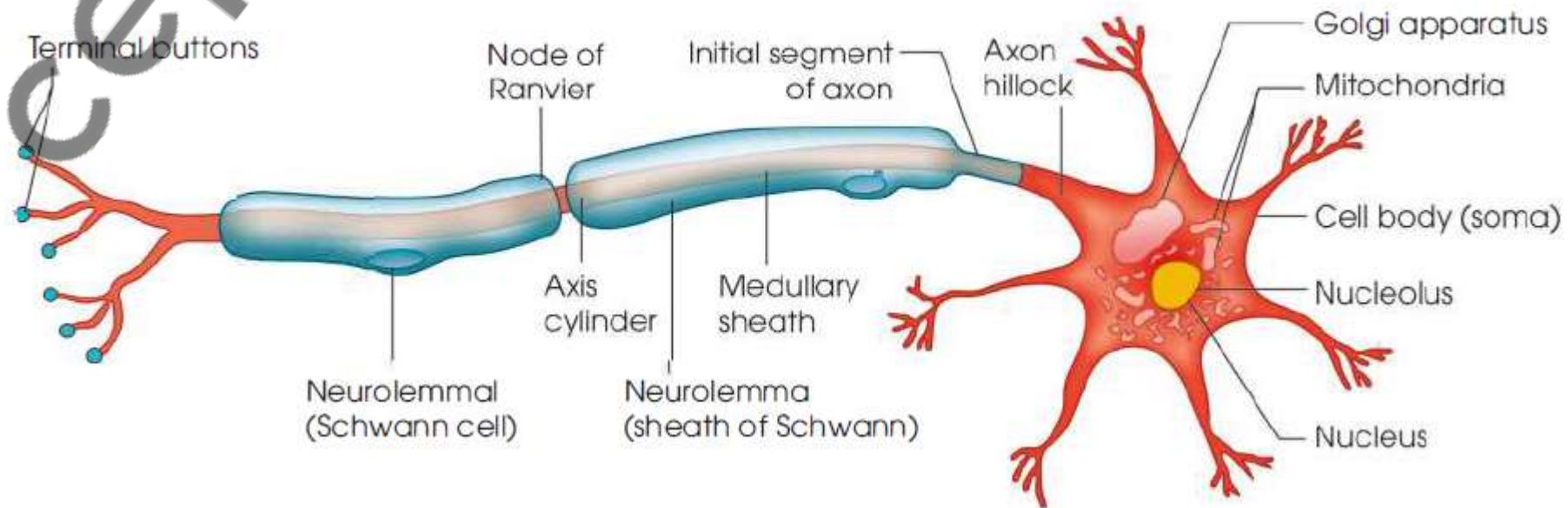
In short, the nervous system has three overall functions:

- **Sensory function** - sensory receptors detect the changes in the external environment.
- **Integrative function** - The CNS integrates this information.
- **Motor function** - effectors (muscles and glands) bring about a response.

Example: You are riding a bicycle and see that the traffic light has turned **RED** (sensory function). Your CNS integrates the information (**RED** light means '**STOP**' - integrative function), and you use your muscles to apply your brakes to stop the bicycle (motor function).

Neuron

The nervous system is composed of nervous tissue. The nervous tissue consists of two elements: Nerve cell or neuron and neuroglia. **The neurons are the structural and functional units.** They communicate with each other directly or indirectly through synapses. They are the electrically excitable cells and they transmit impulses via chemical and electrical signals.





Nerve Impulse and Muscle Contraction

Neurons are a special type of cell for information transfer around the body.

- ❖ *Dendrites*: Receive signals from neighboring neurons (like radio antenna).
- ❖ *Axon*: Transmit Signals over a distance (like telephone wire)
- ❖ *Axon Terminal*: Transmit Signals to other neuron dendrites (like a radio transmitter).
- ❖ *Myelin Sheath*: Speeds up signal transmission along axon.

Structure of Neuron

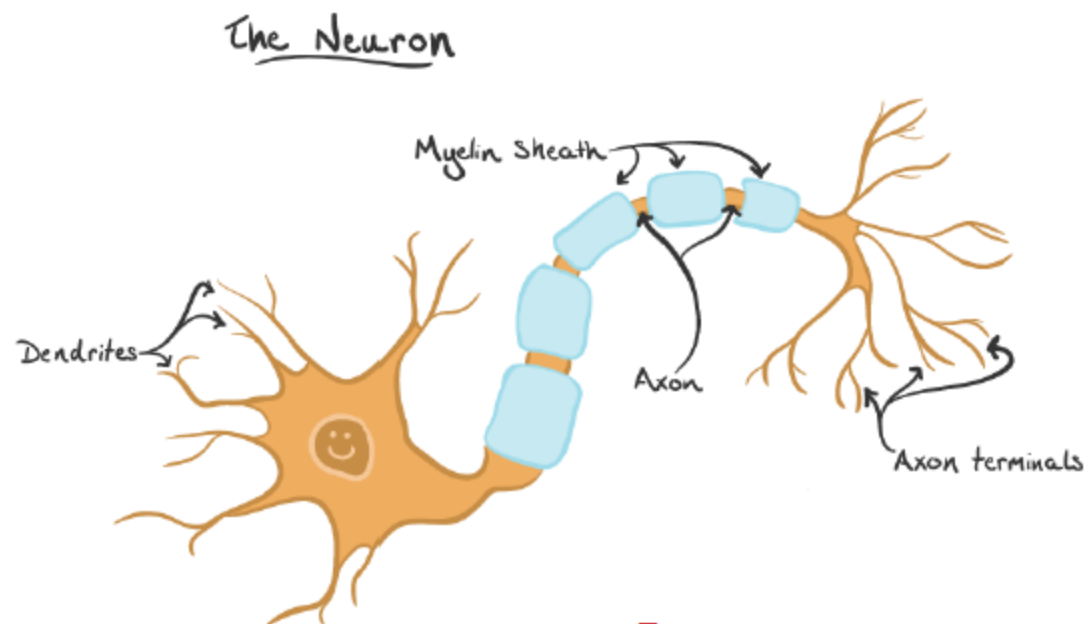
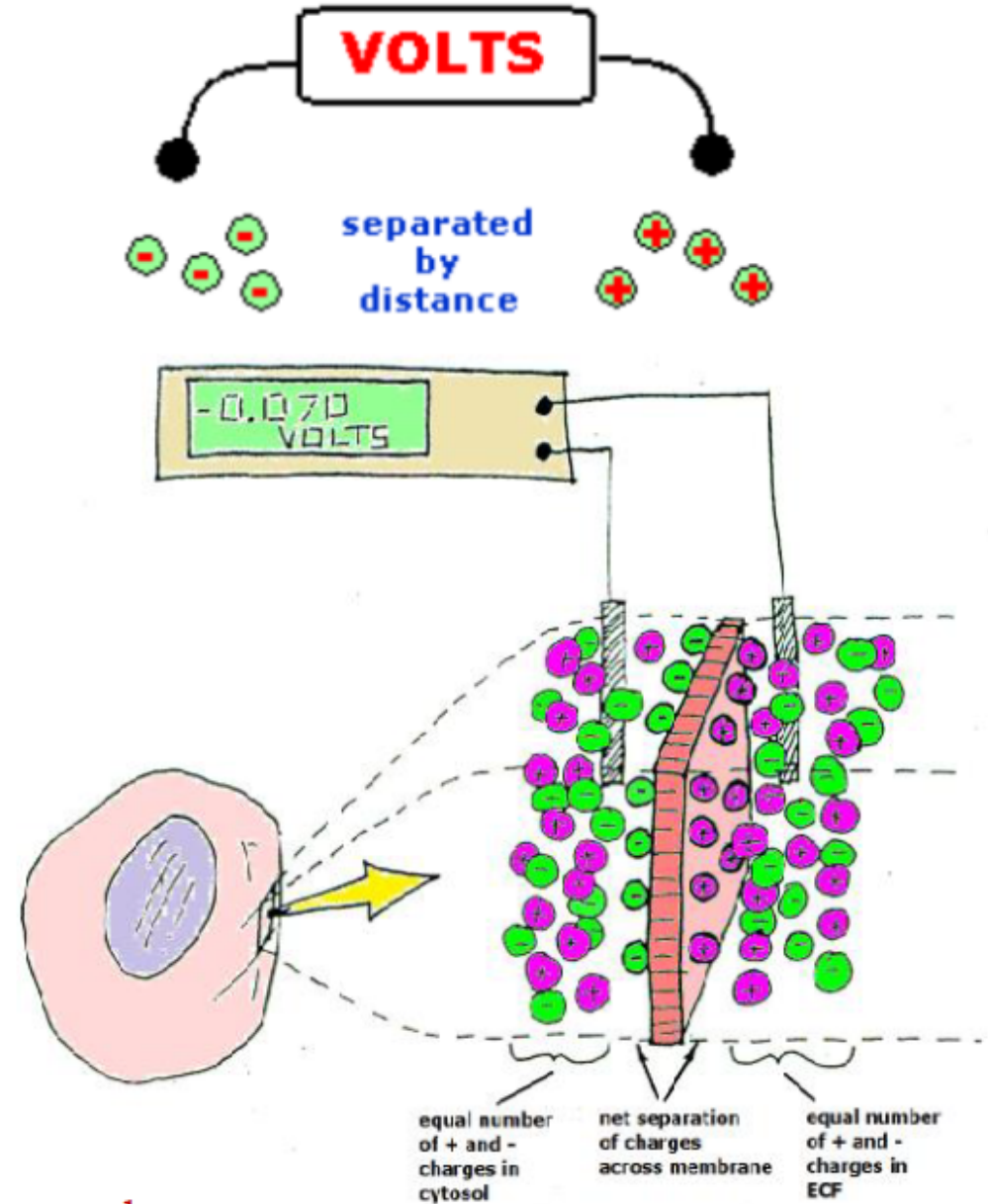


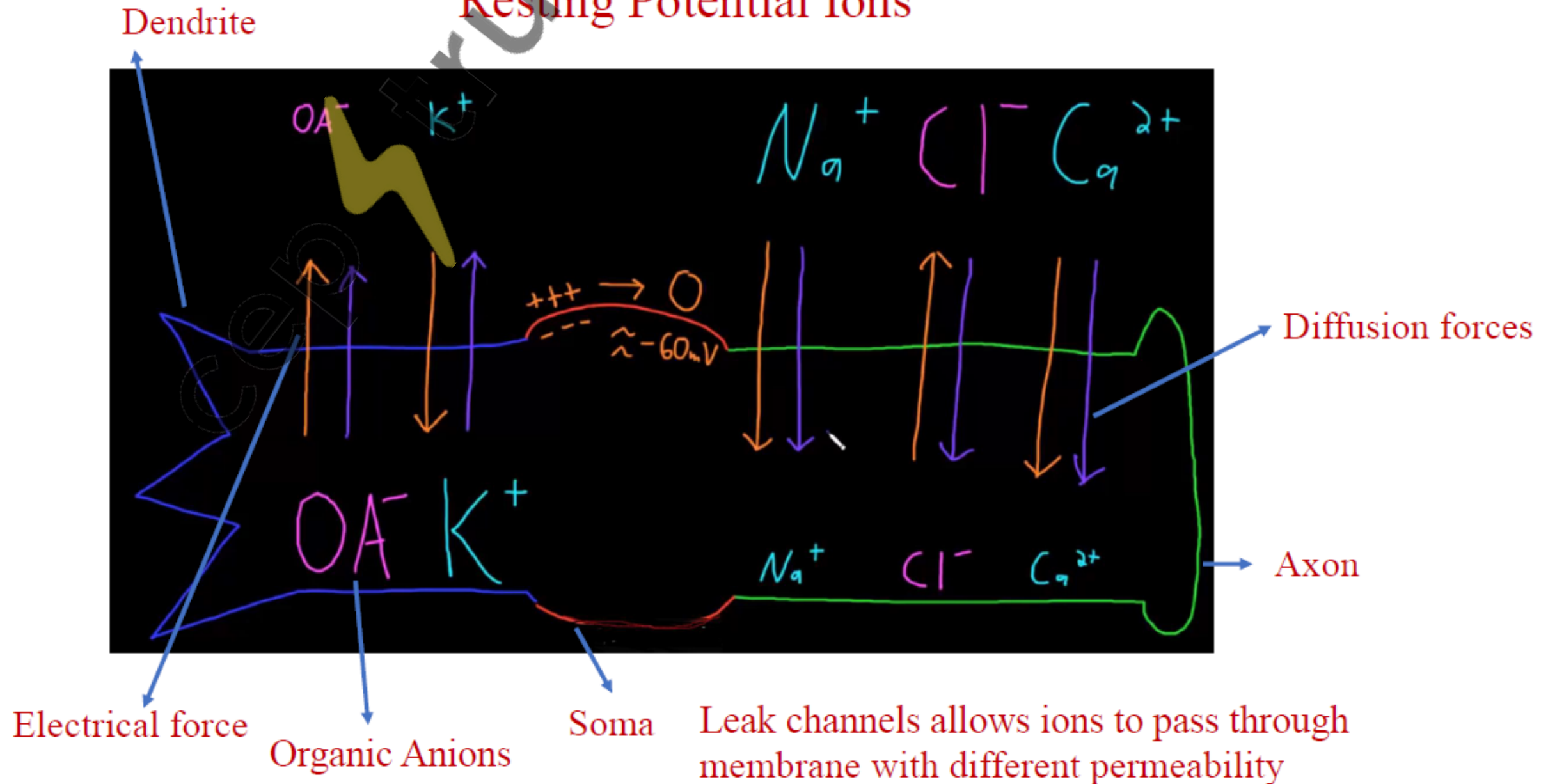
Image: cnx.org

Potential Difference/Resting Potential

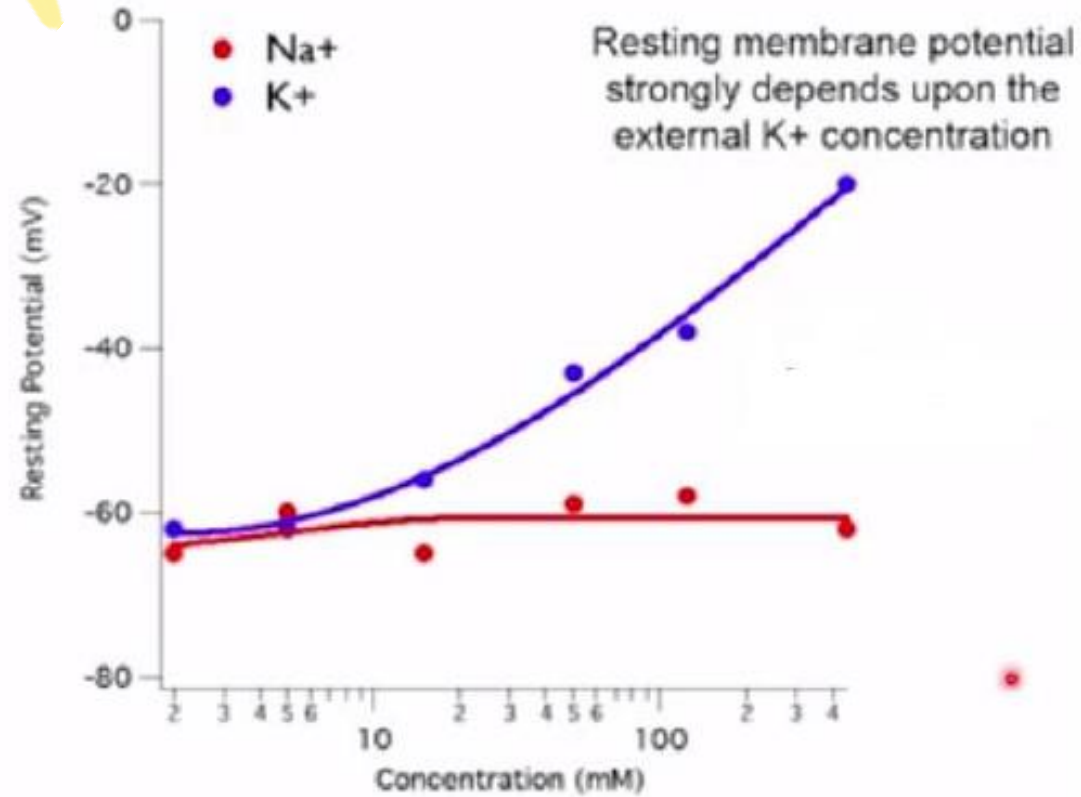
- **Electrical potential difference** between the inside and the surrounding extracellular fluid of cell is termed the membrane potential.
- When a nerve or muscle cell is at "rest", its membrane potential is called the **Resting membrane potential**



Resting Potential Ions



Resting Potential (Important Ions)



Resting Potential Expression

- *Electrical force*: Each ion will be attracted to the side of membrane with opposite charge.

$$M'_s = -u_s c_s \frac{d\psi}{dx}$$

Diagram illustrating the components of the electrical force equation:

- M'_s is the force.
- $-u_s$ is the Mobility of charge particle.
- c_s is the Total number of charges.
- $\frac{d\psi}{dx}$ is the Elec field gradient.

- *Chemical Diffusion force* related to concentration gradients across neuron membrane (Particles move from higher to lower concentration).

$$M'_s = -D_s \frac{dc_s}{dx}$$

Diagram illustrating the components of the chemical diffusion force equation:

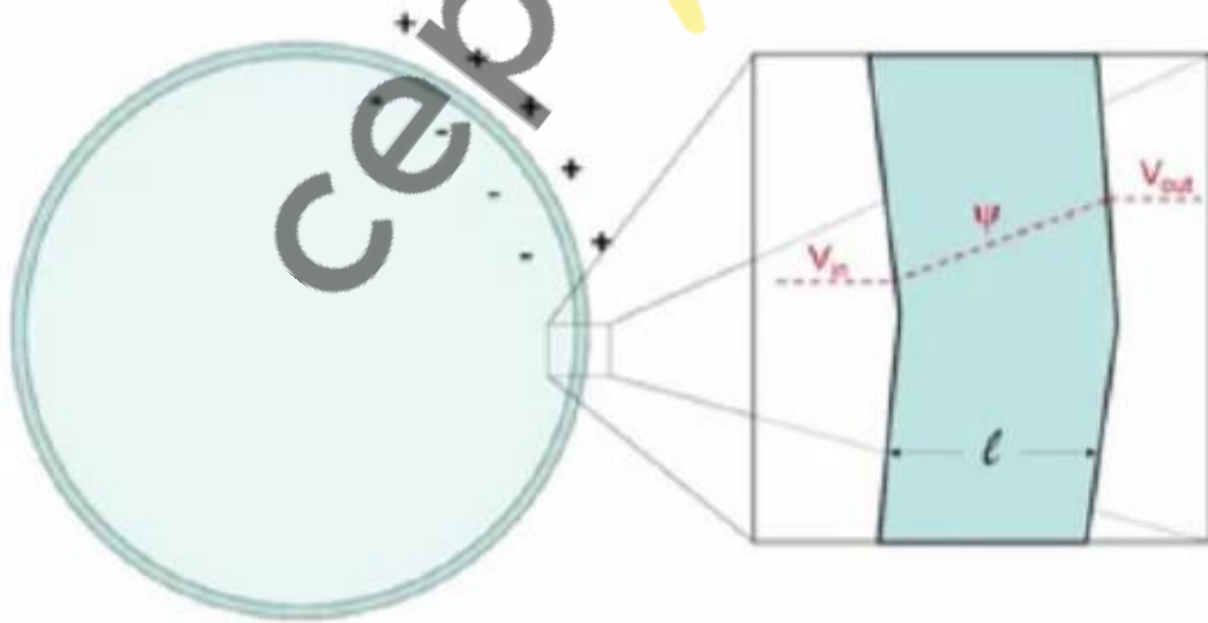
- M'_s is the force.
- $-D_s$ is the Diffusion constant.
- $\frac{dc_s}{dx}$ is the Concentration gradient.

- Each of ions are acted by two forces combinedly called *Electrochemical forces* which enable performing various functions.

Resting Potential (Nernst Equation)

Combining diffusion and electrophoresis we get:

$$M_s = -D_s \frac{dc_s}{dx} - u_s c_s \frac{d\psi}{dx}$$



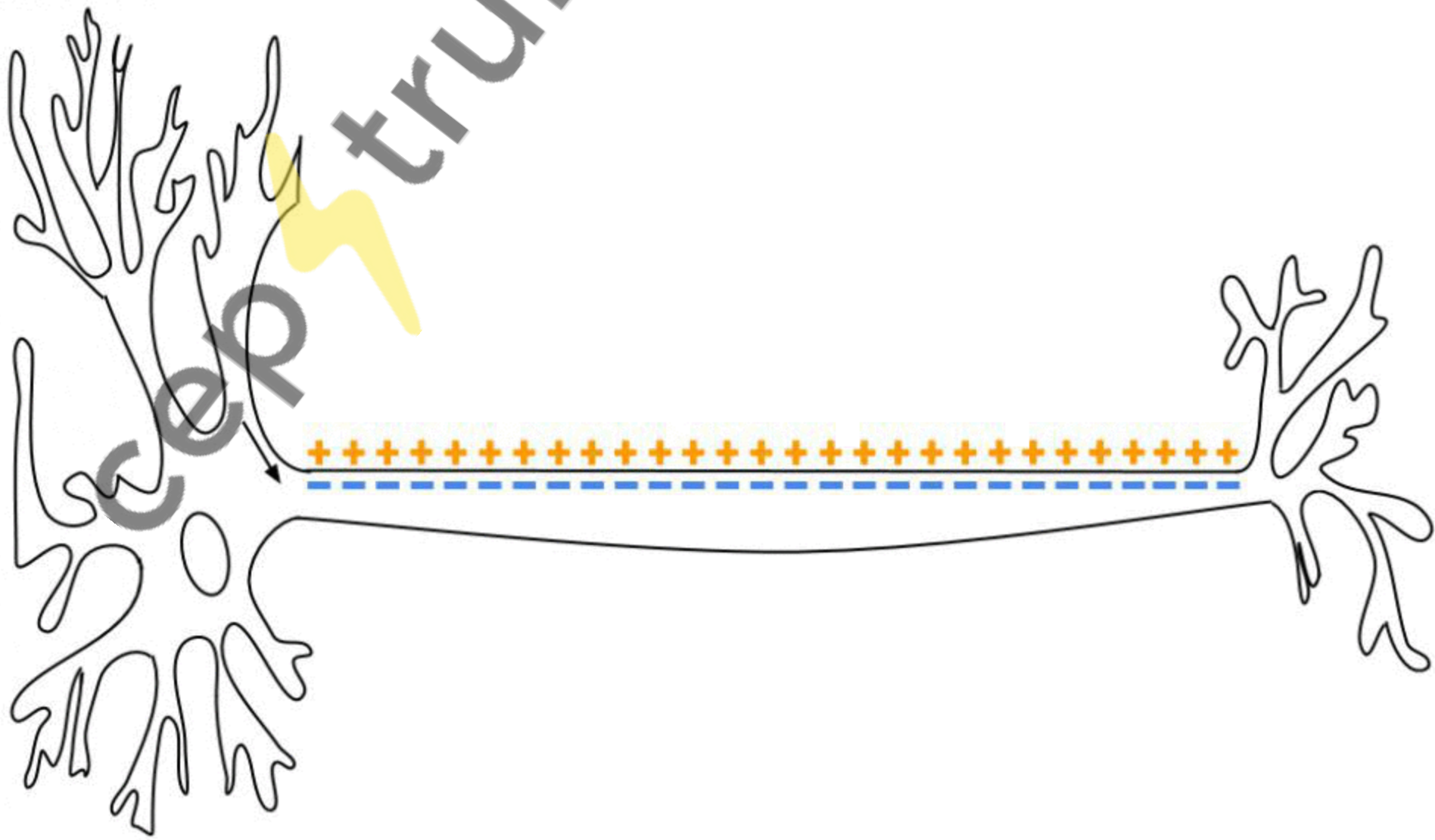
- At equilibrium when chemical and electrical energy are balanced (Net movement)

$$M_s = 0$$

- Simplifying we arrive at the membrane potential (Nernst Equation)

$$E_x = \frac{RT}{zF} \ln \frac{[X]_o}{[X]_i}$$

T= room temperature, z=charge ,
F=Faraday constant



Study how 100 billion Neurons are firing of nearly 50 action potentials per second thereby controlling what we do, how we think, move muscles, listen to lecture classes 😊

Signal Propagation Between Neurons

- The process of sending signals takes place in two steps: along the cell (action potential) and between cells (neurotransmitters)
- Action potentials travel in neurons cell as an electrochemical cascade, allowing a net inward flow of positively charged ions into the axon.

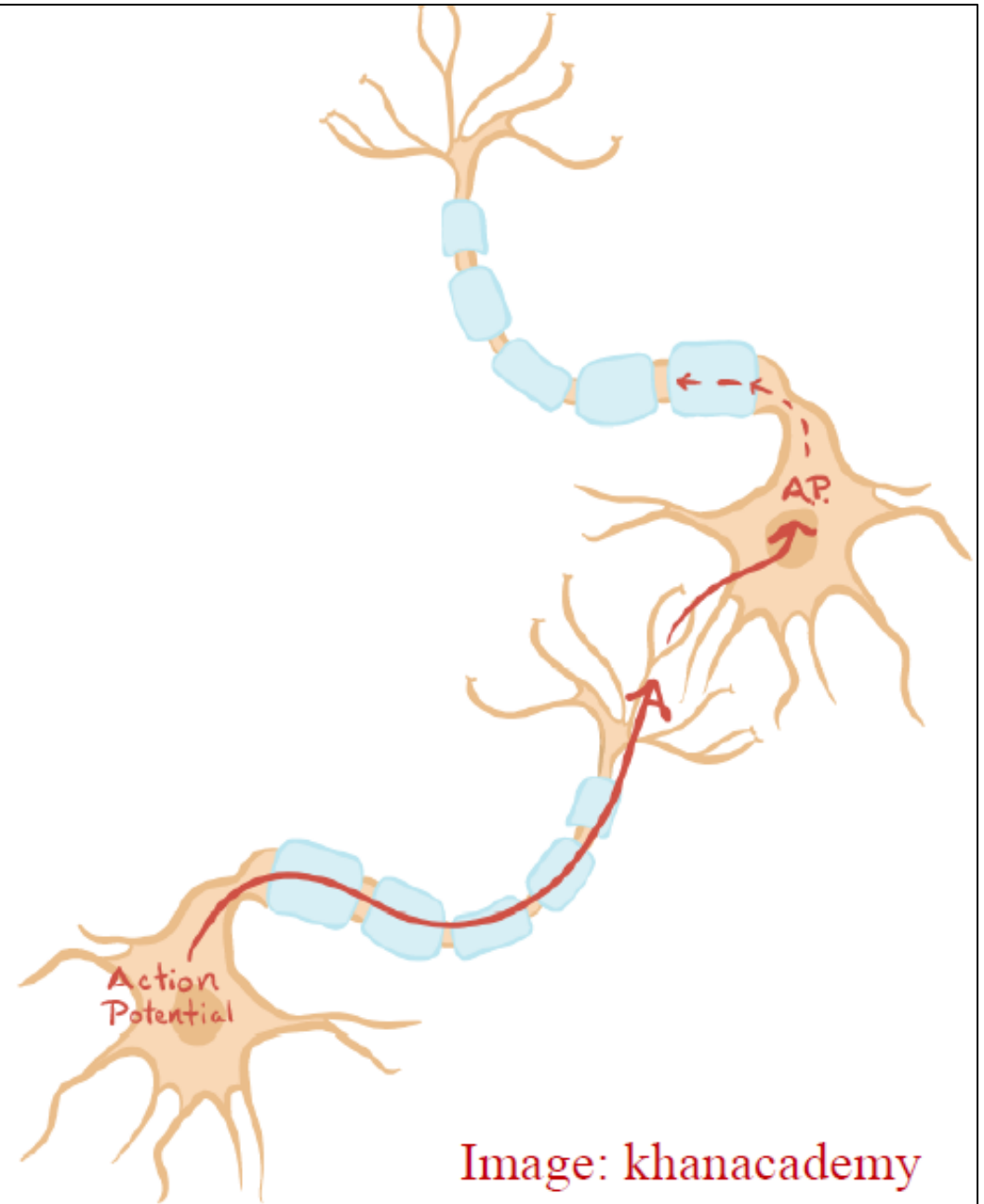
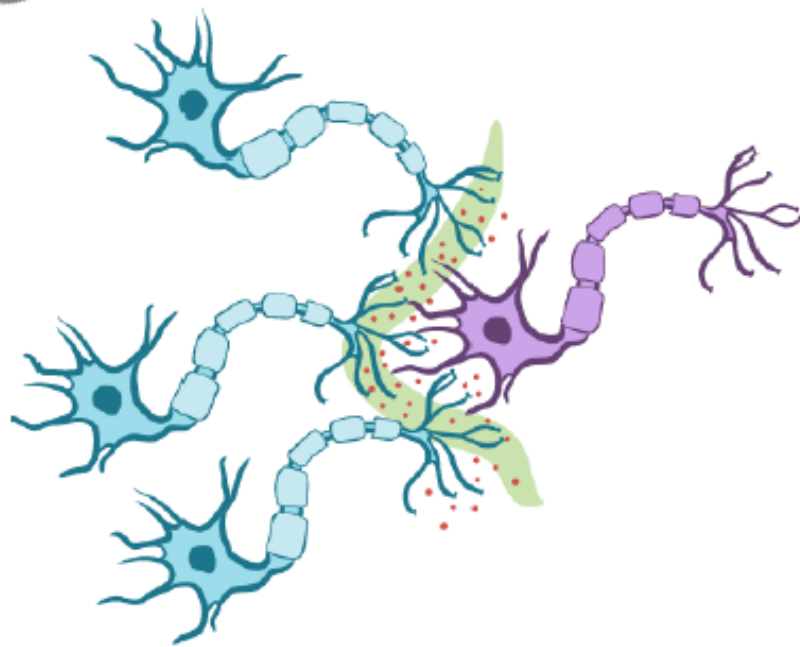


Image: khanacademy

- Chemicals known as neurotransmitters (glutamate, dopamine) are stored in membrane-bound vesicles at the axon terminal of neurons.
- Get released when Ca^{2+} enters the axon terminal and act by binding to receptors on the membrane of the postsynaptic cell.
- Neurotransmitters are “excitatory” firing a target neuron (Glutamate)
“inhibitory” making a target neuron less likely to fire (GABA).

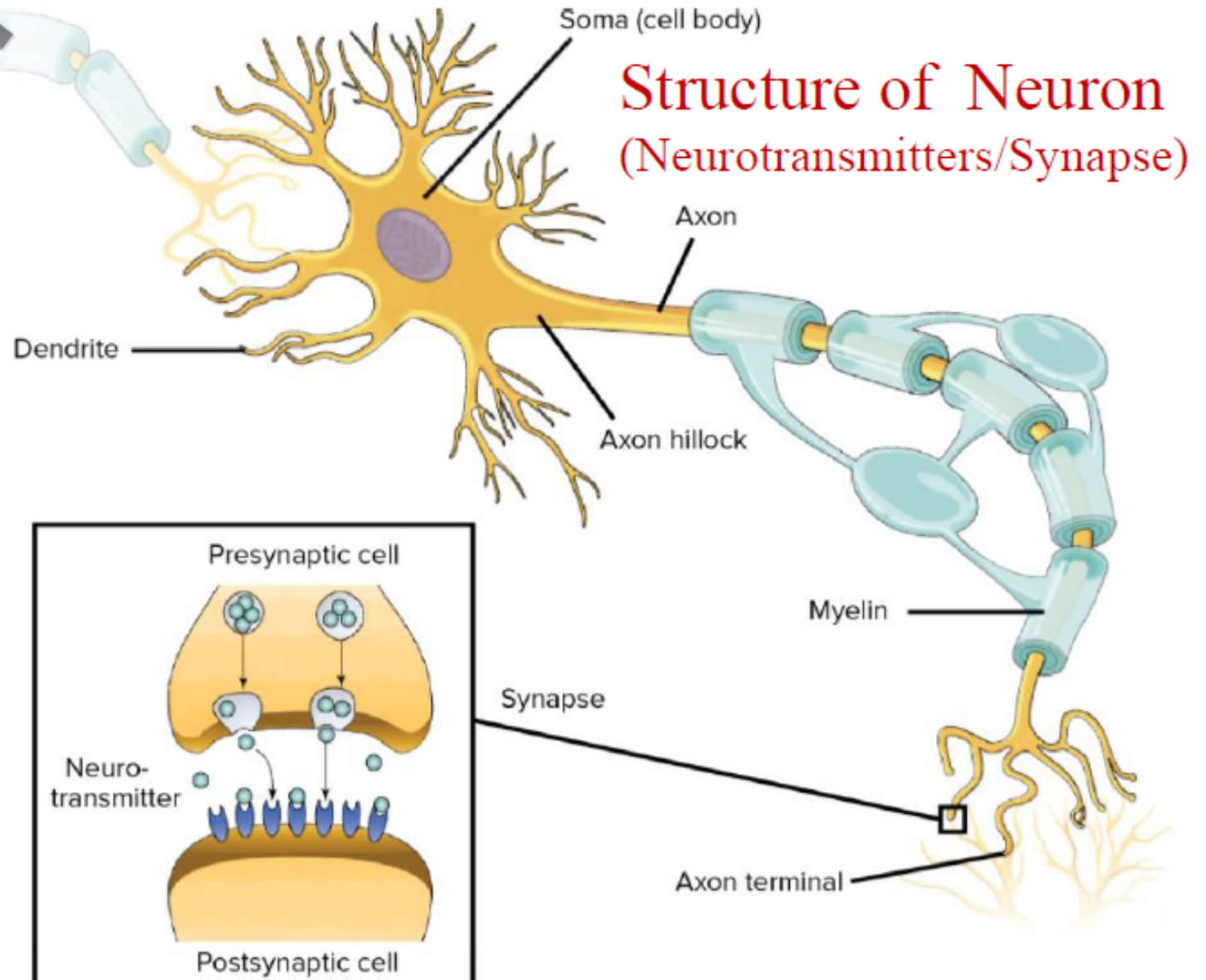
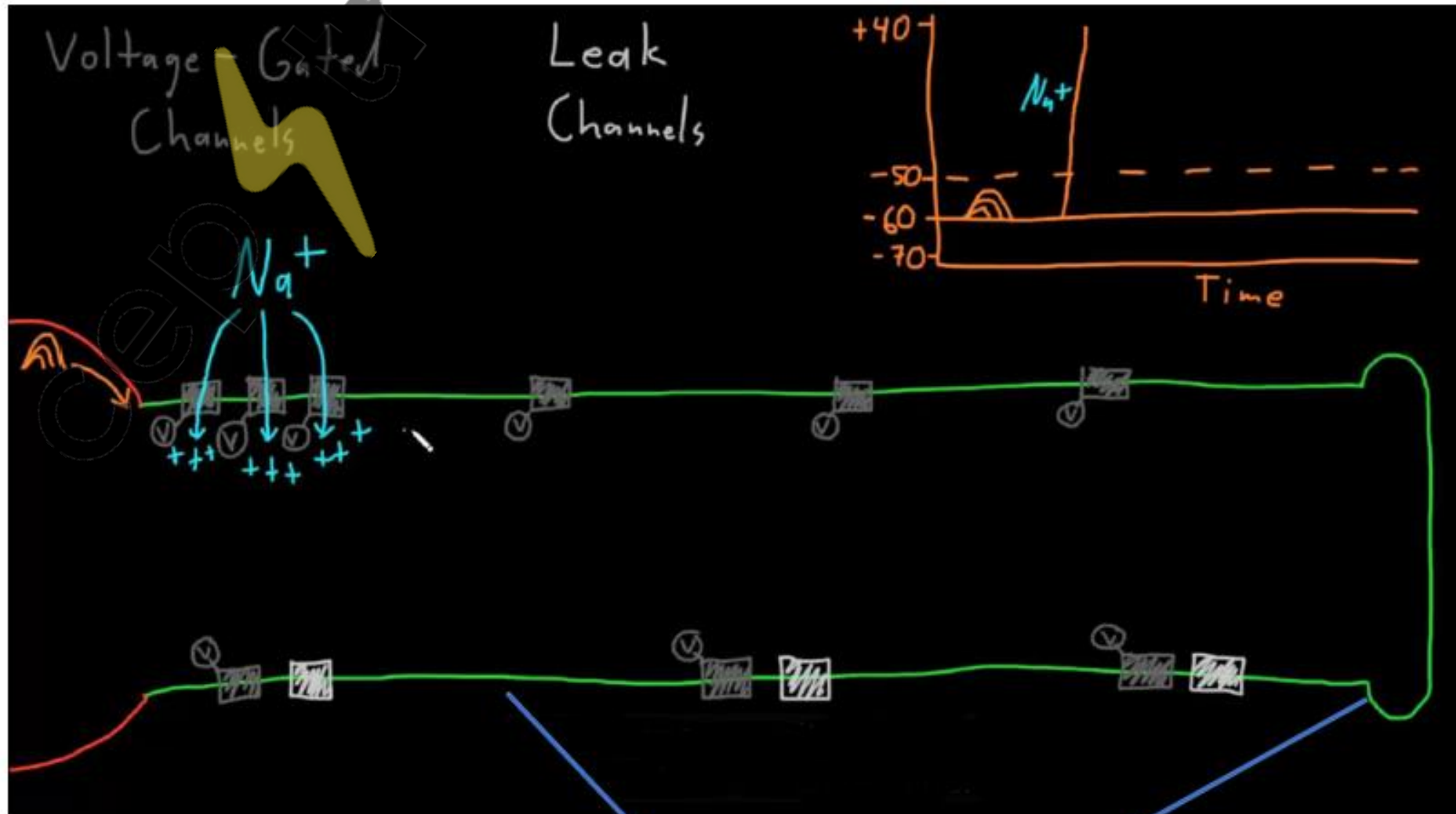
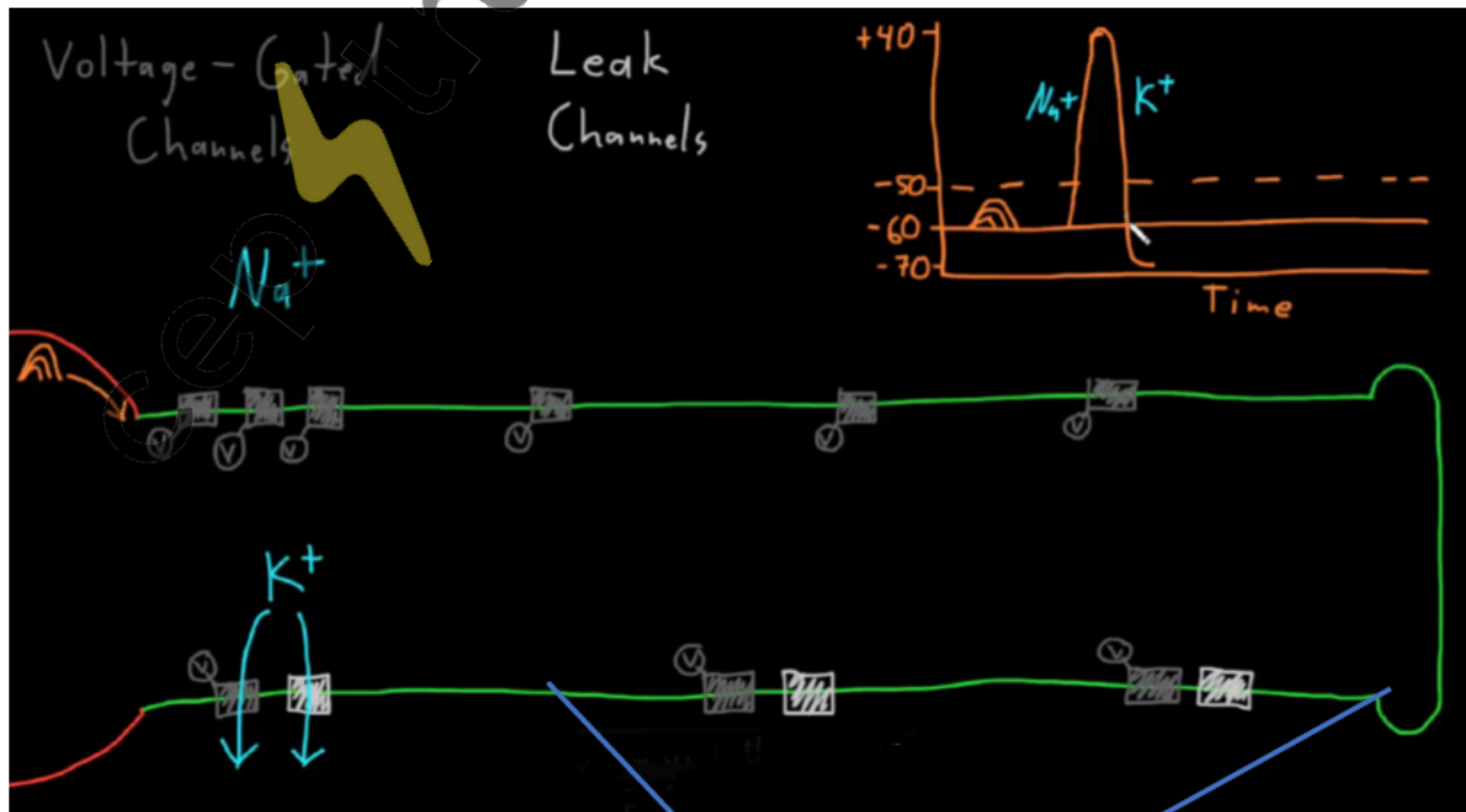


Image: cnx.org

Action Potentials(Raising- Na^+)



Action Potentials (Falling-K⁺)



Axon Enlarged

Action Potential/Nerve Impulse

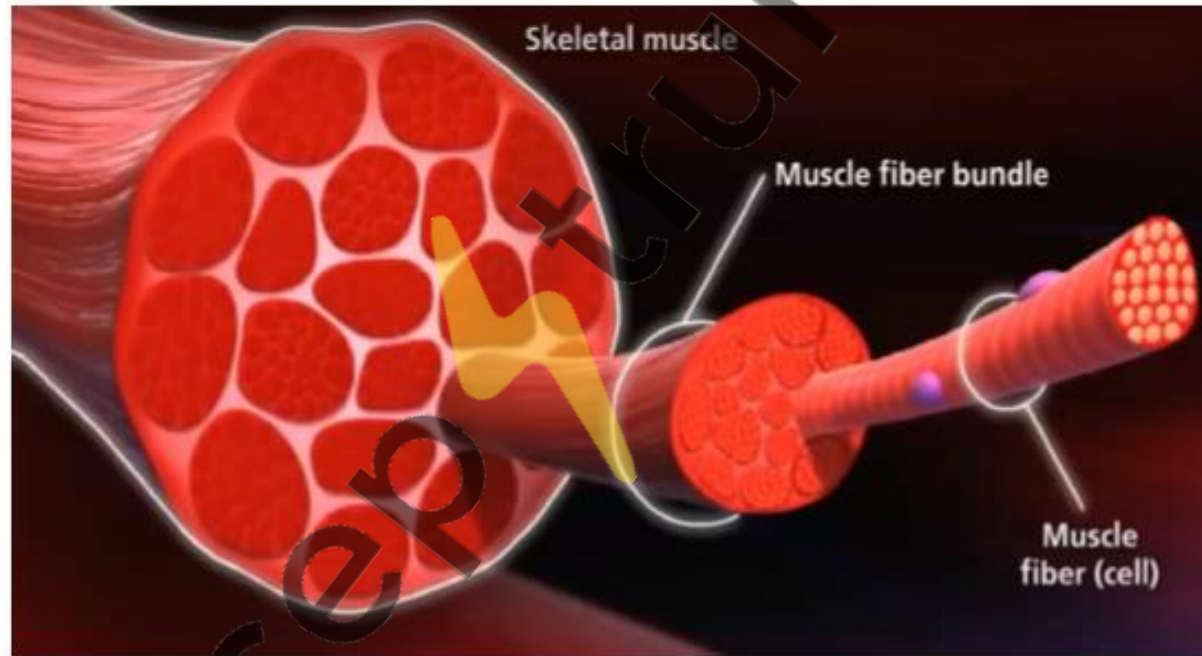
- *Action potential (AP)* occur when the combined effect of graded potentials in time brings the membrane of trigger zone over threshold potential (-50mv)
- They occur in *axons* of neuron.
- AP do not decay with time and distance and remain constant in duration.
- Myelinated axons conduct action potentials pass faster.
- AP speed in axons is around 1 to 100 metre/sec
- Size of AP varies from neuron to neuron

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STRUCTURE OF MUSCLES

Muscle Innervation

What is the fundamental unit of Skeletal Muscle?



SARCOMERE

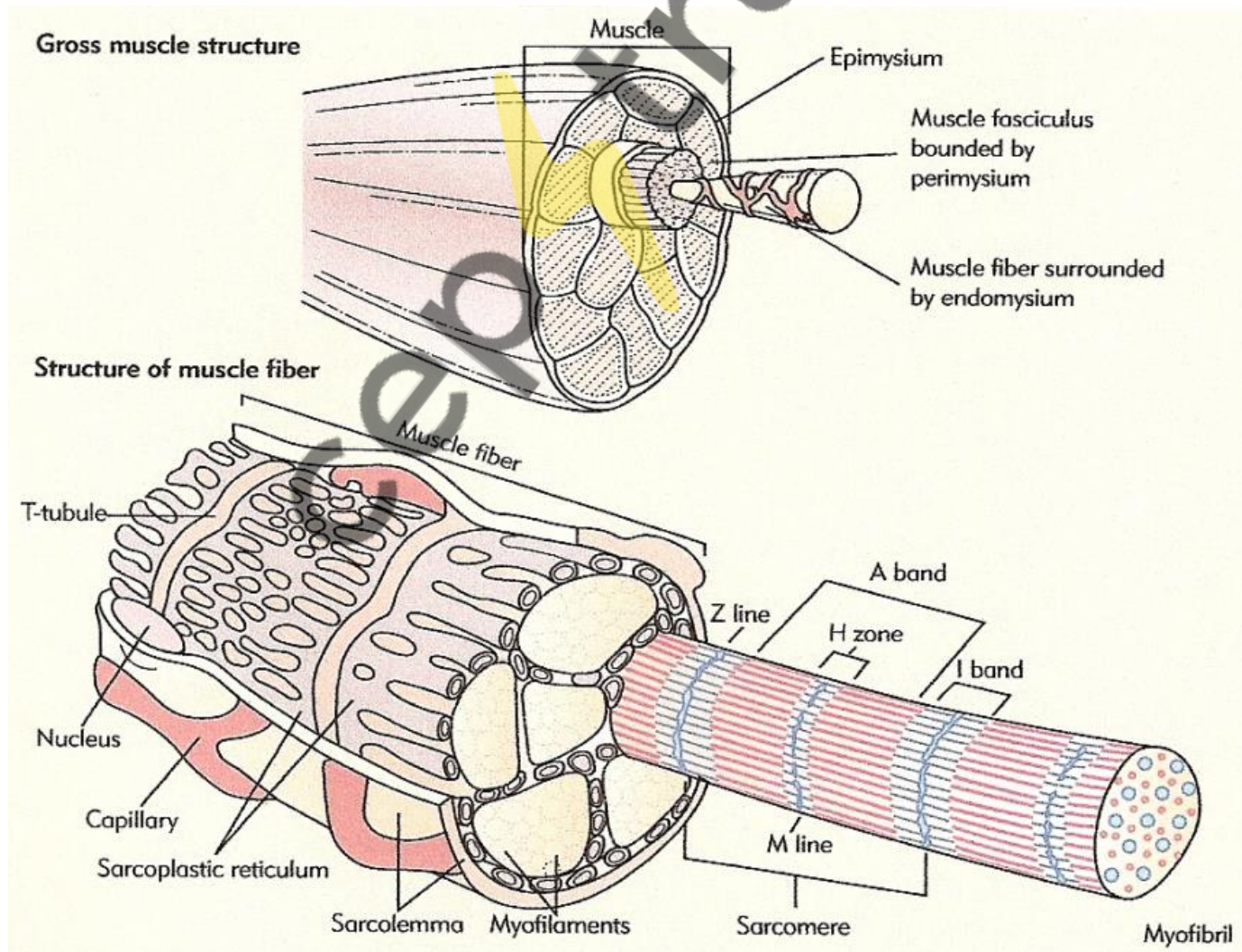
Thick and thin protein filaments in myofibril or muscle cell.

Myosin anchored at the M-line



Actin anchored to Z-line

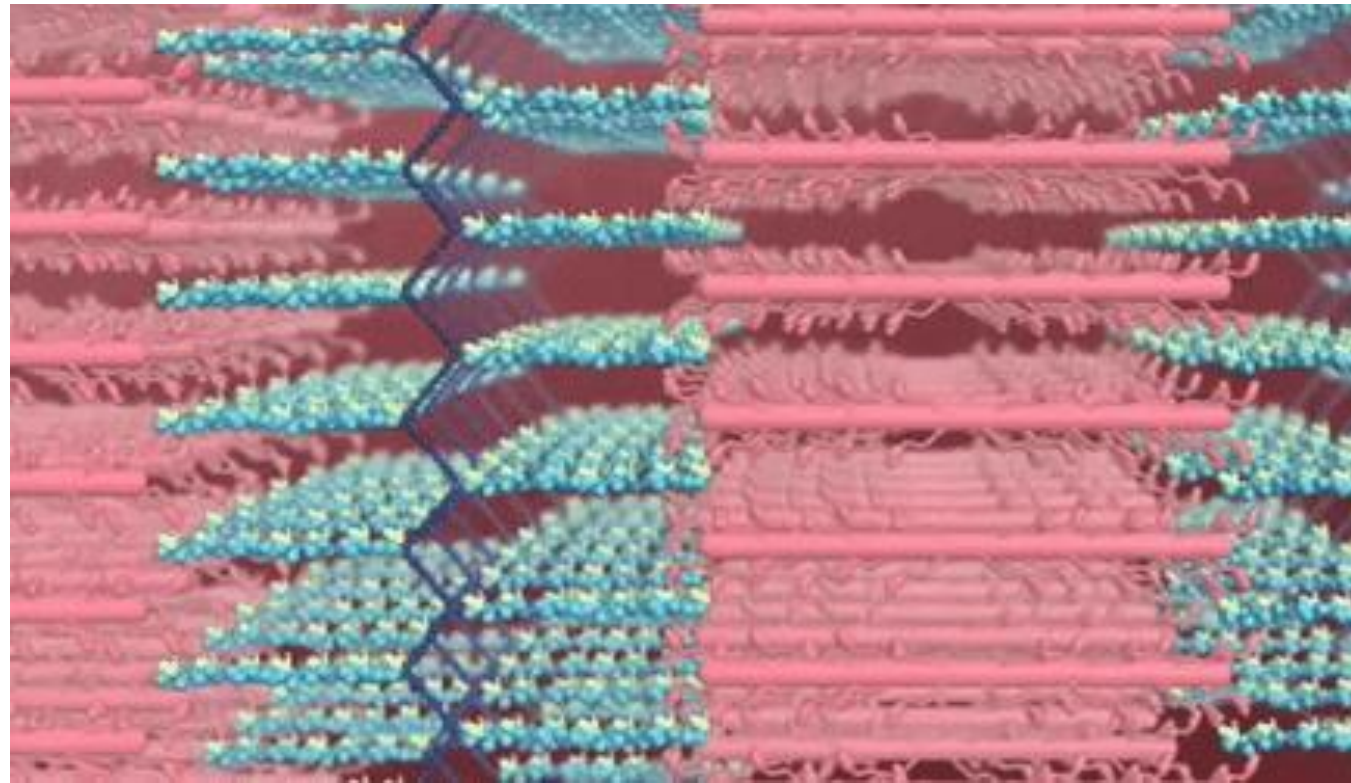
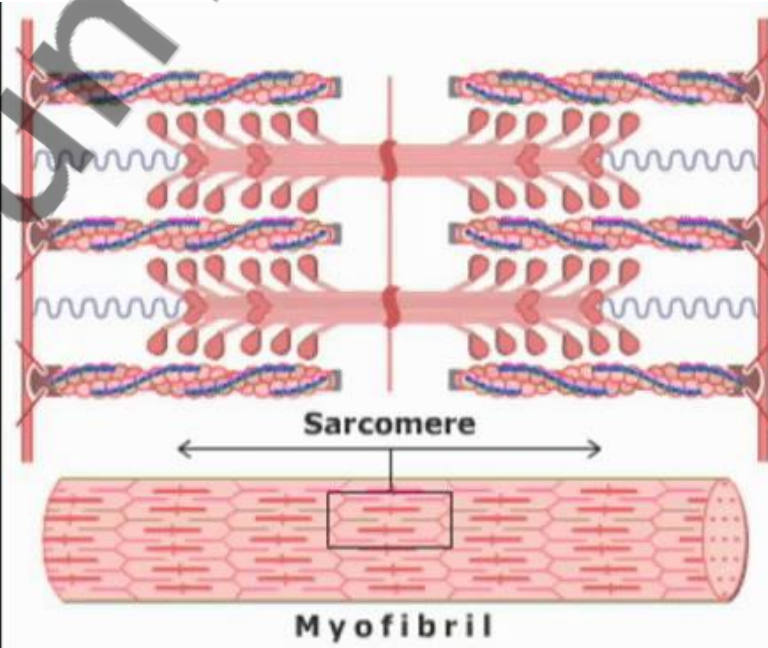
Skeletal muscles consist of numerous subunits or bundles called fascicles (or fascicles). Fascicles are also surrounded by connective tissue (called the perimysium) and each fascicle is composed of numerous muscle fibers (or muscle cells).



Muscle cells, ensheathed by endomysium, consist of many fibrils (or myofibrils), and these myofibrils are made up of long protein molecules called myofilaments. There are two types of myofilaments in myofibrils: thick myofilaments and thin myofilaments.

Skeletal muscles are usually attached to bone by **tendons** composed of connective tissue. This connective tissue also draws the entire muscle & is called epimysium.

*Interaction
between actin
and myosin*



Receptors

Mechanoreceptors: Detect mechanical compression or stretching of the receptor of tissues adjacent to the receptor

Thermoreceptors: Detect changes in temperature (cold/warmth)

Nociceptors: Pain receptors

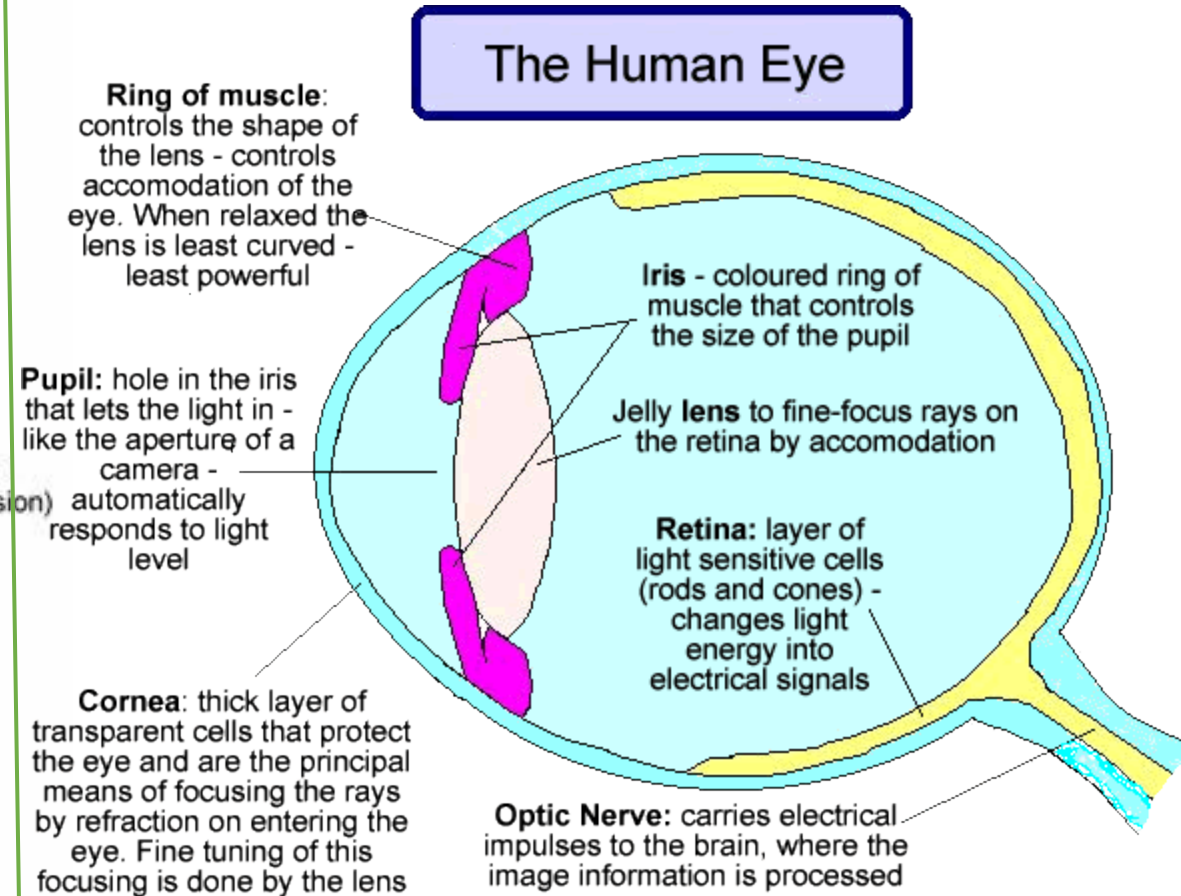
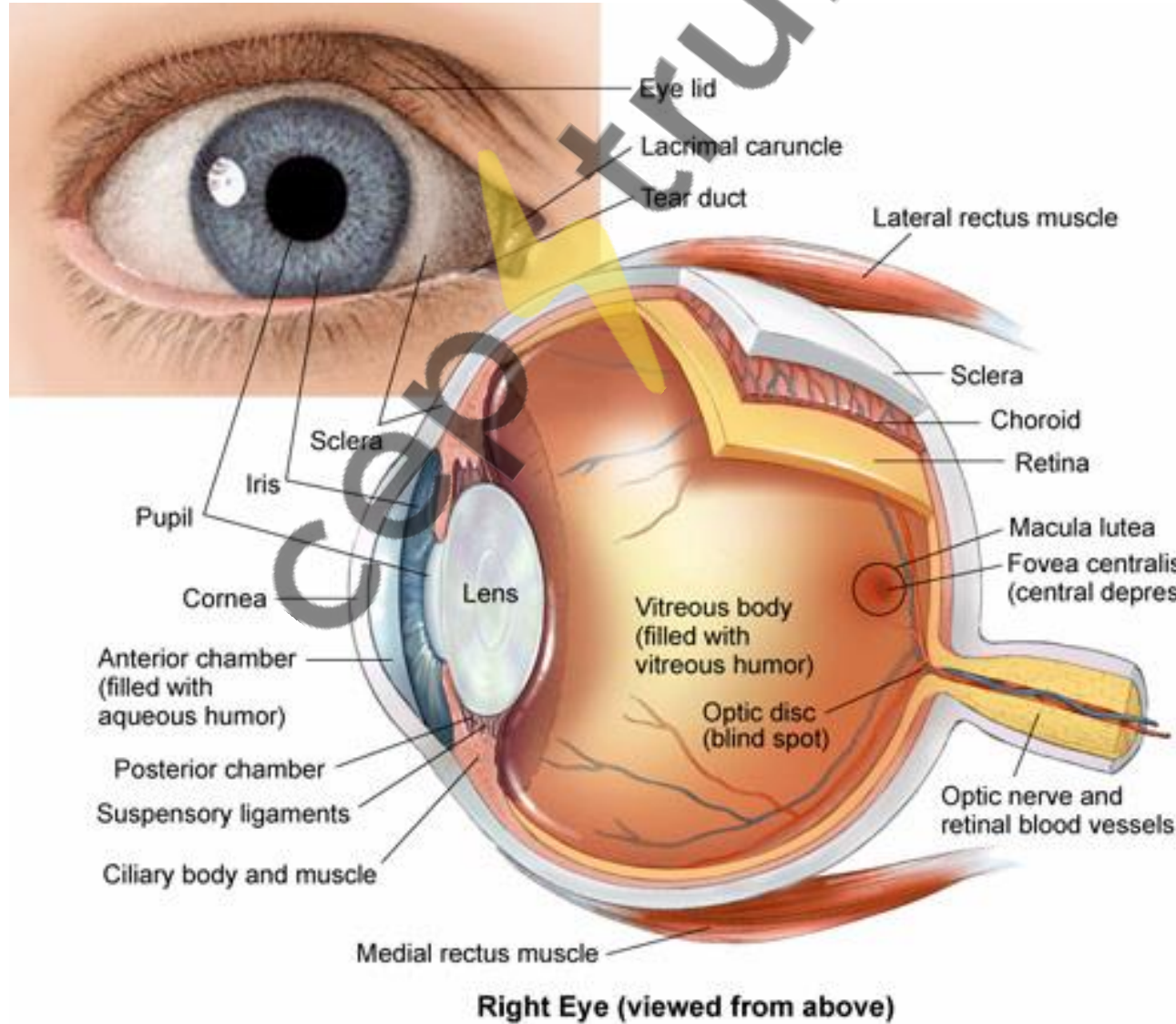
Electromagnetic receptors: Detect light

Chemoreceptors: Detect taste, smell, oxygen level in blood, osmolality (electrolyte-water balance) of the fluids, CO₂ concentrations etc.

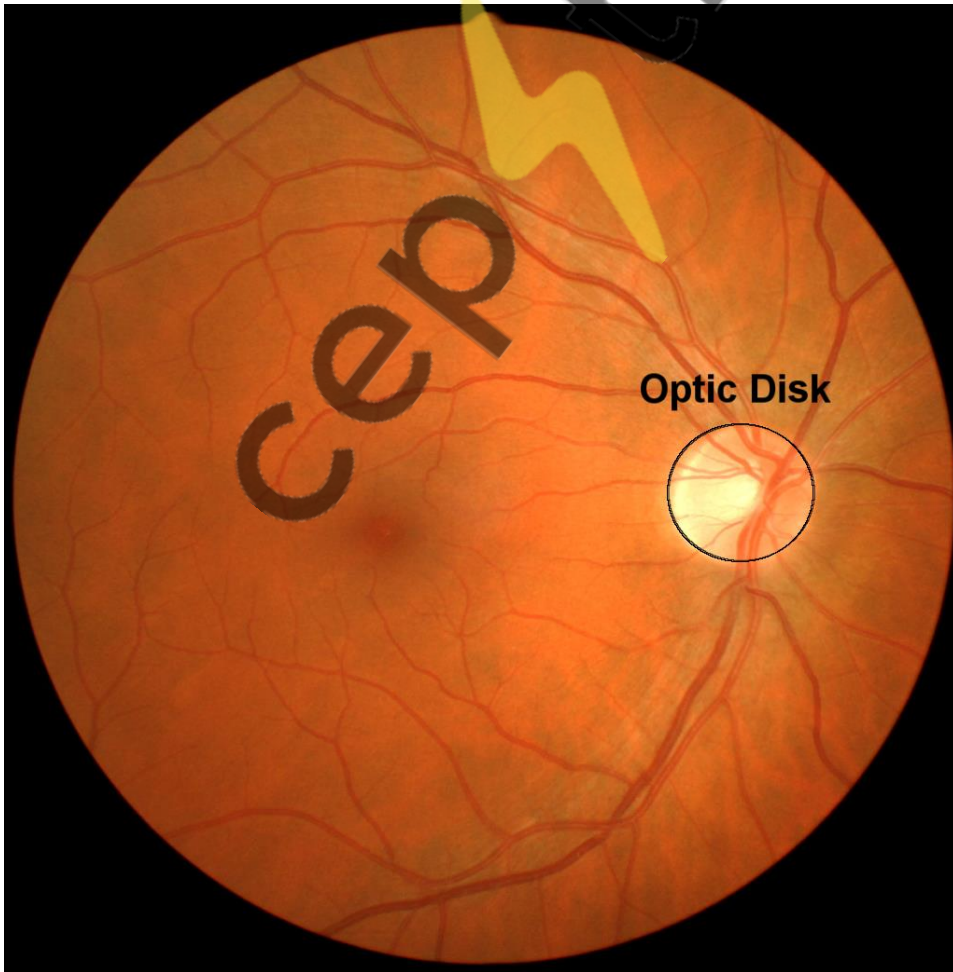
Vision and visual receptors

- ☐ Basic structure of the eye
- ☐ Blind spot
- ☐ Formation of an image on retina
- ☐ Farsightedness (hyperopia/hypermétropia) and Nearsightedness (myopia)
- ☐ Intraocular fluid, tonometer, and glaucoma
- ☐ Photochemistry of vision by rods and cones cells
- ☐ Night blindness
- ☐ Colour vision in reptiles and birds

Human Eye: Structure & Functions:



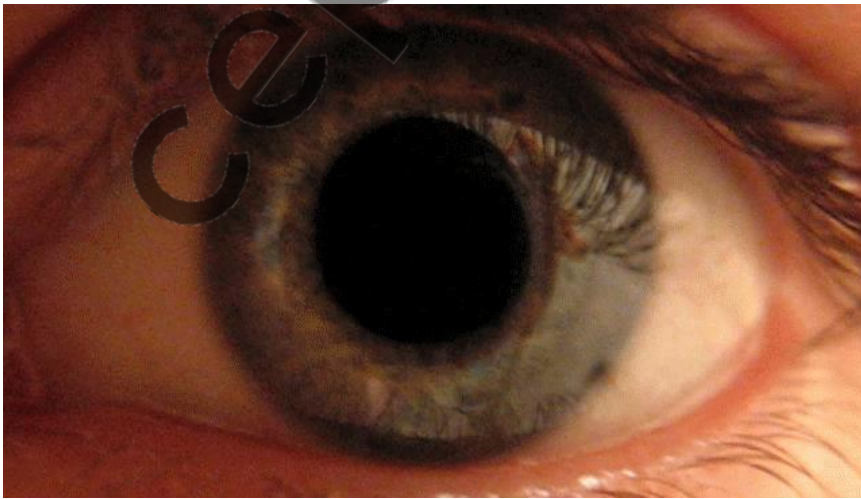
Blind Spot (Optic Disc)



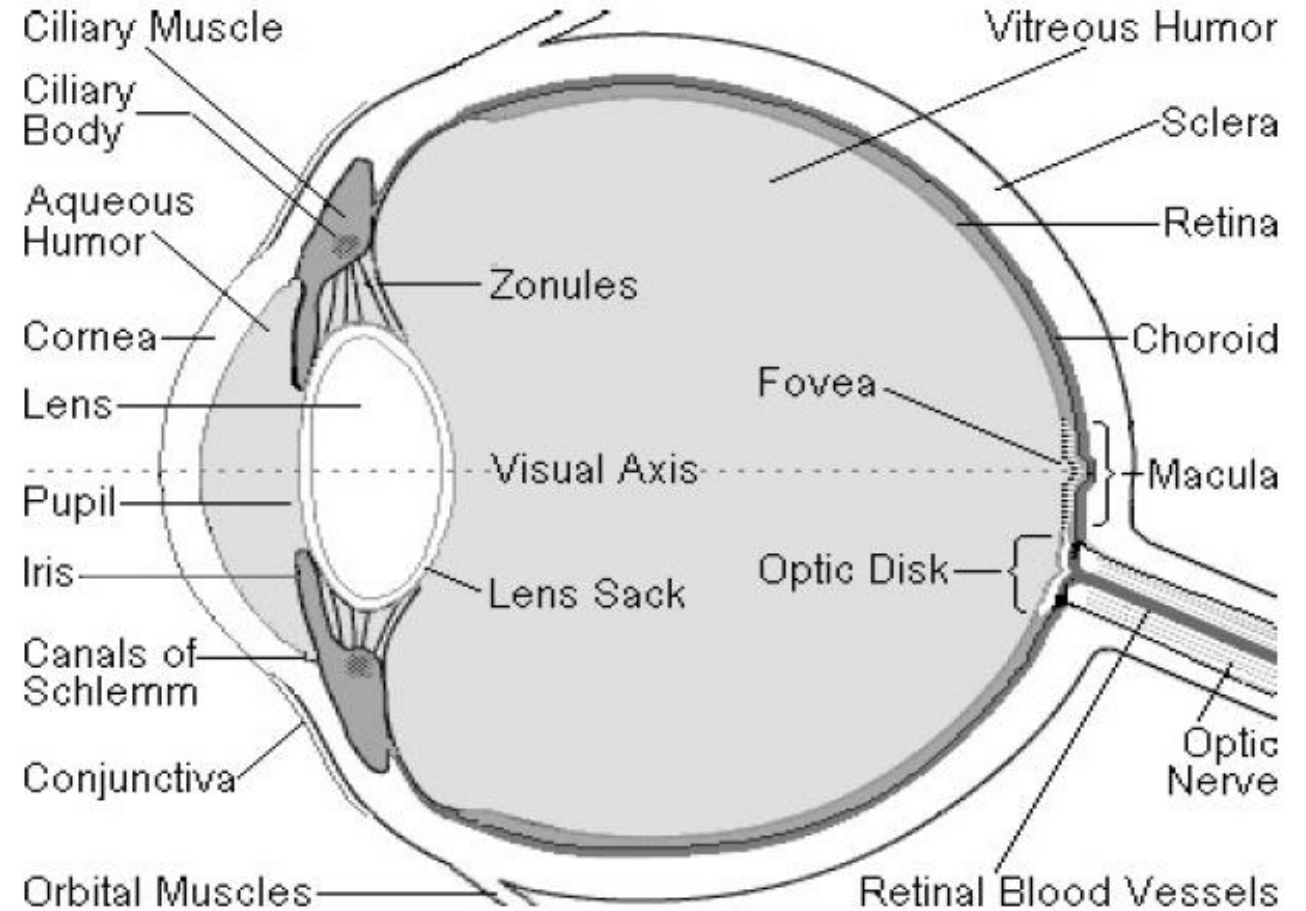
- contains no photoreceptors, i.e. no rods or cones overlying the optic disc.
- entry point for the major blood vessels that supply the retina.
- represents the beginning of the optic nerve.
- carries 1–1.2 million afferent nerve fibres from the eye towards the brain.

The eye as a camera:

- ☐ A lens system,
- ☐ A variable aperture system (the pupil)
- ☐ A film (the retina)

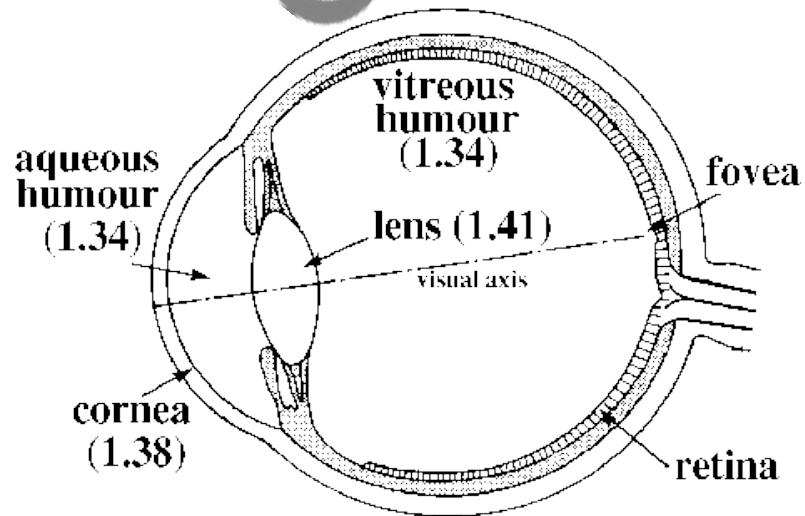


Pupil works like aperture



The lens system of eye is composed of four refractive interfaces:

1. The interface between air and the anterior surface of the cornea,
2. The interface between the posterior surface of the cornea and the aqueous humor,
3. The interface between the aqueous humor and the anterior surface of the crystalline lens of the eye, and
4. The interface between the posterior surface of the lens and the vitreous humor.



Refractive Index (RI)

Air: 1
Cornea: 1.38
Aqueous humor: 1.34
Crystalline lens (on average): 1.41
Vitreous humor: 1.34

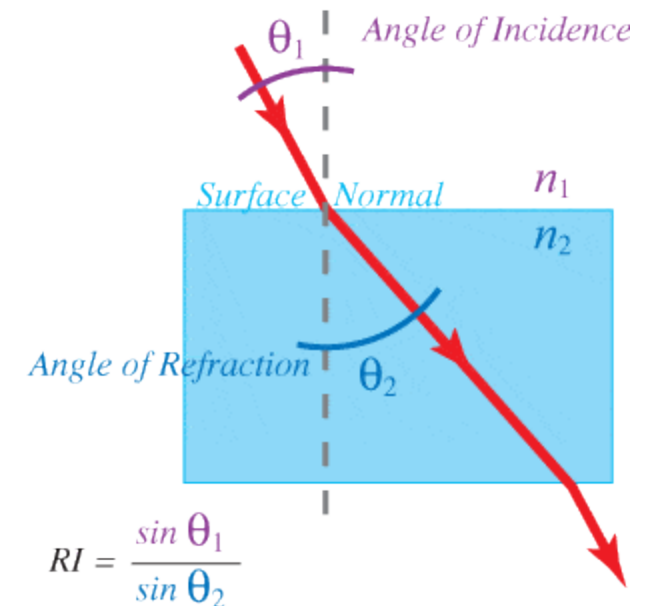
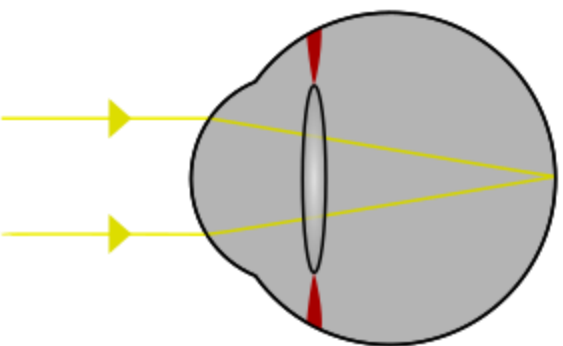
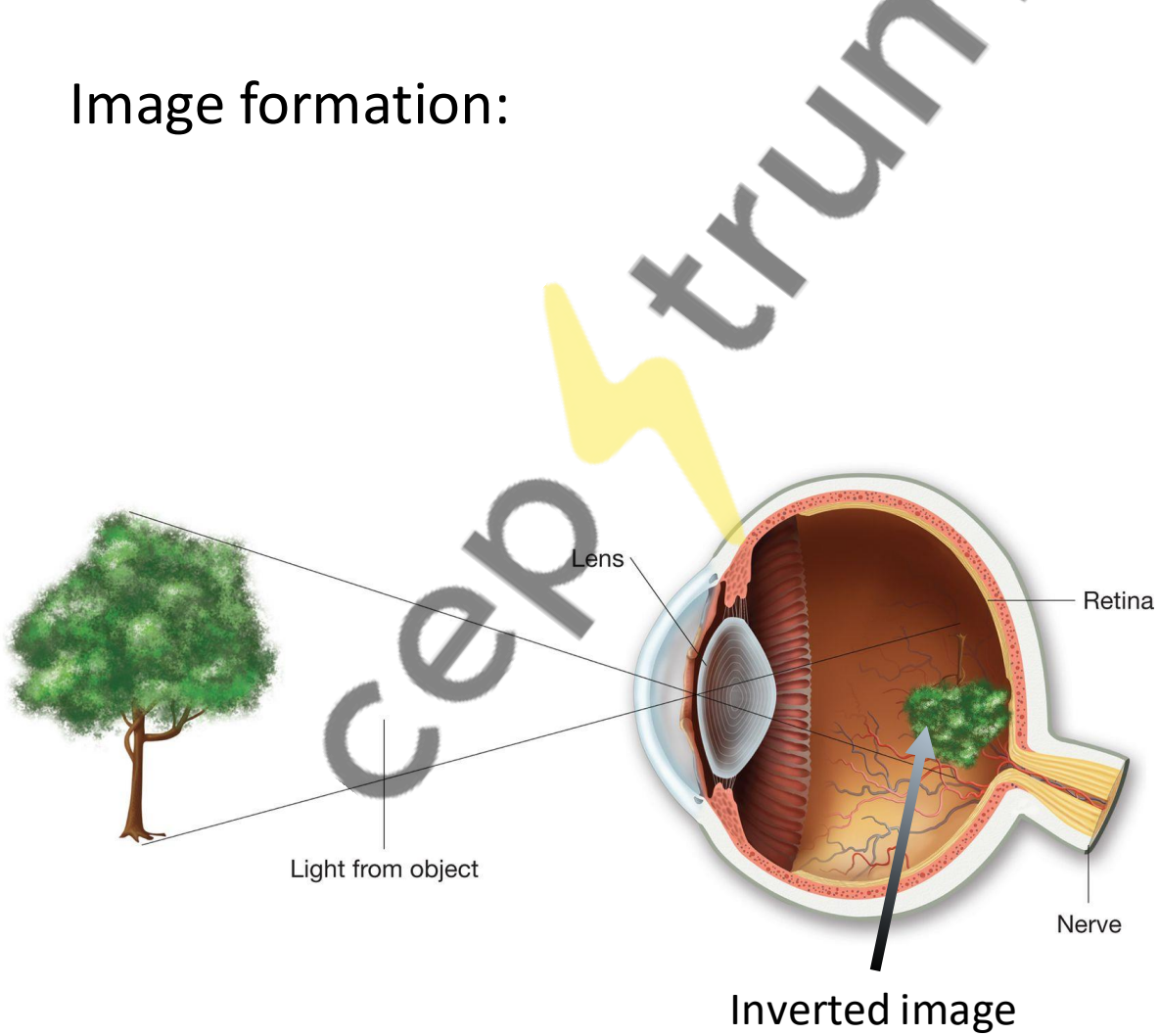
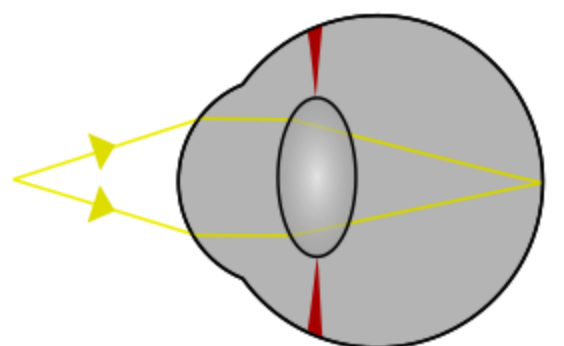


Image formation:

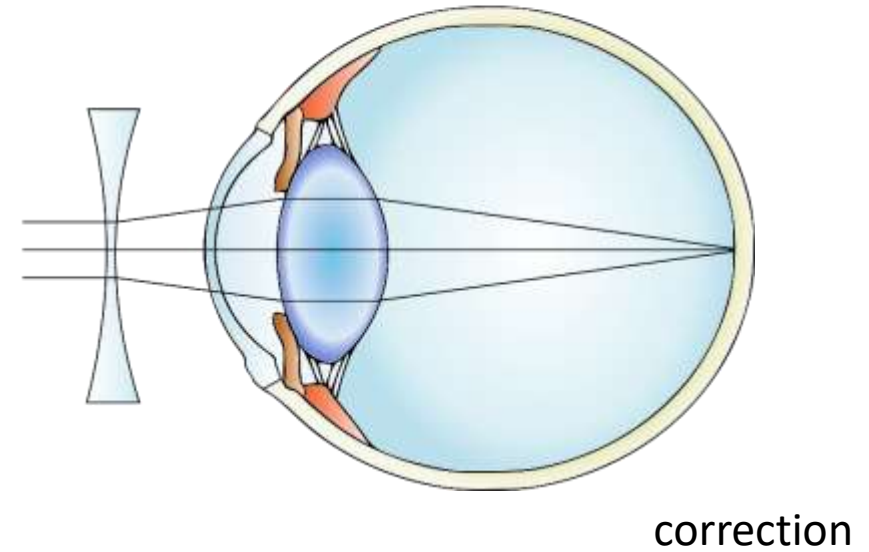
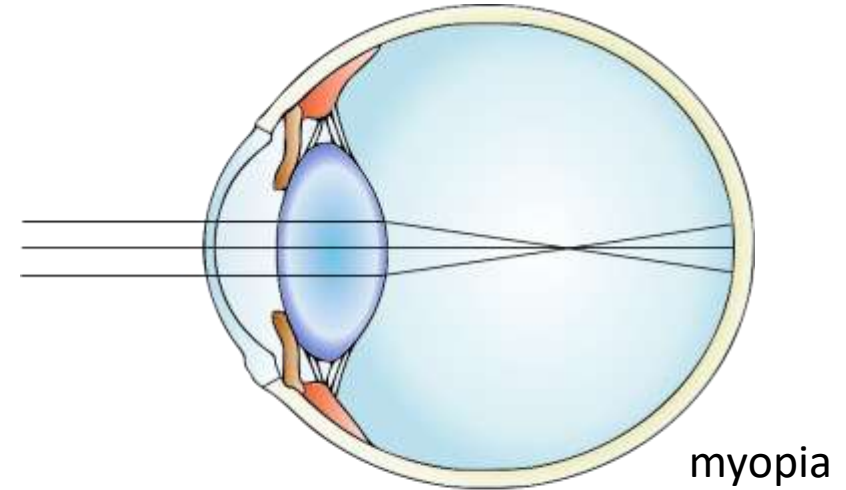
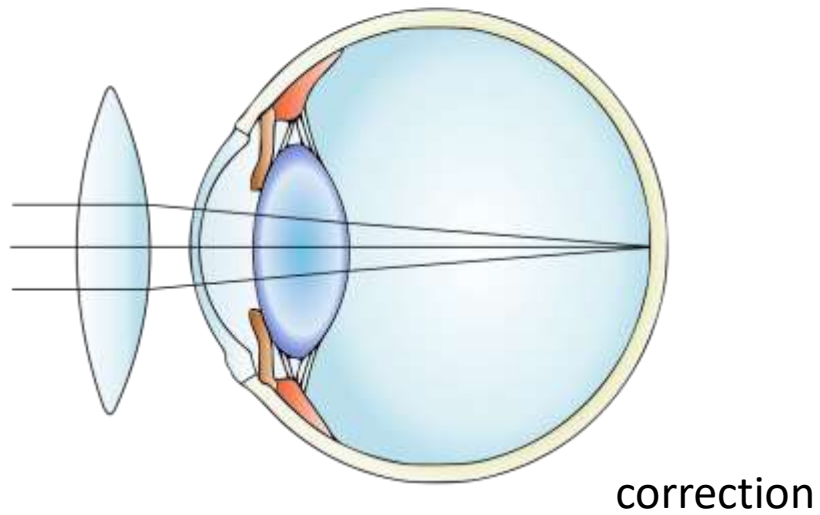
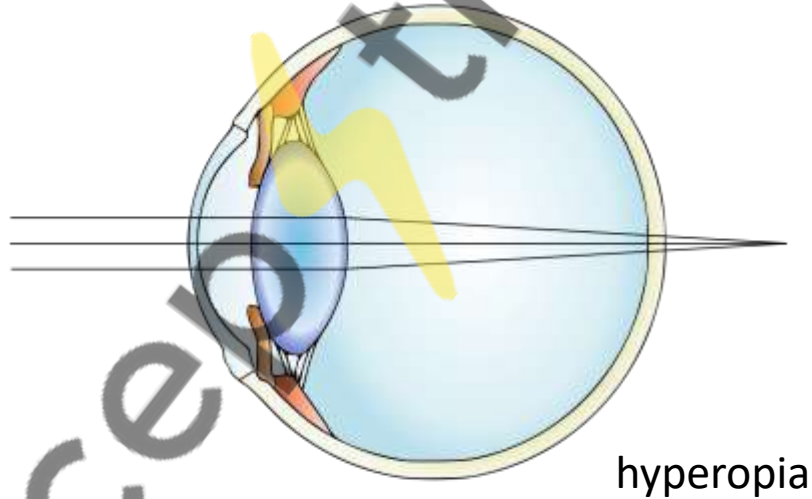


Far object



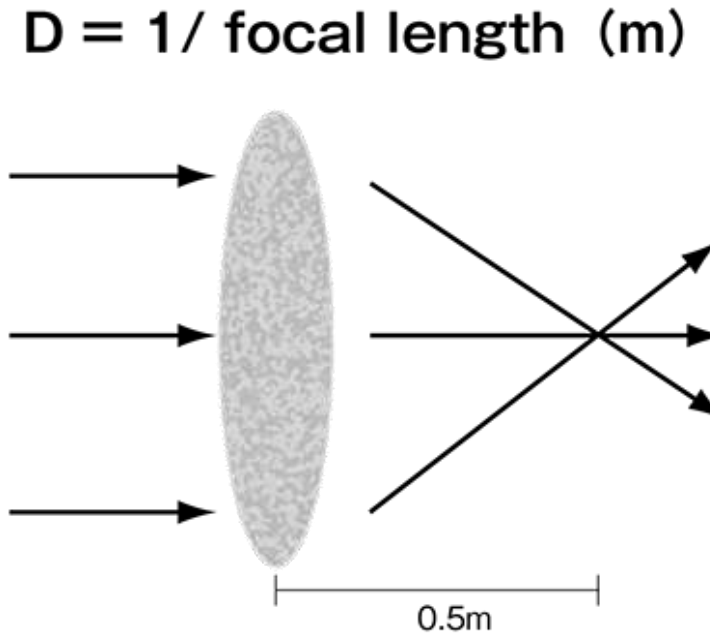
Near object

Farsightedness (Hyperopia) and Nearsightedness (Myopia)



Refractive Power (RP) of lens

Dioptr (D) is the reciprocal number of the focal length (measured in meter). It represents the refractive power (RP) of the lens. Convex lenses have a +ve dioptr, whereas concave lenses have –ve dioptr.



$$D = 1 / 0.5\text{m} = 2$$

What is emmetropic vision?

Definitions

What is hyperopia?

Hyperopia, also known as farsightedness, is a common type of refractive error where distant objects may be seen more clearly than objects that are near. For people with significant hyperopia, vision can be blurry for objects at any distance, near or far.

What is myopia?

Otherwise known as nearsightedness, myopia occurs when the eye grows too long from front to back. Instead of focusing images on the retina, the lens of the eye focuses the image in front of the retina. People with myopia have good near vision but poor distance vision.

What is binocular vision???

Fluid System of the EYE – Intraocular Fluid

- ✓ Keeps the eyeball round and distended by maintaining pressure.
- ✓ Two chambers – Aqueous and Vitreous.
- Aqueous humor – free flowing fluid in front of the lens.
- Vitreous humor – gelatinous mass with little flow of fluid at the back of the lens.

VITREOUS HUMOR

- A clear avascular gel which occupies the posterior compartment of the eye.
- ***It has the following composition:***
 - Water (99%)
 - Network of collagen fibrils
 - Large molecules of hyaluronic acid
 - Peripheral cells (hyalocytes)
 - Inorganic salts, sugar and ascorbic acid

Aqueous Humour

- It is a clear, colourless, watery fluid filling the anterior chamber and posterior chamber of the eyeball.
- Volume: 0.31 ml
 - Anterior chamber– 0.25ml
 - Posterior chamber– 0.06ml
- Refractive Index: 1.336
- Density : 1.025–1.040 (greater than water)
- pH : 7.2 (acidic)
- Rate of formation: 2.3µl/min

Ocular tonometry

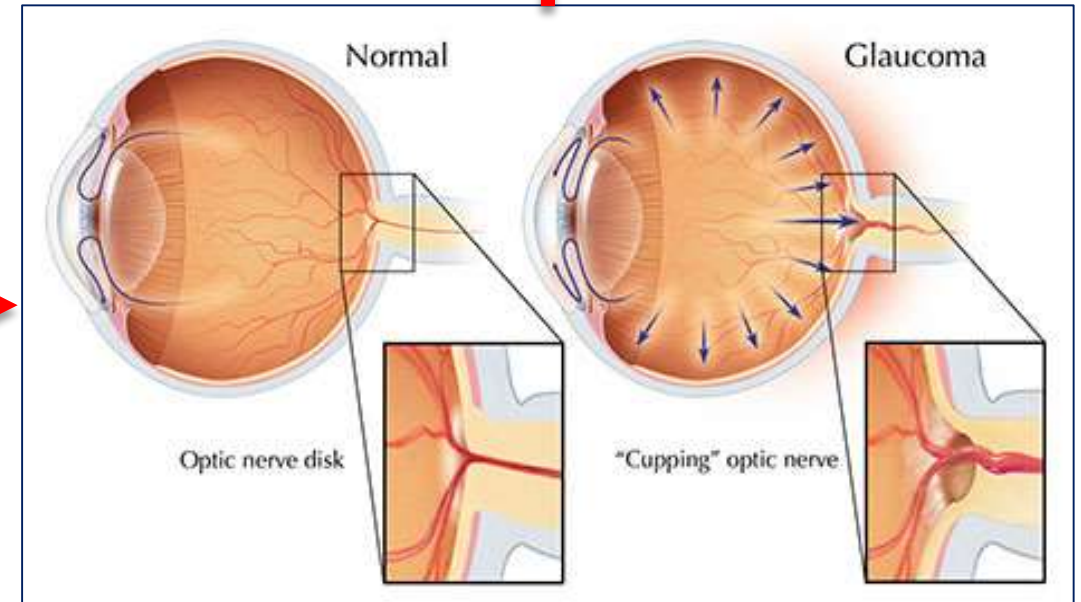
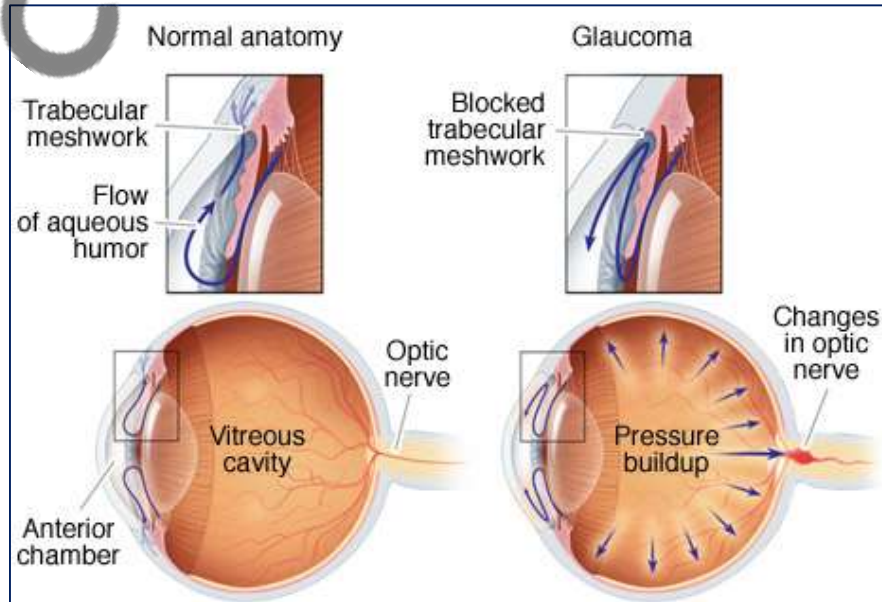
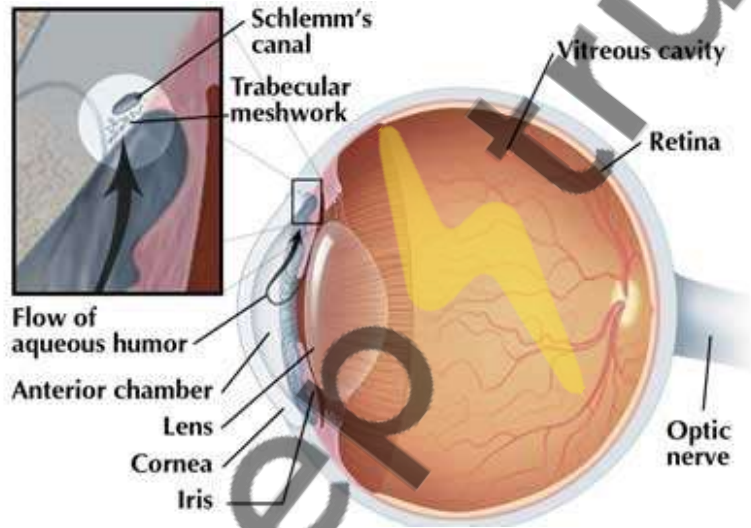
- ❑ Procedure to measure intraocular pressure (IOP), i.e. the fluid pressure inside the eye.
- ❑ Important test to evaluate the risk of glaucoma.
- ❑ Different types of tonometers are available based on various techniques.



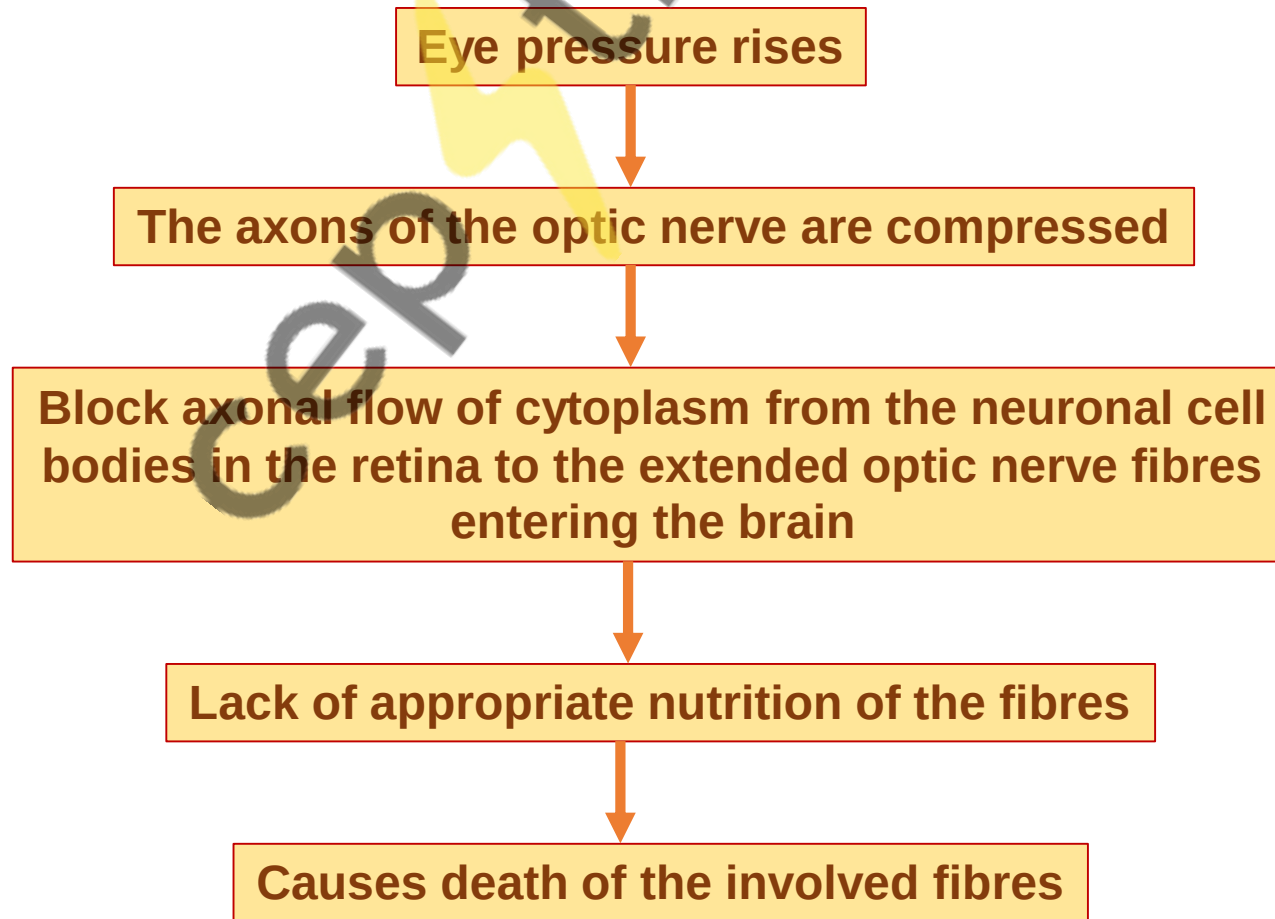
What is tonometer?

Tonometer is a device that measures intraocular pressure (IOP).

Glaucoma

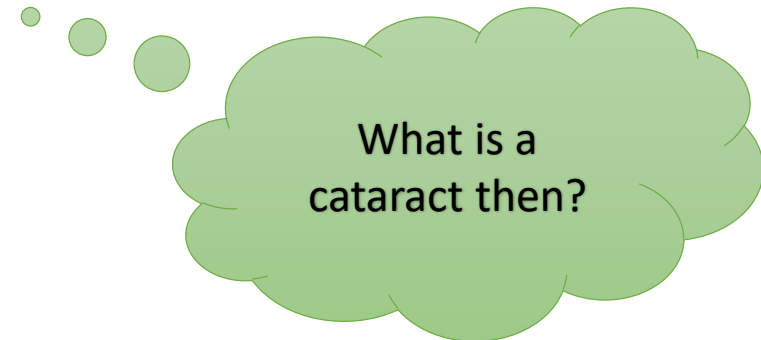


Glaucoma – the process

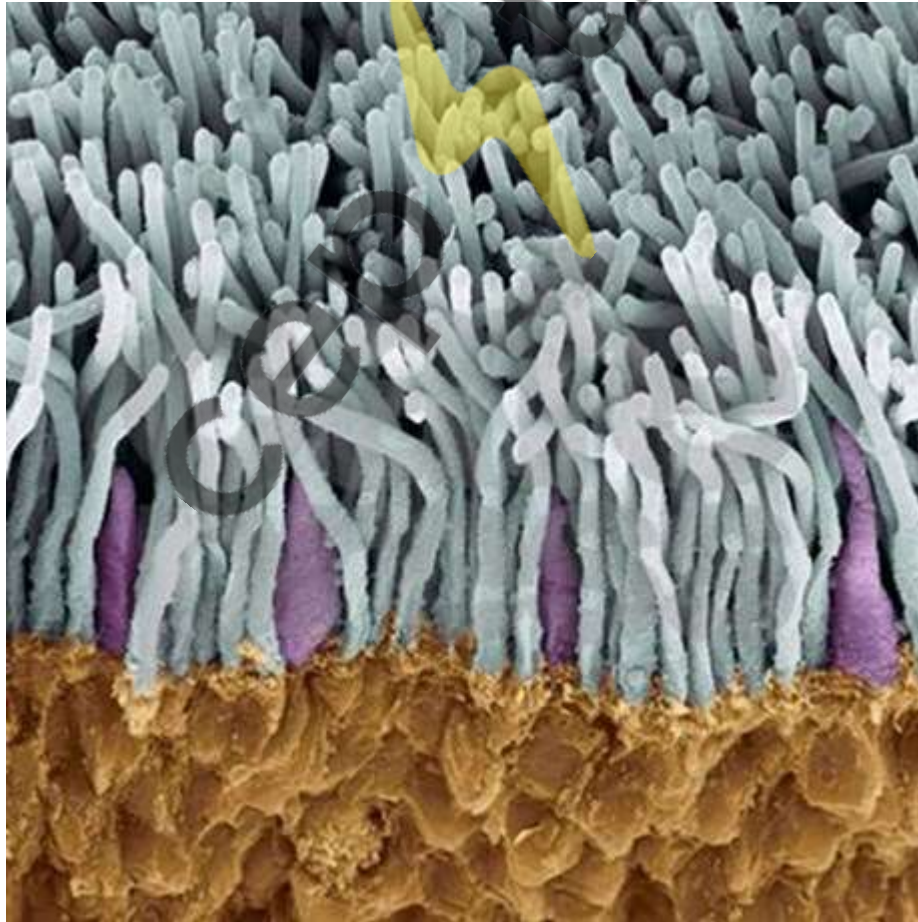


Surgical treatments:

1. Laser Trabeculoplasty
2. Trabeculectomy



Photoreceptors – Rods and Cones



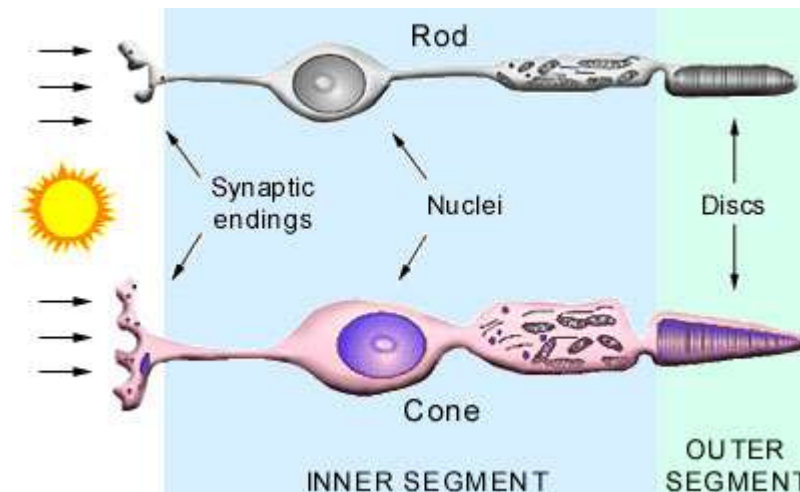
SEM image of rods and cones

6 million cones and 120 million rods

Rods are responsible for night vision; Cones provide colour vision in brighter light.

Why no colour in dim light!!!

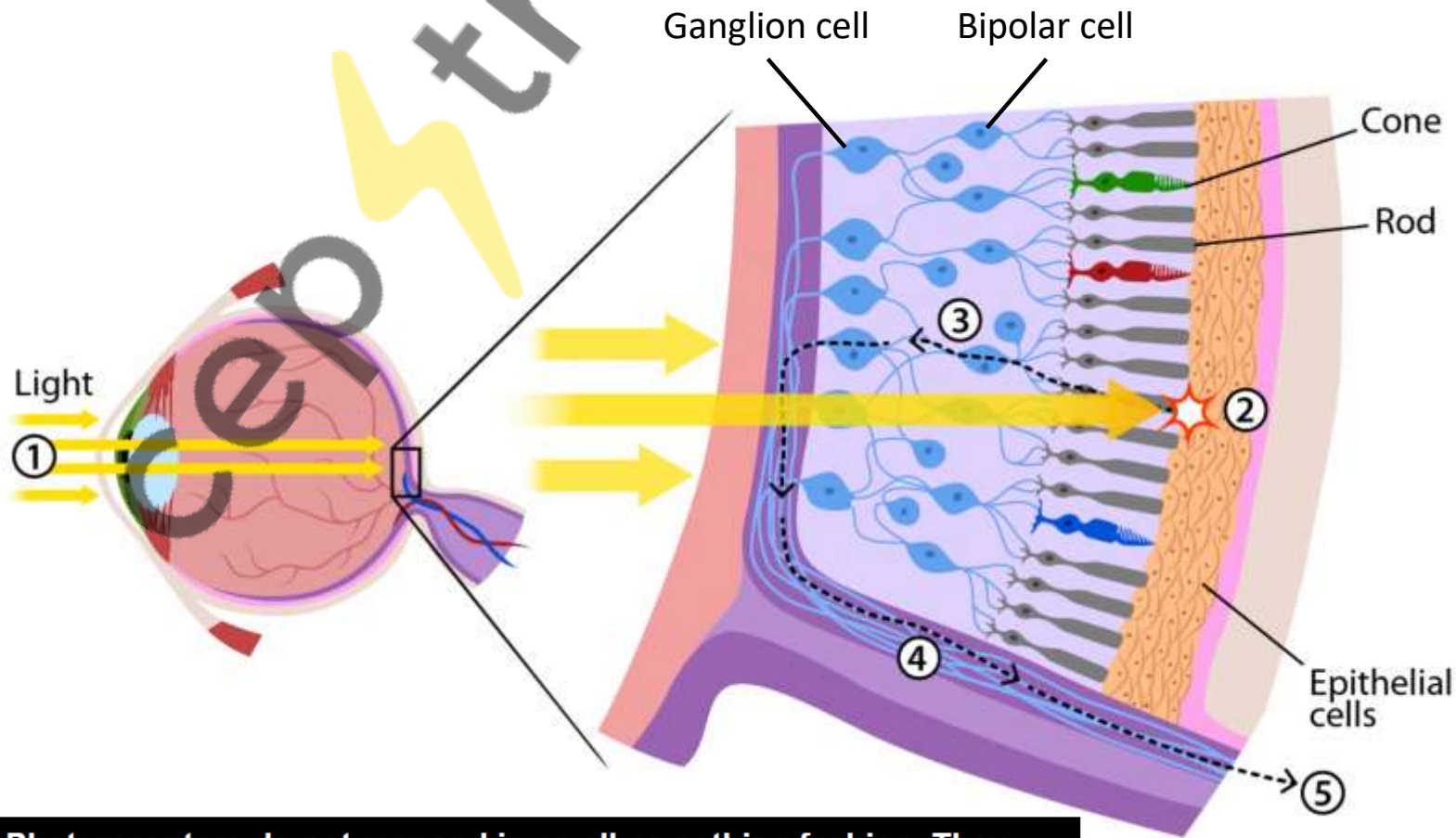
Blue
Green
Red



Lost rod vision?
Should not drive
at night.

Lost cone vision?
Should not drive
at all !

Flow of image information



Chromophore: The light-catching visual pigments of the retina found in cones.

Rhodopsin: The visual pigment found in rods.

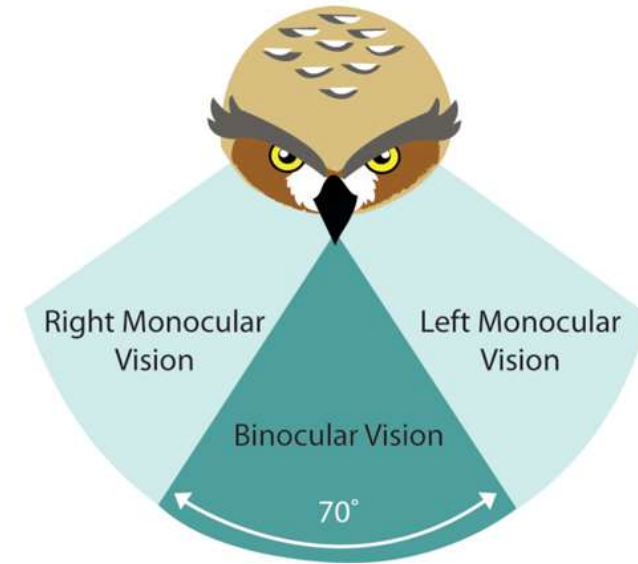
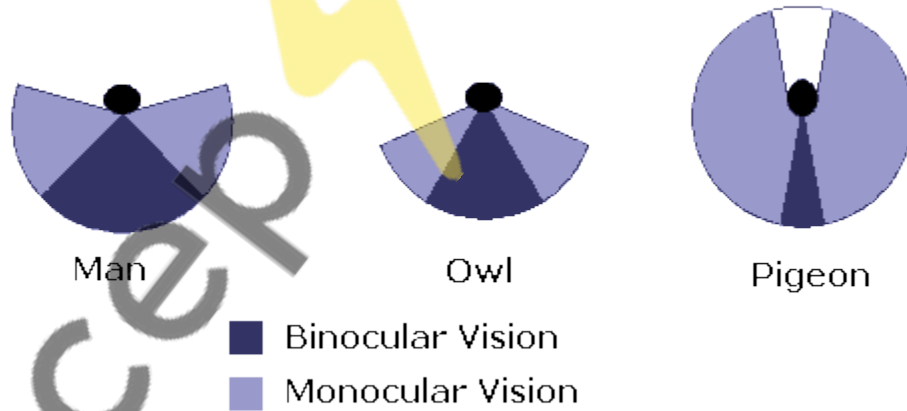
Photoreceptors do not respond in an all-or-nothing fashion. They pass their information on to the bipolar cells via graded potentials instead of action potentials.

Graded potential: An electrical potential that can vary continuously in amplitude.

What is Night Blindness?

Bird vision

Good eyesight is essential for safe flight!



- Avian eye resembles that of a reptile, with ciliary muscles that can change the shape of the lens rapidly and to a greater extent than in the mammals.
- Birds have the largest eyes relative to their size in the animal kingdom.
- Avian eye is protected by a third transparent movable membrane.
- Birds of prey have a very high density of receptors to maximise visual acuity.
- Nocturnal birds have tubular eyes, low no. of colour detectors, but a high density of rod cells to function well in poor light.

Let's try something interesting ...



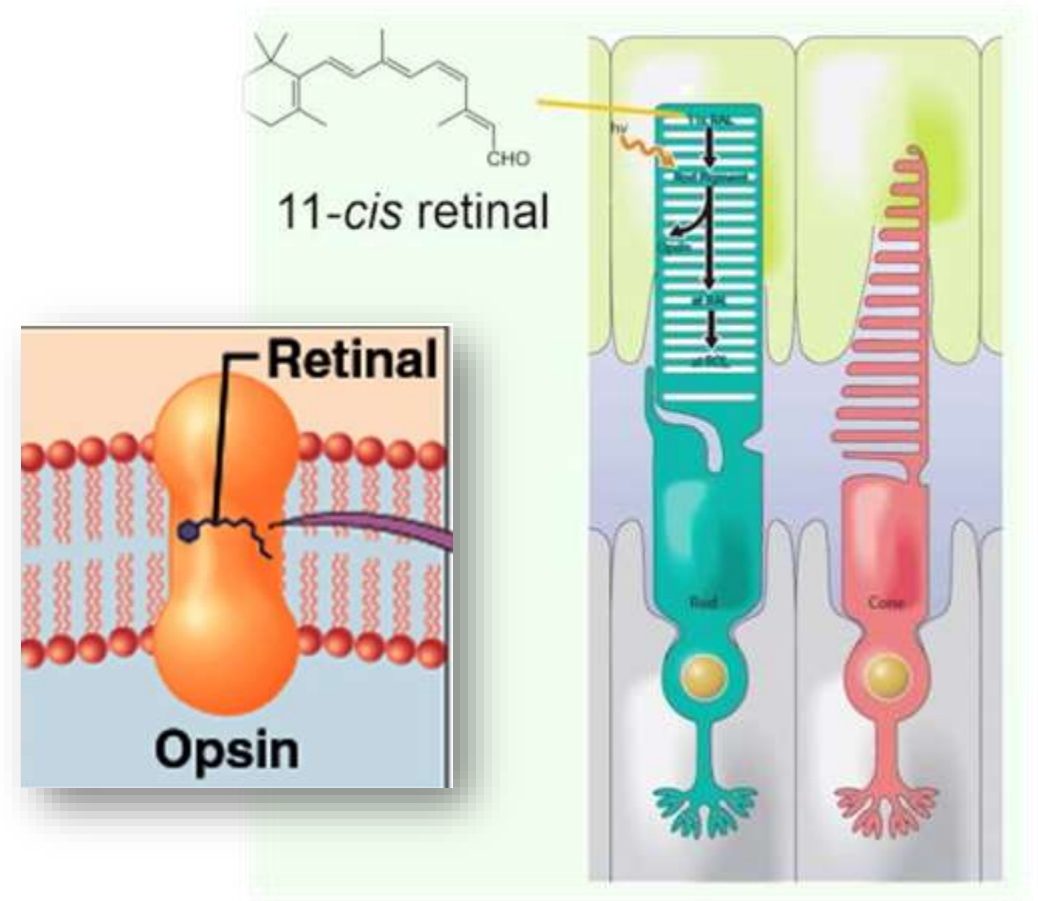
Did you feel the blind spot in your right eye ???

Visual Cycle

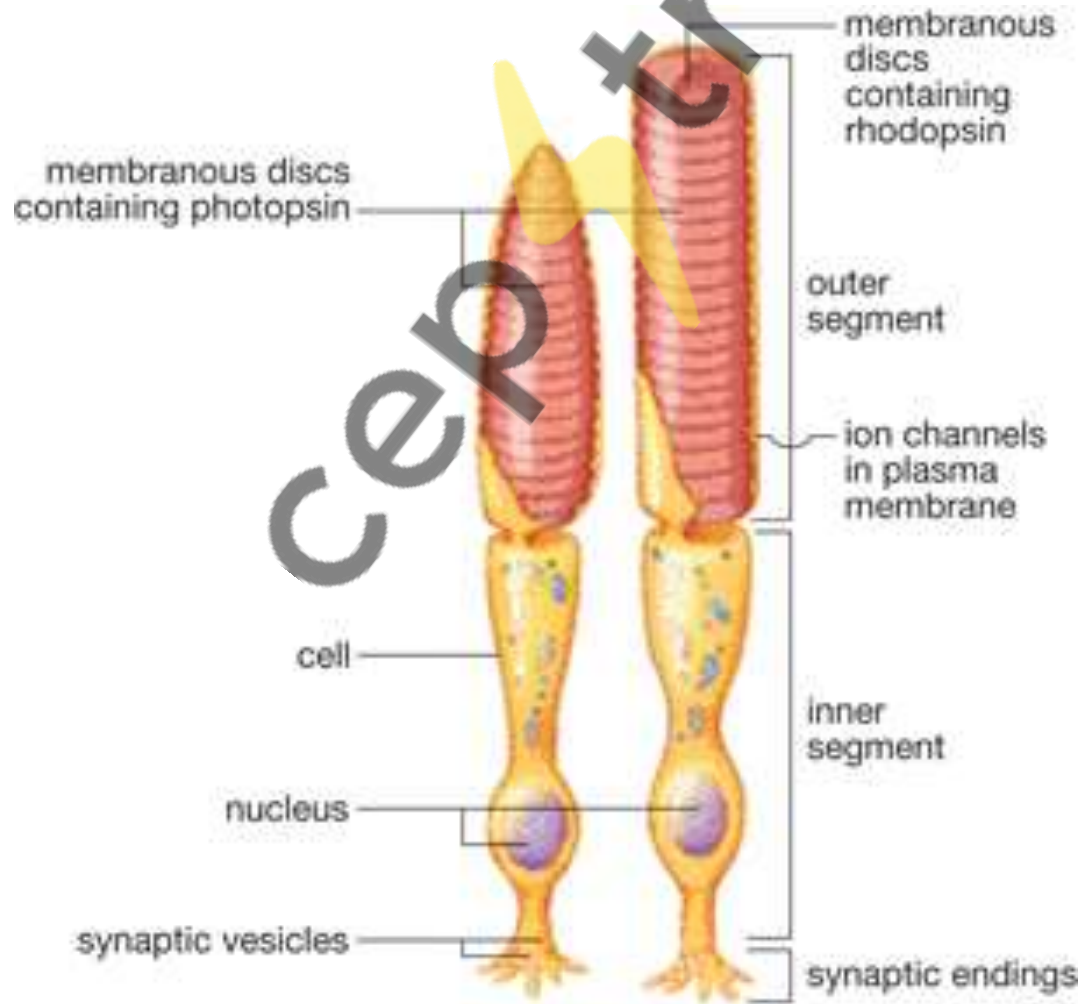
- ❑ It is a process by which light (or photon) is converted into electrical signals in the rod, cone and photosensitive ganglion cells of the retina of the eye.
- ❑ This cycle was elucidated by **George Wald** (1906–1997) for which he received the Nobel Prize in 1967.
- ❑ It is called '**Wald's Visual Cycle**' after him.
- ❑ It is also known as **visual phototransduction**.

Photopigments

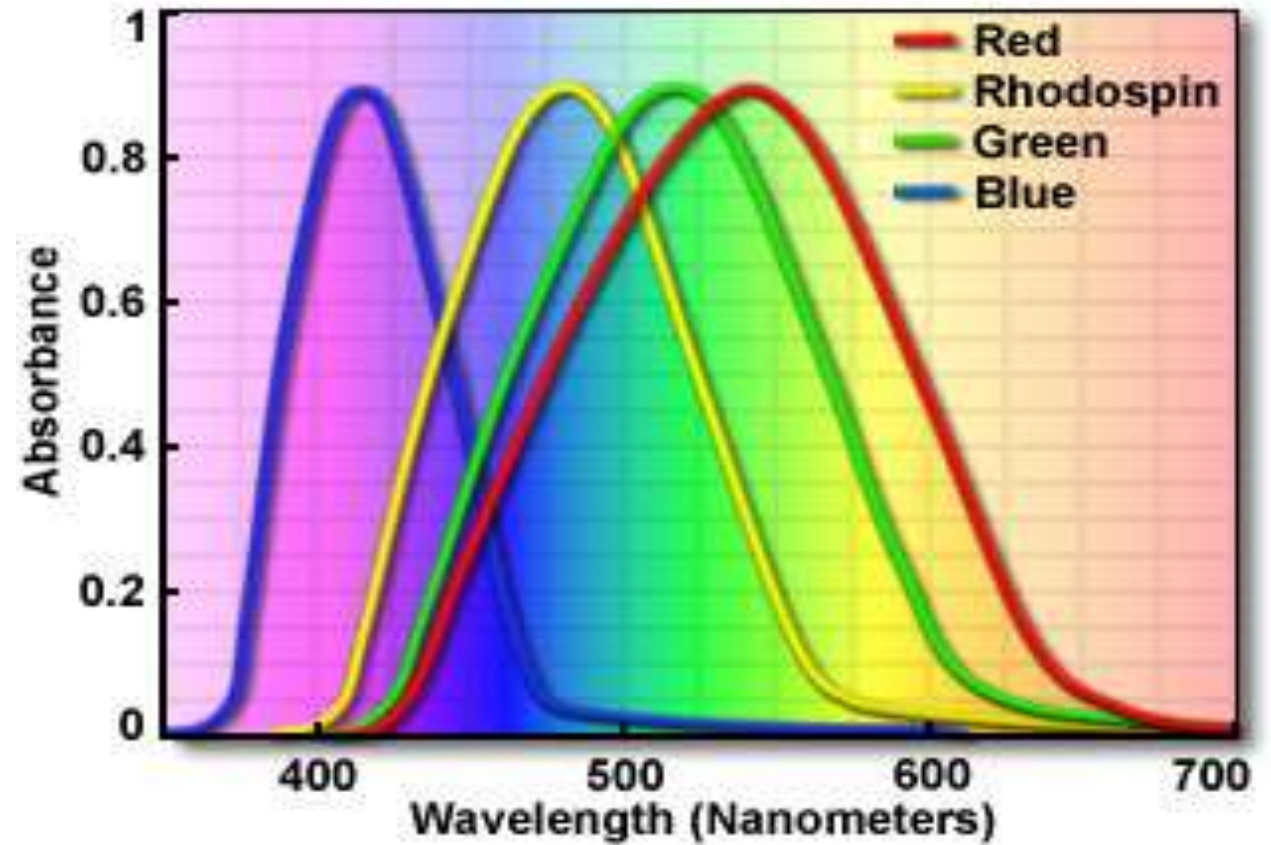
- All photopigments associated with vision contain two parts:
 1. A glycoprotein known as **opsin** and
 2. derivative of vitamin A called **retinal**.
- Vitamin A derivatives are formed from carotene, the plant pigment that gives carrots their orange colour.
- Good vision depends on adequate dietary intake of carotene-rich vegetables such as carrots, spinach, broccoli, or foods that contain vitamin A, e.g. liver.
- Retinal is the light absorbing part of all photopigments. There are 4 types of opsins in human retina, 3 in the cones and 1 in the rods (Rhodopsin), depending on small variations in the amino acid sequences.



Photopigments

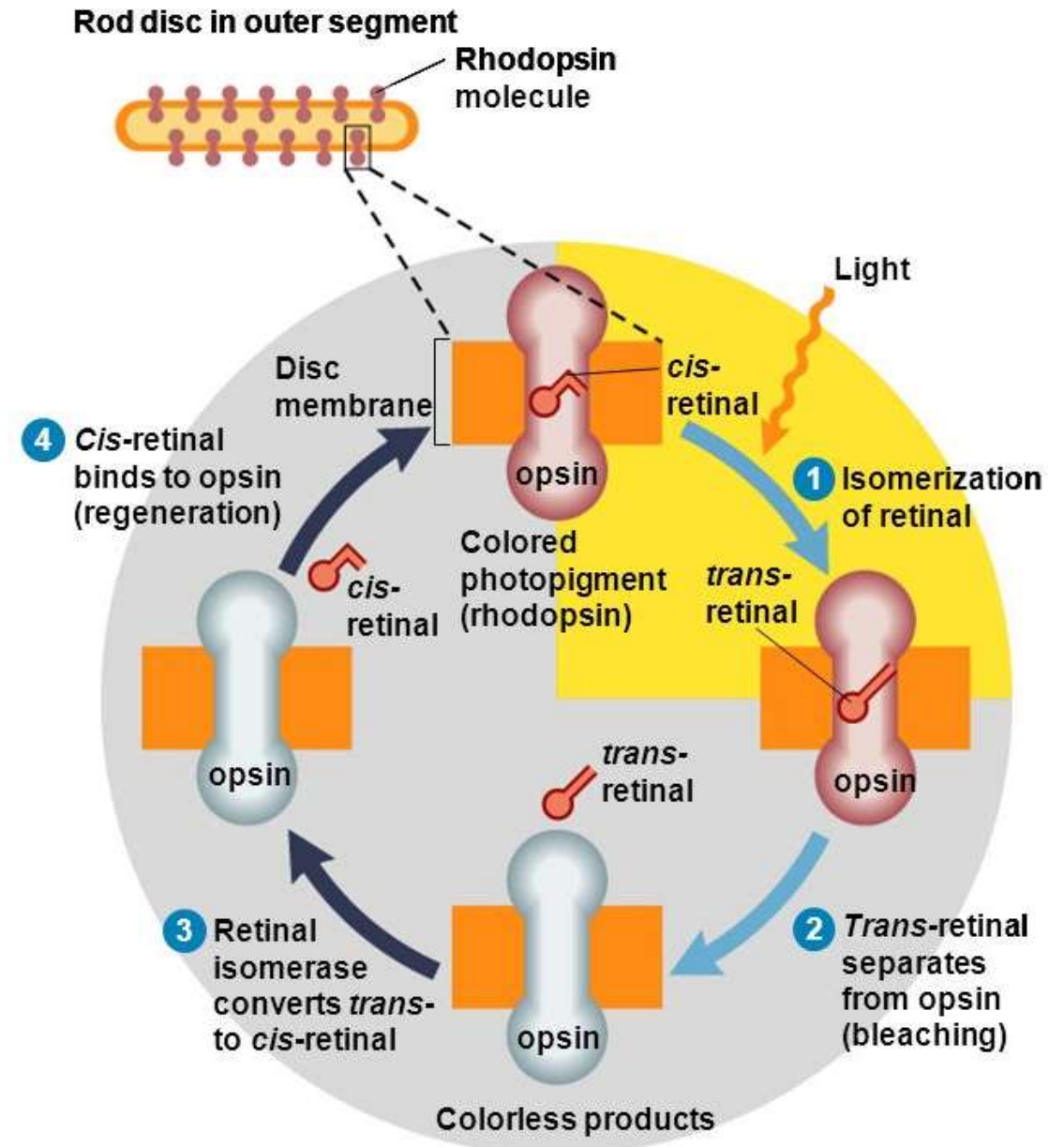


Absorption Spectra of Human Visual Pigments

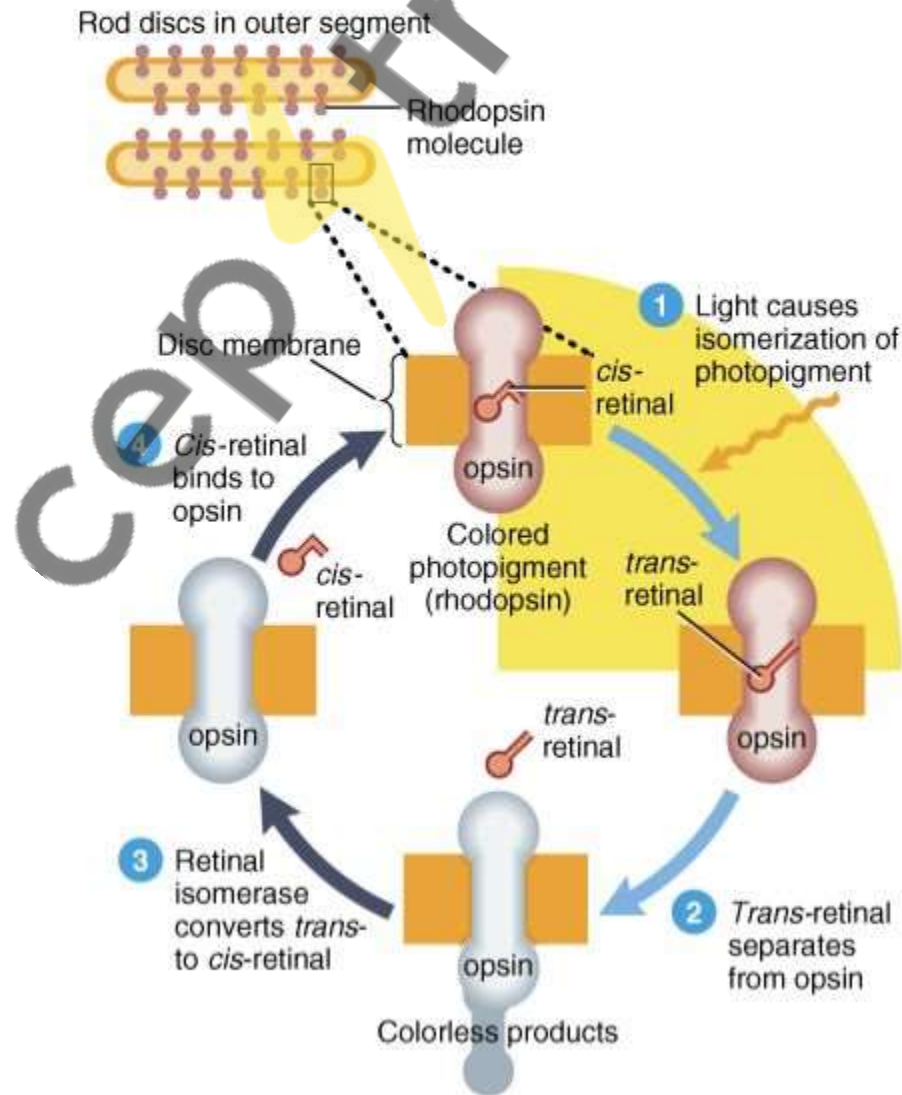


Visual cycle – the process ...

1. In darkness, retinal has a bent shape, called **cis-retinal**, which fits snugly into the opsin. When **cis-retinal** absorbs a photon of light, it straightens out to a shape called **trans-retinal**. This cis-to-trans conversion is called **isomerization** and is the first step in visual cycle. Several unstable chemical intermediates (e.g. ions) form and disappear in this stage, leading to production of a receptor potential.
2. In about a minute, **trans-retinal** completely separates from opsin. This part of the cycle is called **bleaching** of photopigment.
3. An enzyme called **retinal isomerase** converts **trans-retinal** back to **cis-retinal**.
4. The **cis-retinal** then can again bind to opsin, reforming a functional photopigment. This part of the cycle is called **regeneration**.

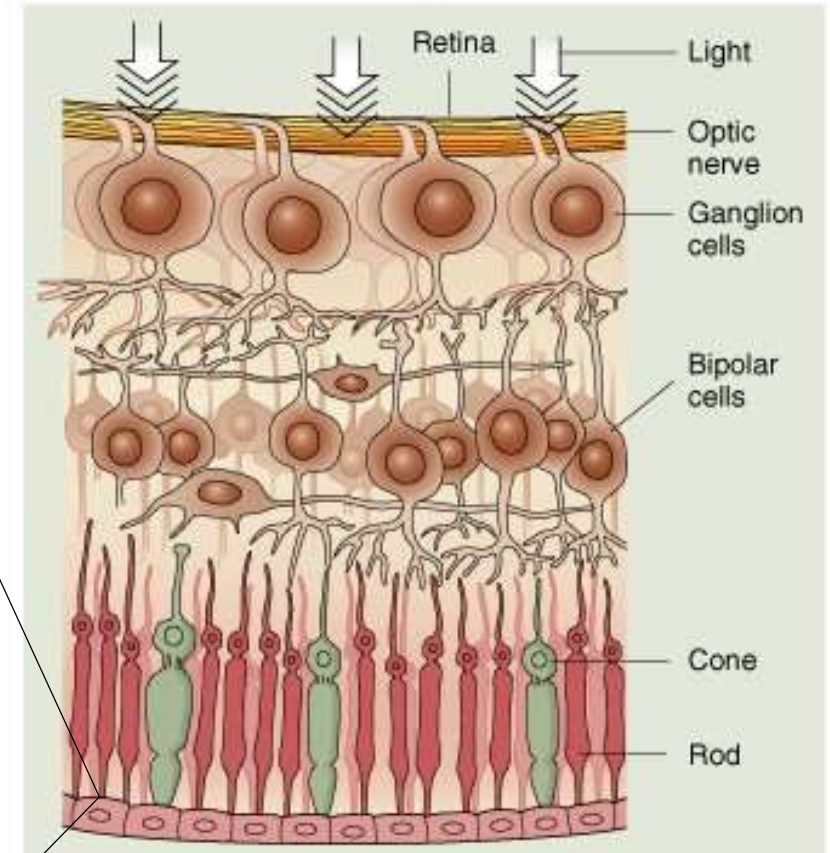
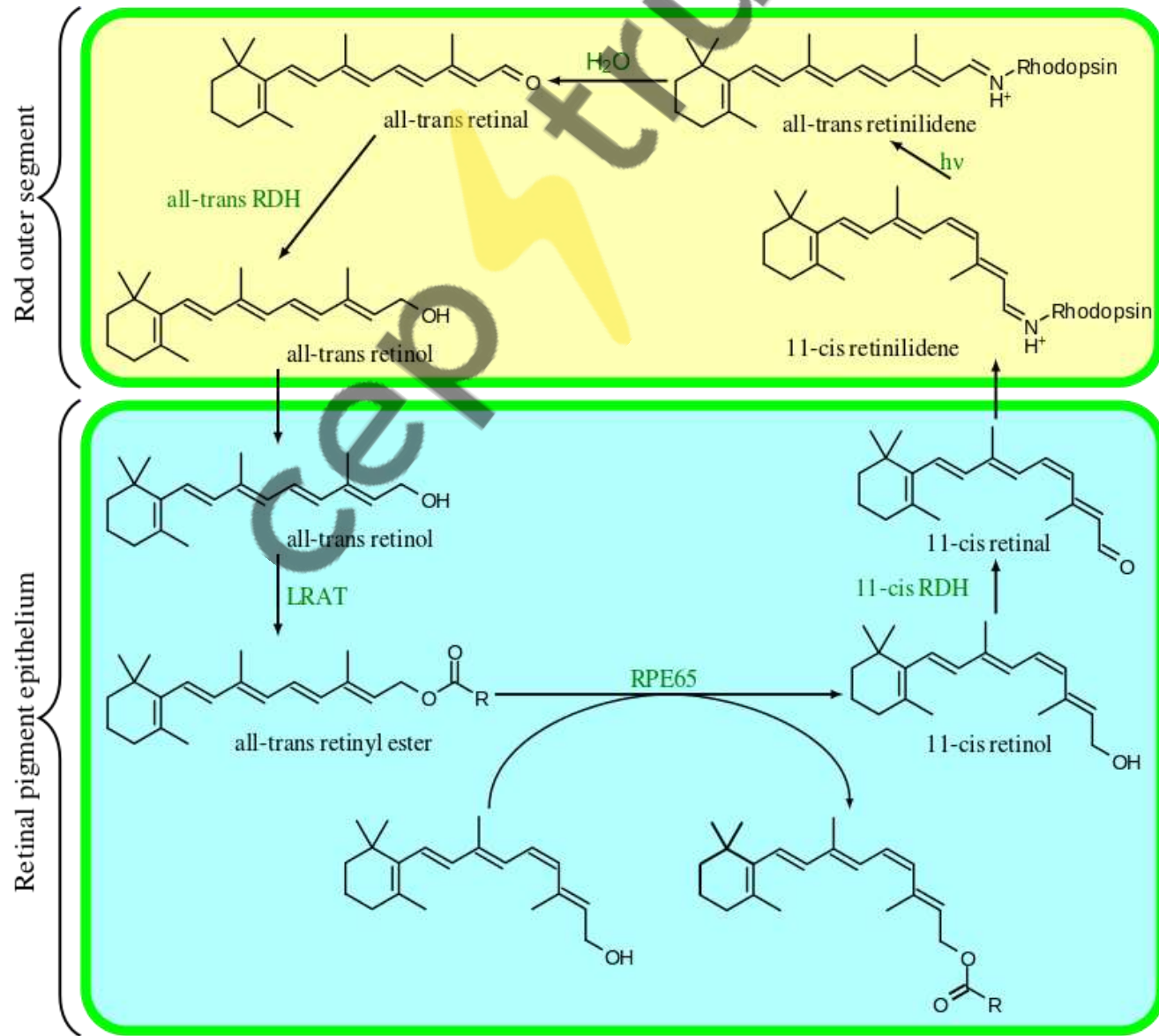


Photopigments



- Isomerization
 - light cause *cis*-retinal to straighten & become *trans*-retinal shape
 - lead to production of receptor potential
- Bleaching
 - enzymes separate the *trans*-retinal from the opsin
 - colorless final products
- Regeneration
 - in darkness, an enzyme converts *trans*-retinal back to *cis*-retinal (resynthesis of a photopigment)

Visual cycle – the process ...



Some important notes ...

- ☐ The pigmented layer of the retina adjacent to the photoreceptors stores a large quantity of vitamin A and contributes to the regeneration process in rods.
- ☐ The extent of rhodopsin generation decreases drastically if the retina detaches from the pigmented layer.
- ☐ Cone photopigments regenerate much more quickly than rhodopsin in rods and are less dependent on the pigmented layer.

Half of all rod pigments take 5 minutes to regenerate.

Half of all cone pigments take 90 seconds only.

Light and dark adaptation

Light adaptation: Dark → Light

Fast. Happens in seconds

As light level increases, more and more photopigments are bleached.

While light is bleaching some pigment molecules, others are being regenerated. In daylight regeneration of rhodopsin cannot keep up with the bleaching process.

Hence rods contribute little to daylight vision. In contrast, cone photopigments regenerate rapidly enough even in bright light.

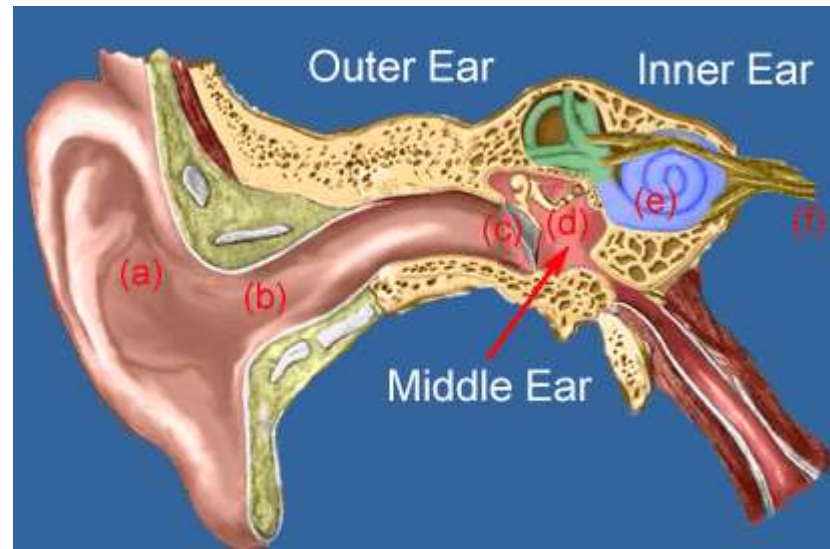
Dark adaptation: Light → Dark

Slow. Takes many minutes

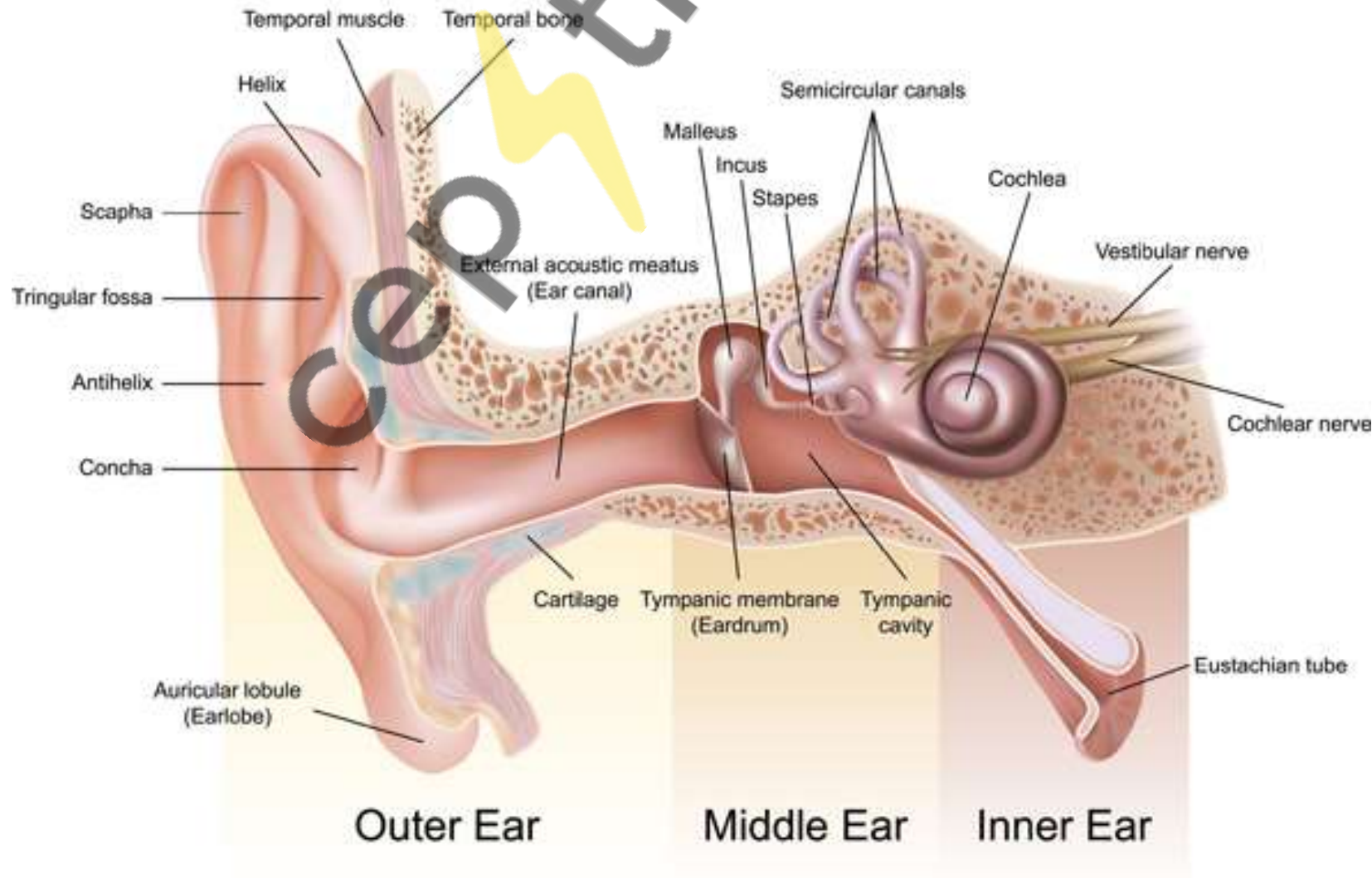
If light level decreases abruptly, sensitivity increases rapidly at first and then more slowly. In complete darkness, full regeneration of cone pigments occurs during the first 8 minutes of dark adaptation. During this time, a threshold light flash is seen as having colour. Rhodopsin regenerates more slowly, and our visual sensitivity increases until even a single photon can be detected. At very low light levels, objects appear as shades of grey since only the rods are functioning.

Hearing

- ☐ Hearing is the ability to perceive sound.
- ☐ The ear is an engineering marvel because its sensory receptors can transduce sound vibrations with amplitudes as small as the diameter of a gold atom (0.3 nm) into electrical signals.
- ☐ In the ear, mechanical energy is converted into electrical signals.
- ☐ The signals are 1000 times faster than photoreceptors can respond to light.
- ☐ The ear also contains receptors for equilibrium.



Anatomy of the Ear



The ear is divided into three main regions:

1. The outer or external ear, which collects sound waves and channels them inward;
2. The middle ear, which conveys sound vibrations to the oval window;
3. The inner ear, which houses the receptors for hearing and equilibrium.

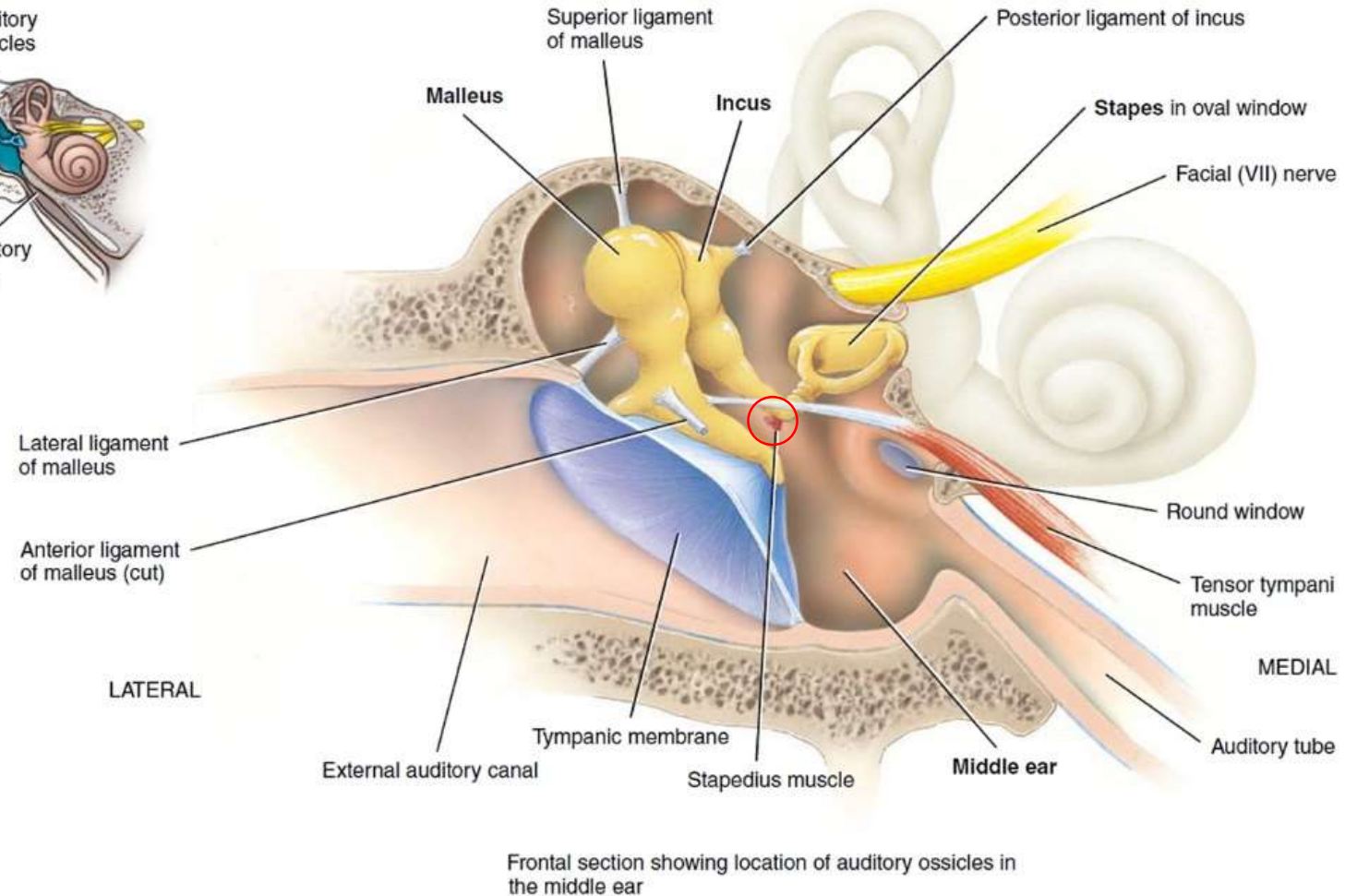
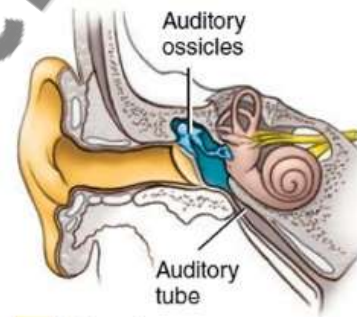
Outer ear

- The outer ear consists of the **auricle**, **external auditory canal**, and **eardrum**.
- The auricle is a flap of **elastic cartilage** shaped like the flared end of a trumpet and covered by skin.
- The rim of the auricle is **helix**; the inferior portion is the **lobule**. Ligaments and muscles attach the auricle to the head.
- The external auditory canal is a curved tube that extends upto the **eardrum**. The eardrum is a thin partition between the external canal and the middle ear.
- The eardrum is covered by **epidermis**.
- The external canal contains a few hairs and specialized sweat glands (**ceruminous glands**) that secrete earwax or **cerumen**.

What are **perforated eardrum** and **impacted cerumen!!!**

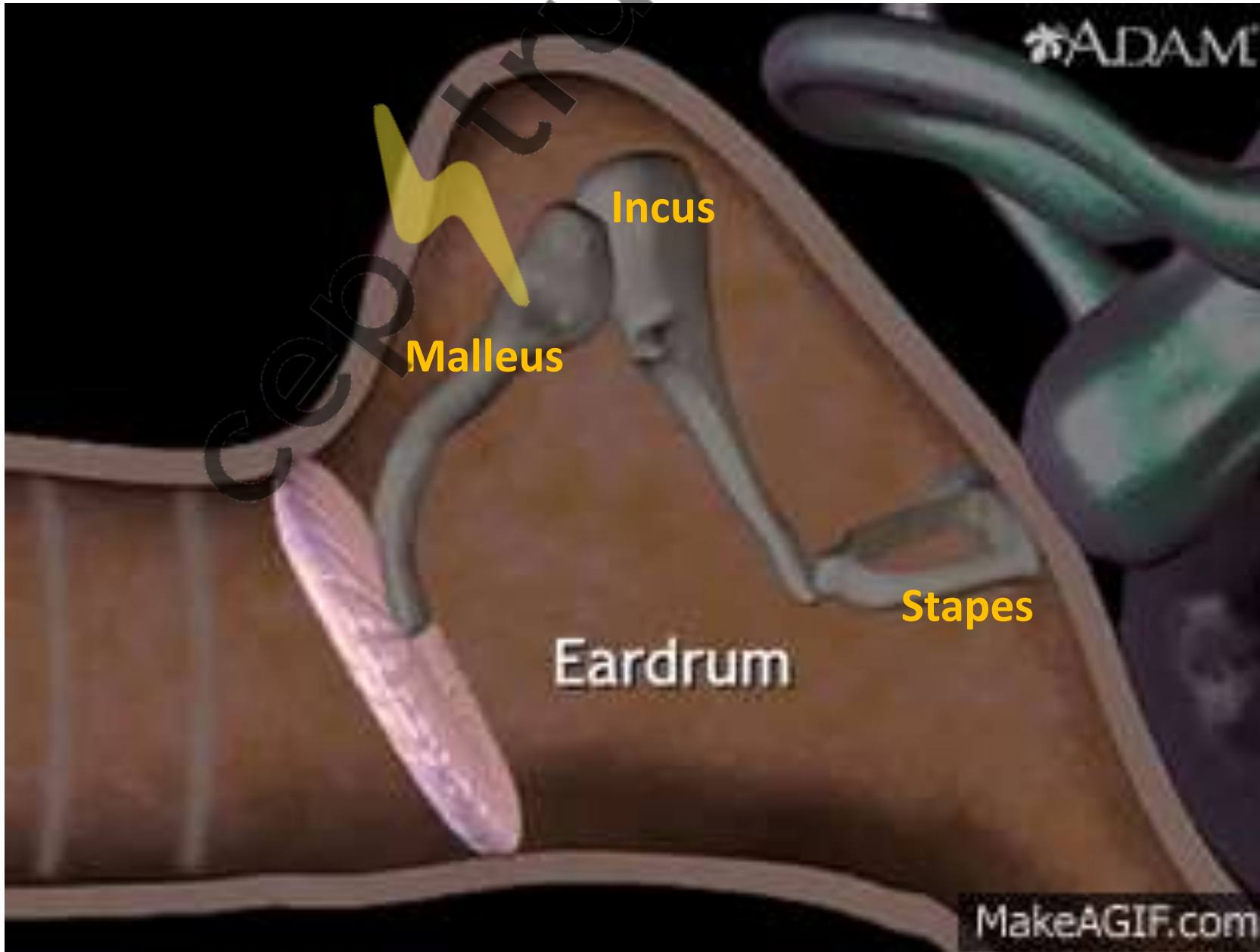
Middle ear

- The middle ear is the small air-filled cavity that is lined by the epithelium.
- Separated from external ear by eardrum.
- **Auditory ossicles: malleus, incus and stapes.**
- The base of the stapes fits into the **oval window**.
- Directly below the oval window is another opening called the **round window**.
- **Muscles/ligaments** limits movements of ossicles to protect inner ear from loud noises.



Smallest skeletal muscle in human body???

Middle Ear – how it works ...



Middle ear
works as
an
amplifier in
two ways

Area difference
(55 to 3.2 sq mm)

Configuration
of ossicles
(Length of
malleus and
incus)

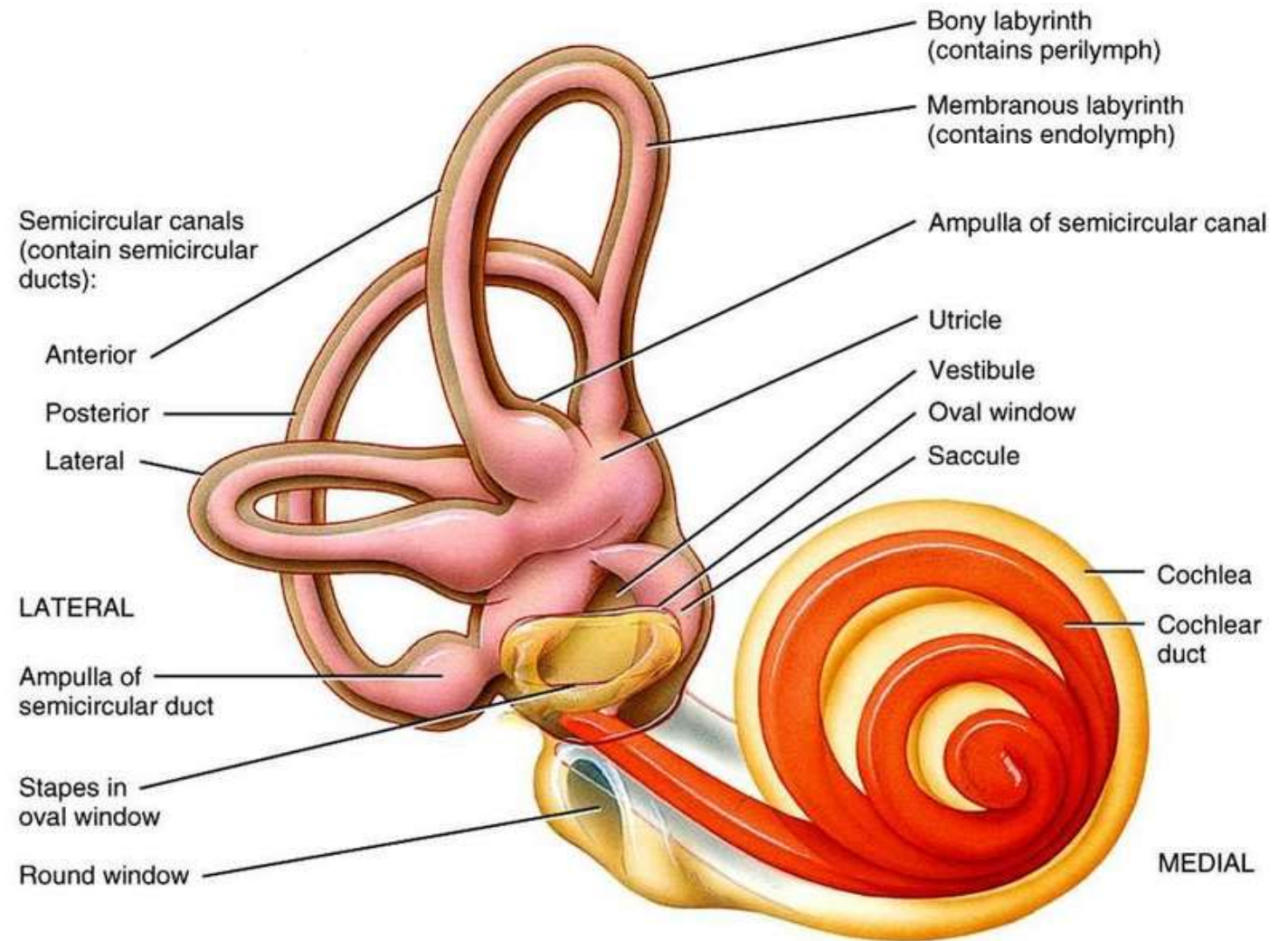
22 times amplification...
But why??



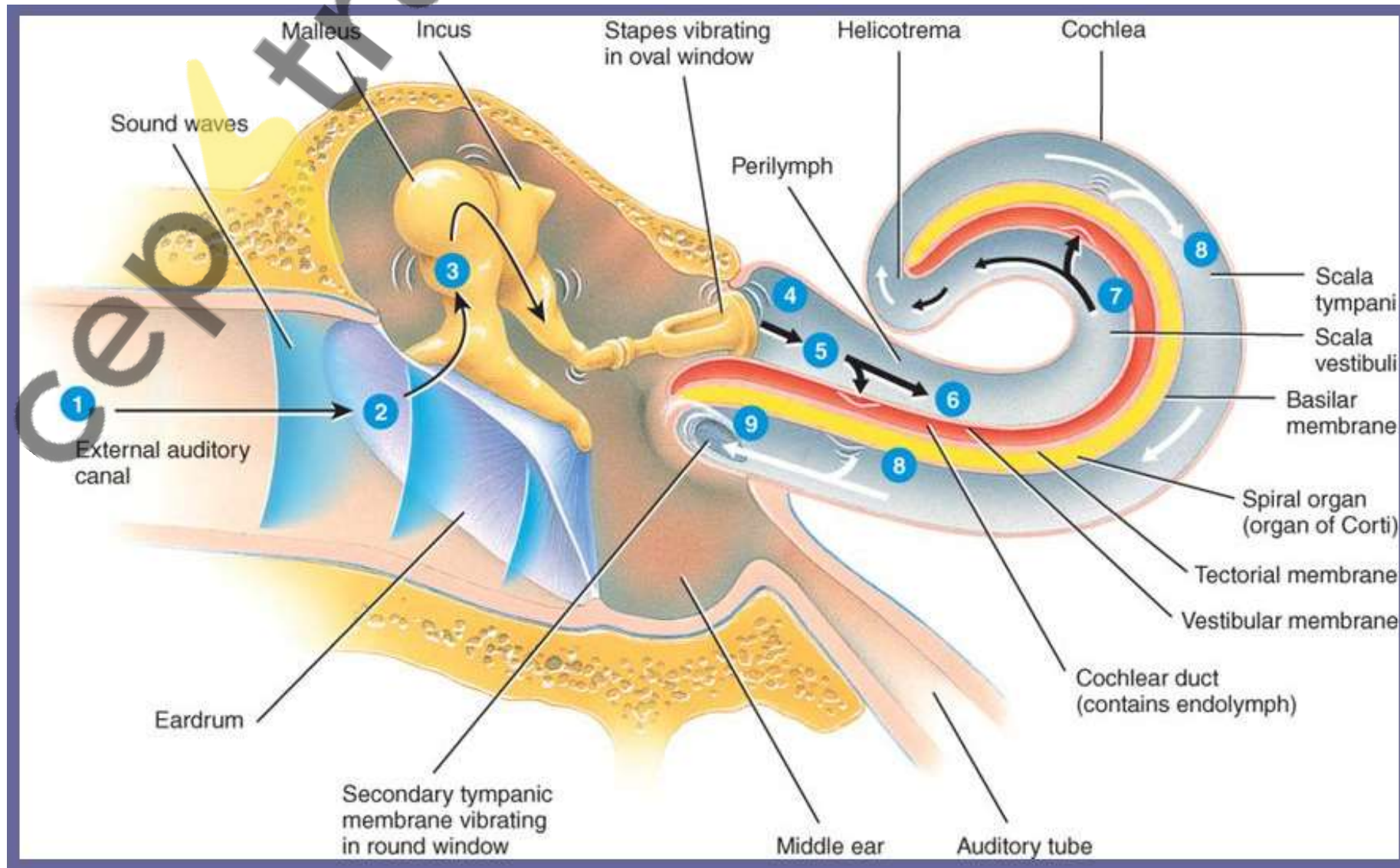
The cochlea in the inner ear conducts sound through a liquid, instead of through air. This fluid has a much higher inertia than air – i.e., it is harder to move. Hence, more pressure is required.

- The Inner ear is also called the **labyrinth**.
- **Bony labyrinth (BL)** and **membranous labyrinth (ML)**.
- BL contains **perilymph** and ML contains **endolymph**.
- The **vestibule** is the **oval** central portion of the BL.
- **Utricle** (little bag) and **saccul** (little sac).
- They are connected to the main duct through **ampulla** (swollen enlargements).
- Vestibule houses sensory as well as motor neurons that synapse with receptors for **equilibrium**.
- Anterior to vestibule is **cochlea**, a bony spiral canal.

Inner ear



Physiology of hearing

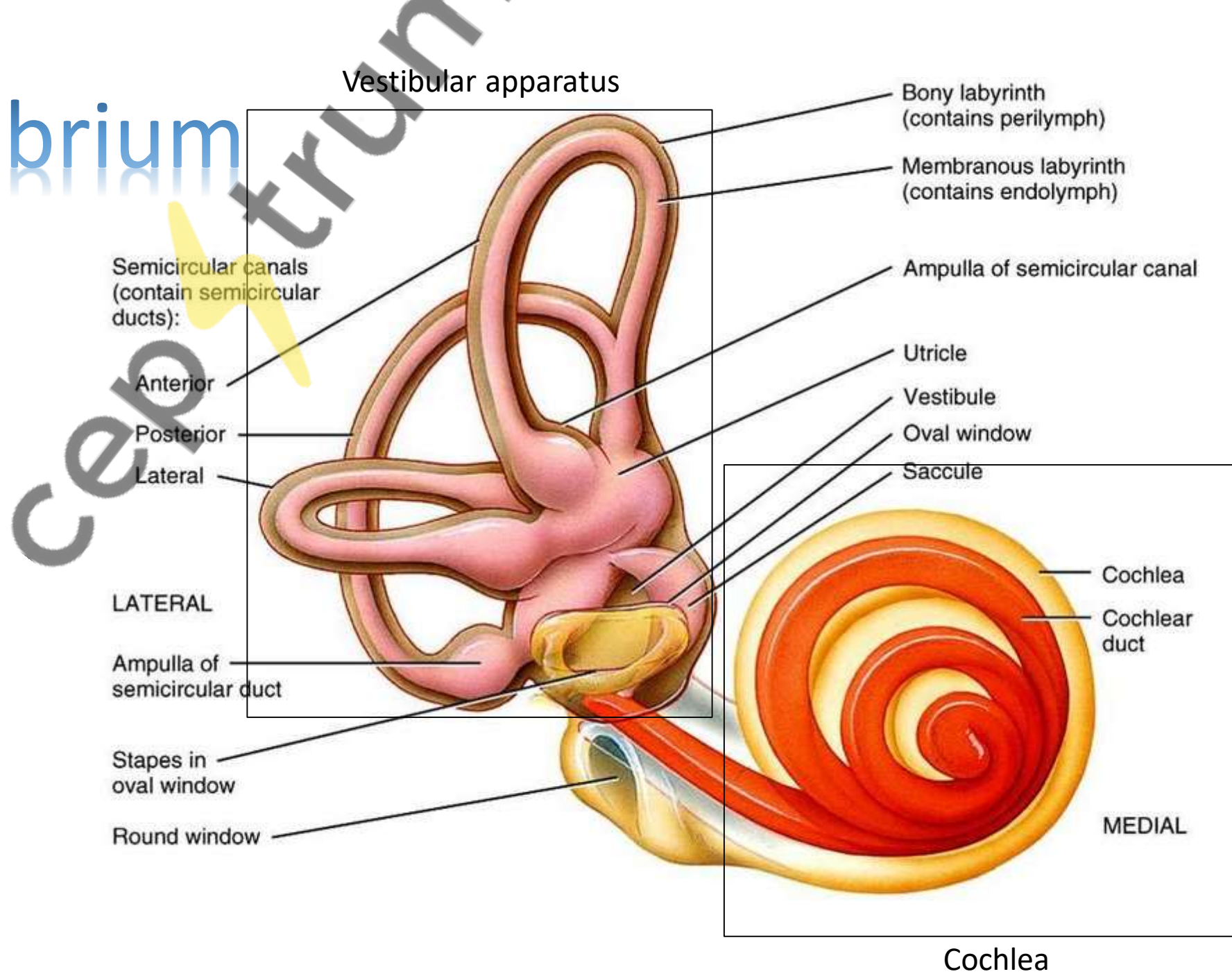


1. The auricle directs sound waves into the external auditory canal.
2. When sound waves strike the eardrum, it vibrates back and forth.
3. The malleus vibrates along with the eardrum, transmitting it further to the incus and then to the stapes.
4. The stapes moves back and forth and makes the oval window vibrate at an intensity ~ 22 times more than the eardrum.
5. This sets up fluid pressure waves in the perilymph of cochlea. As the oval window bulges inward, it pushes on the perilymph of scala vestibuli.
6. Pressure waves are further transmitted to scala tympani and eventually to the round window, making it to bulge outward into the middle ear.
7. Pressure waves push the vestibular membrane and then move into the endolymph of the cochlear duct.
8. The pressure in the endolymph causes the basilar membrane to vibrate. Hair cells are perturbed and electrical signal is generated from mechanical vibrations.
9. Sound waves of various frequencies cause certain regions of the basilar membrane to vibrate more intensely than other regions. Each segment of this membrane is 'tuned' for a particular pitch.



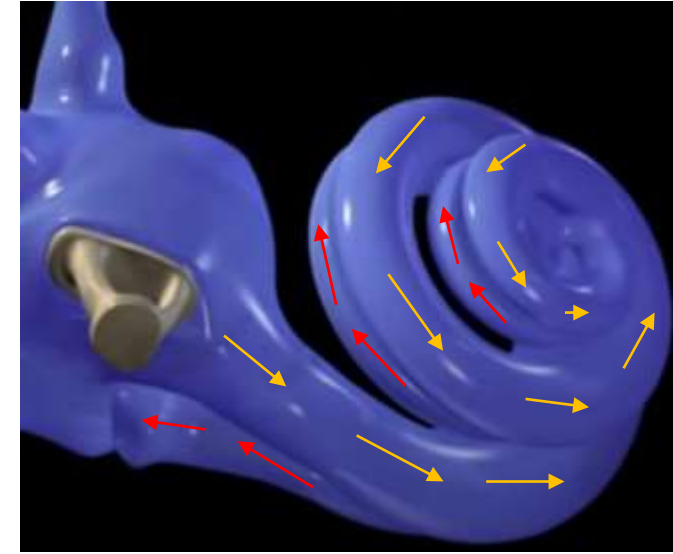
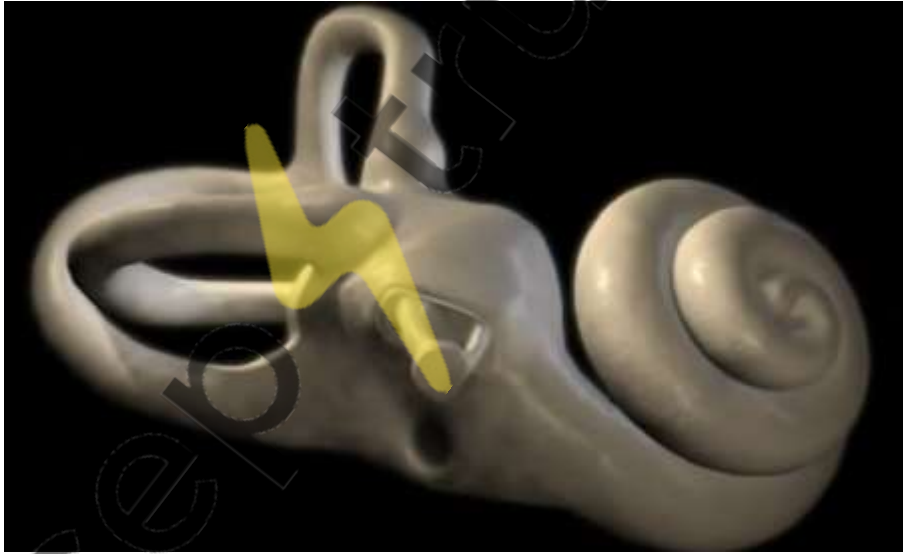
Equilibrium

Inner ear

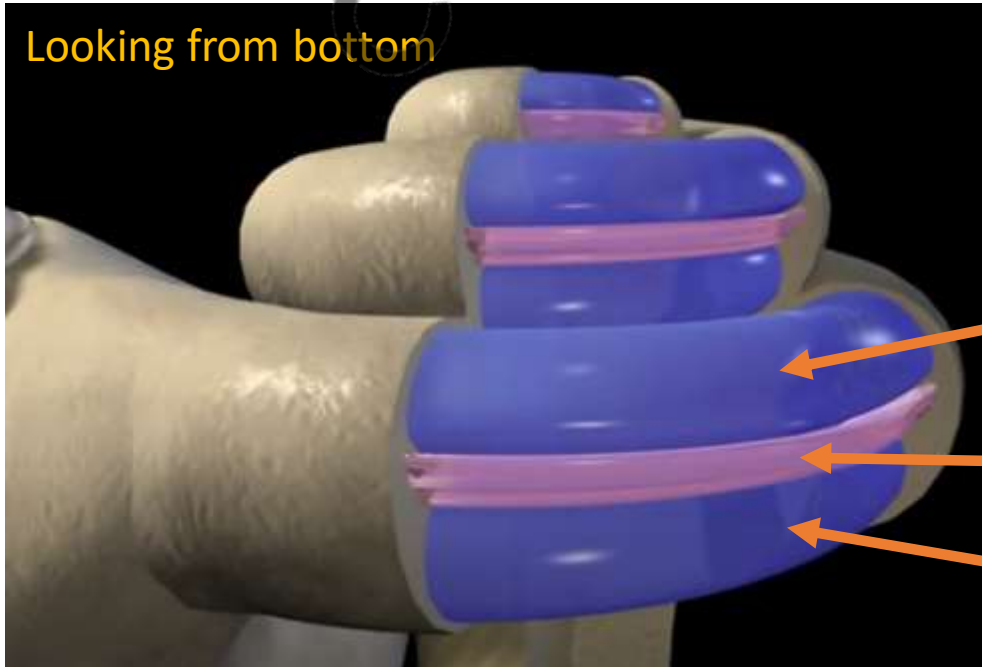


sound

Auditory Transduction



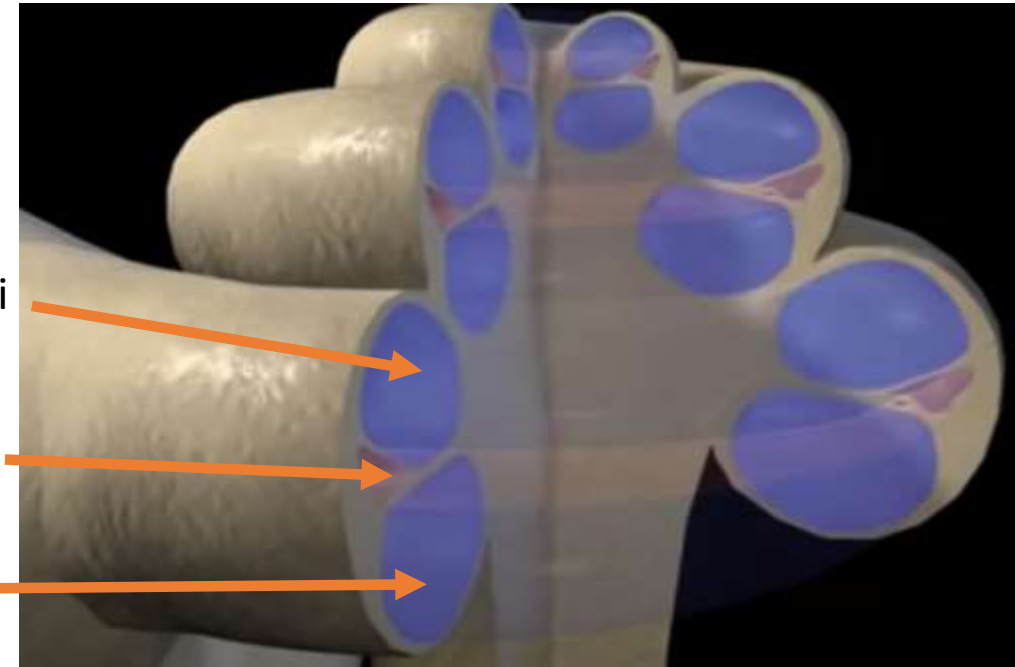
Looking from bottom

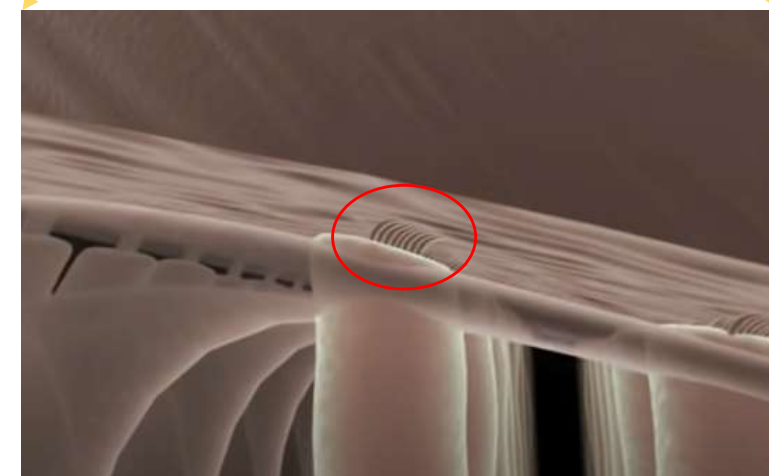
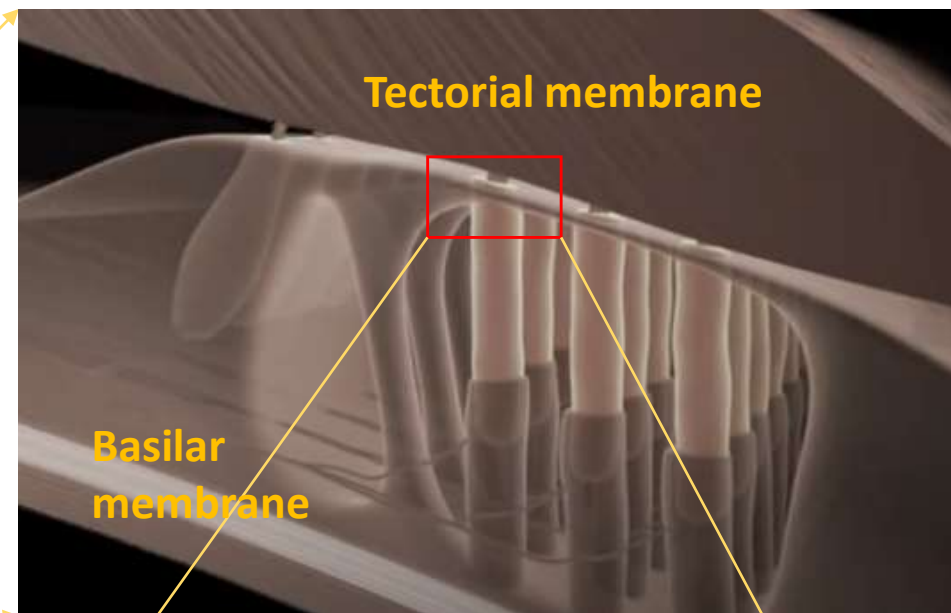
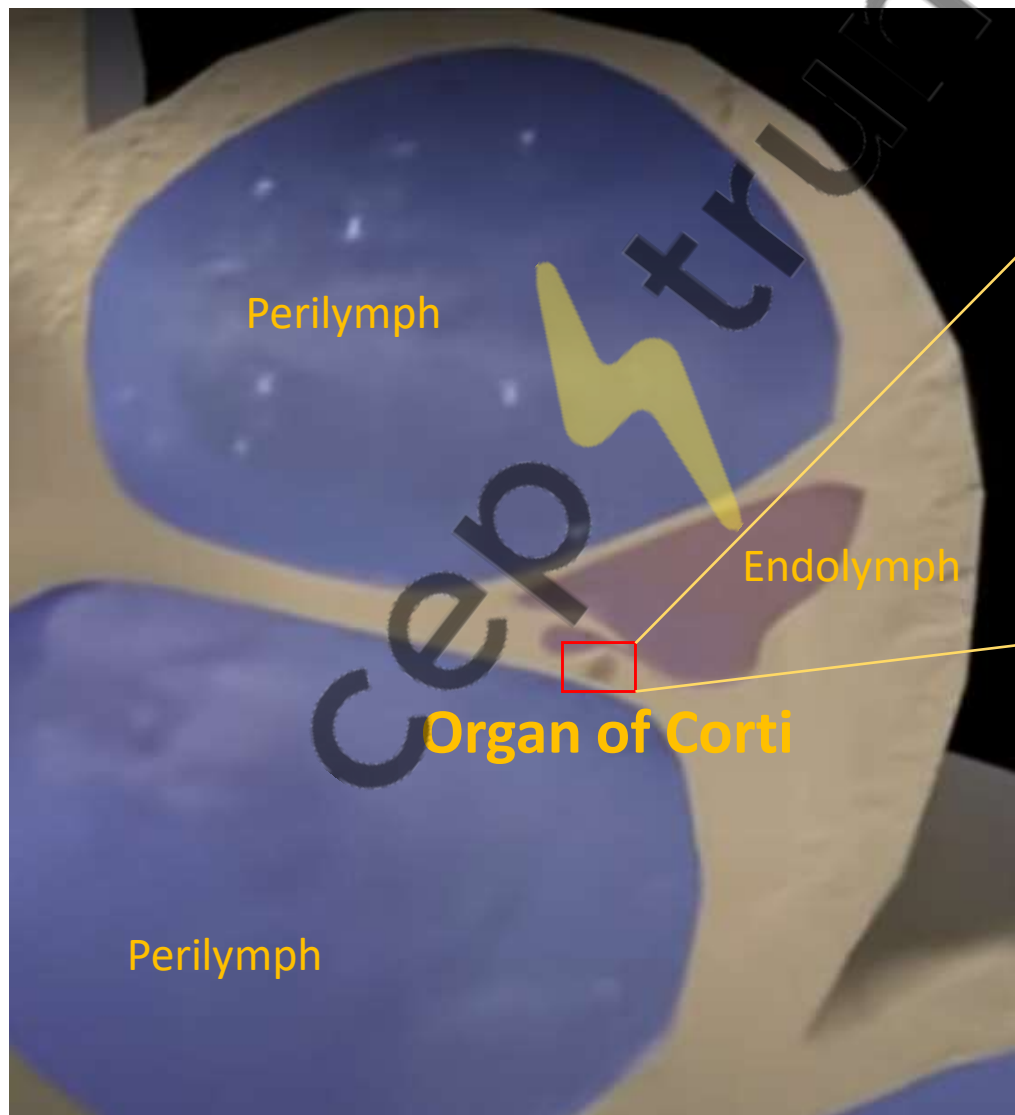


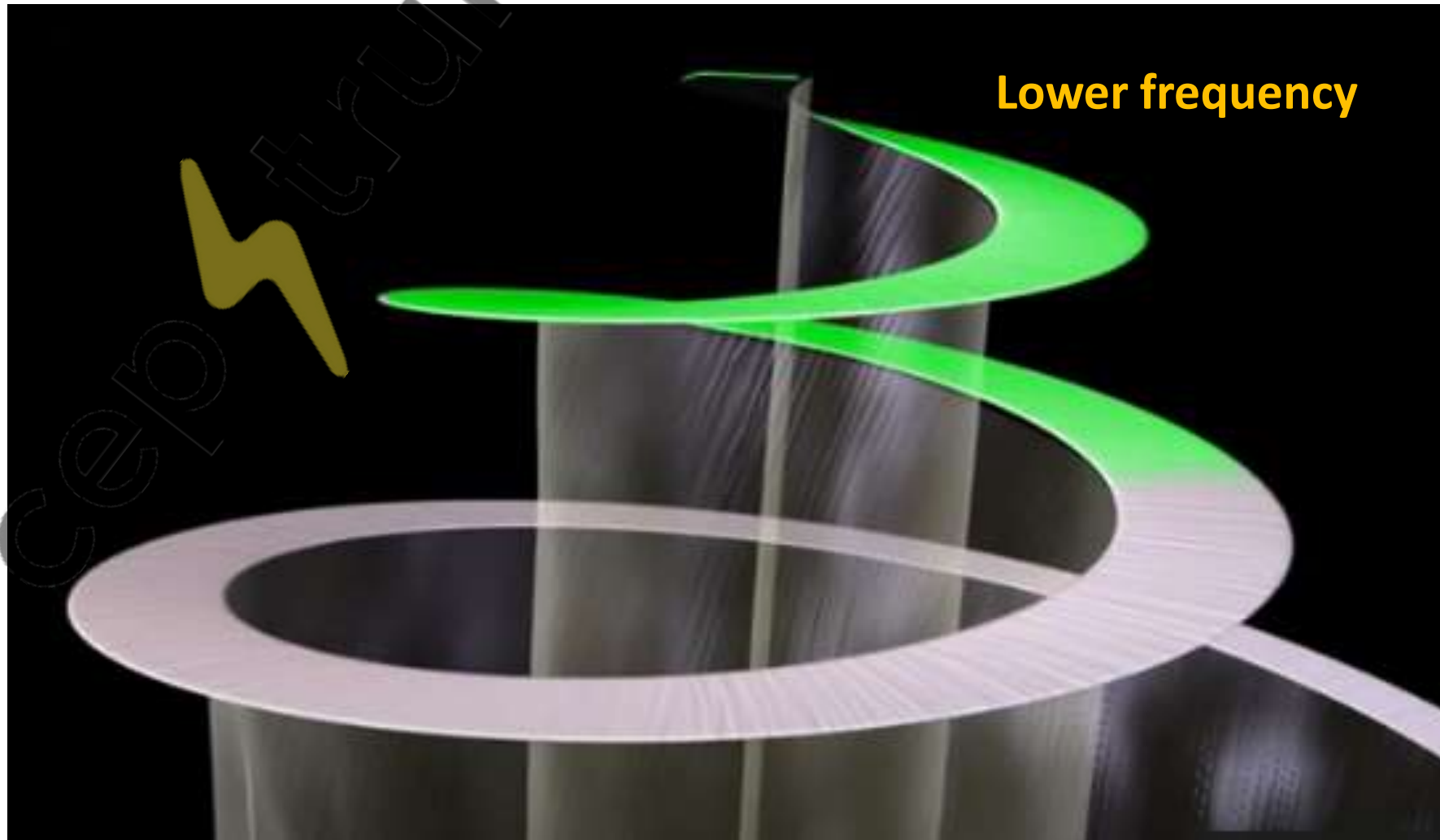
Scala Vestibuli

Cochlear Duct

Scala Tympani







Entire Basilar membrane does not vibrate simultaneously - - Tonotopic organisation

Physiology of Equilibrium

There are two types of equilibrium or balance:

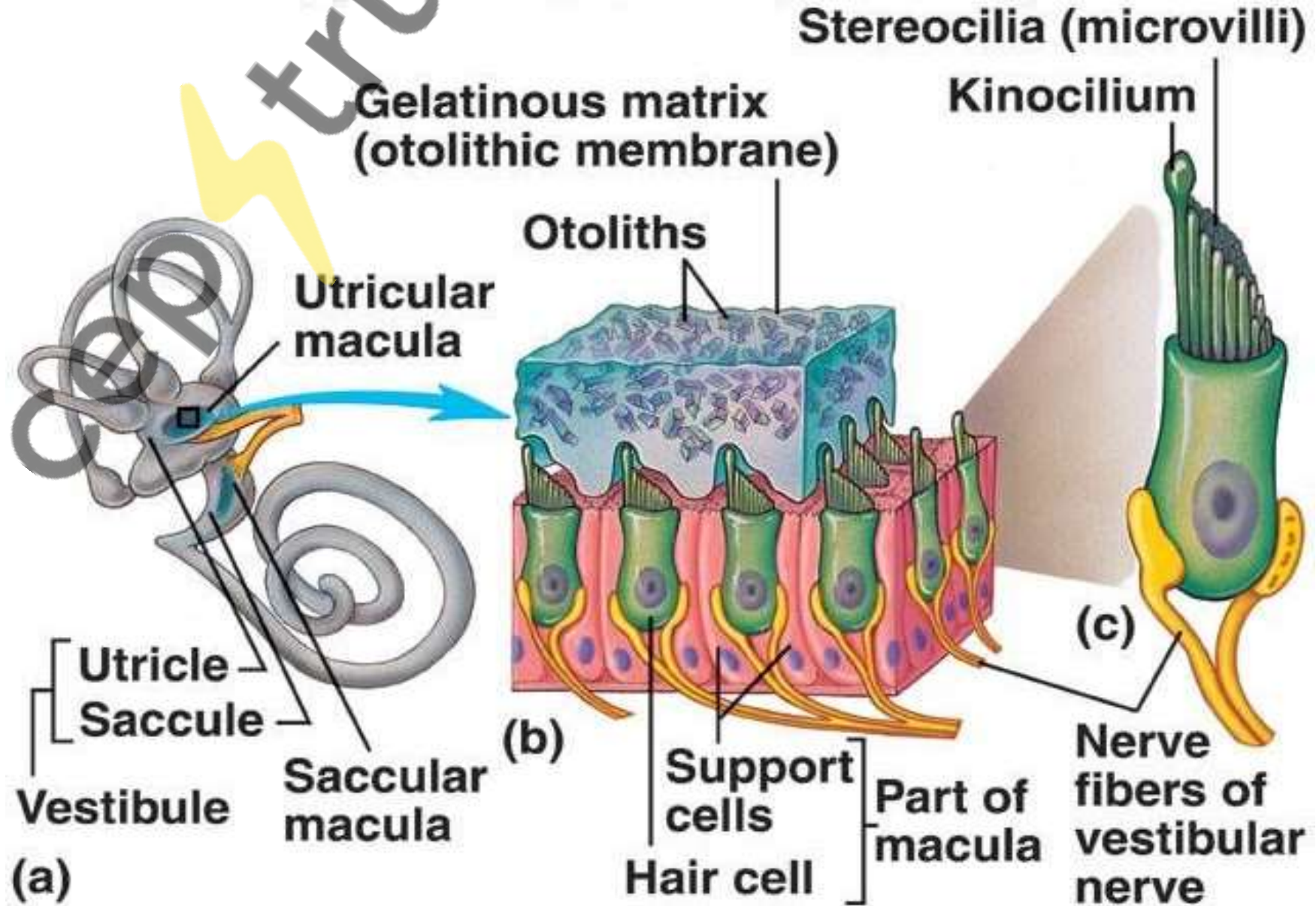
Static equilibrium

- Refers to the maintenance of the position of the body (mainly the head) relative to the force of gravity, including tilting of head and linear

Dynamic equilibrium

- Refers to the maintenance of the position of the body (mainly the head) in response to sudden movements, such as rotational acceleration or deceleration.

Maculae



Maculae: salient features

The walls of both the utricle and the saccule contain a small, thickened region called a **macula**.

The two *maculae* (perpendicular to each other) are the receptors for **static equilibrium**.

They **provide sensory information** on the **position of the head** in space and thus are essential for maintaining appropriate posture and balance.

They also can sense **linear acceleration or deceleration**.

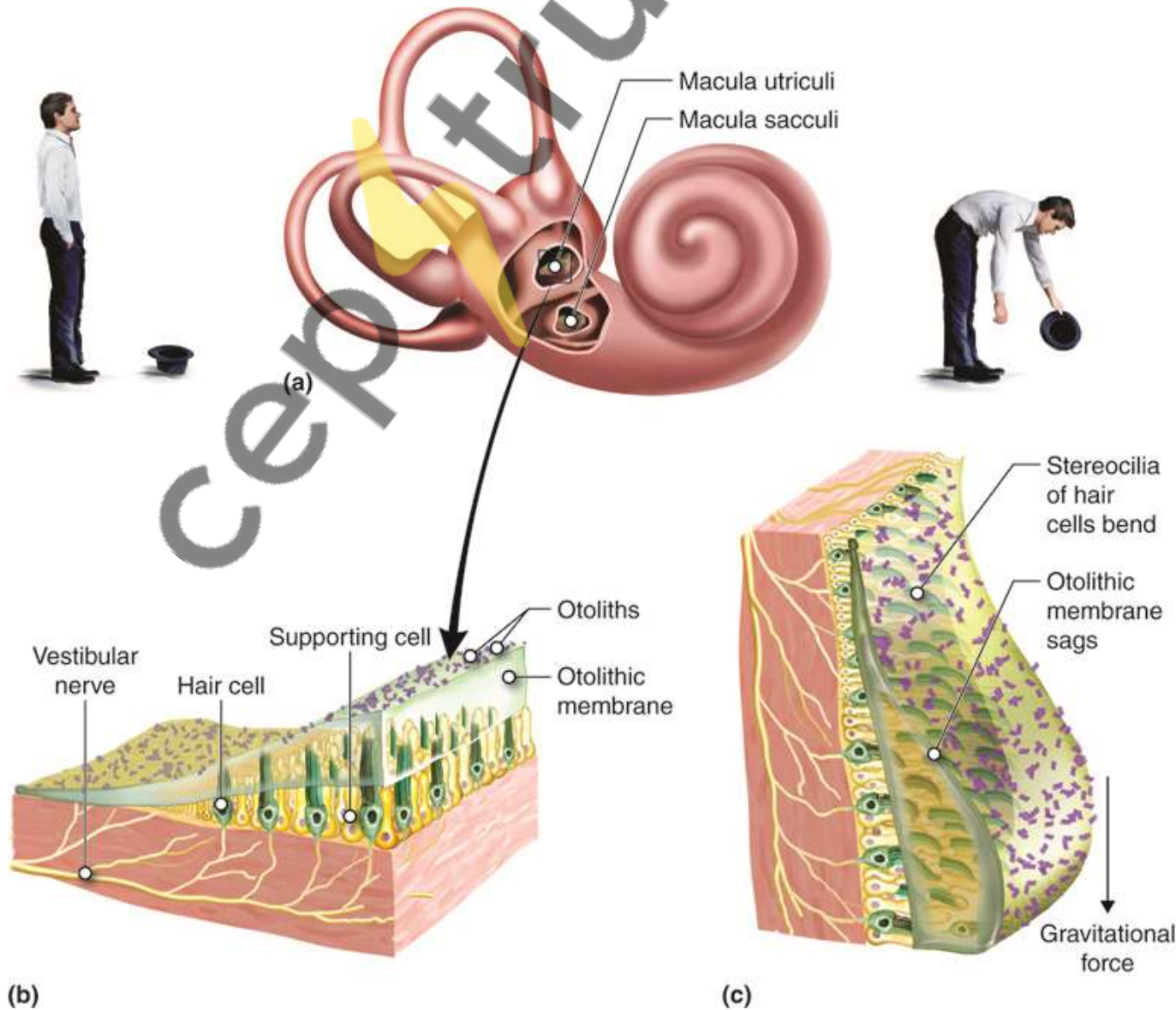
The macula contains two types of cells: **Hair cells** and **Support cells**.

Each hair cell contains **40-80 stereocilia** plus **one kinocilium**, collectively called **hair bundle**.

Support cells secrete the thick, gelatinous, G-protein layer, called **the otolithic membrane**, that rests on the hair cells.

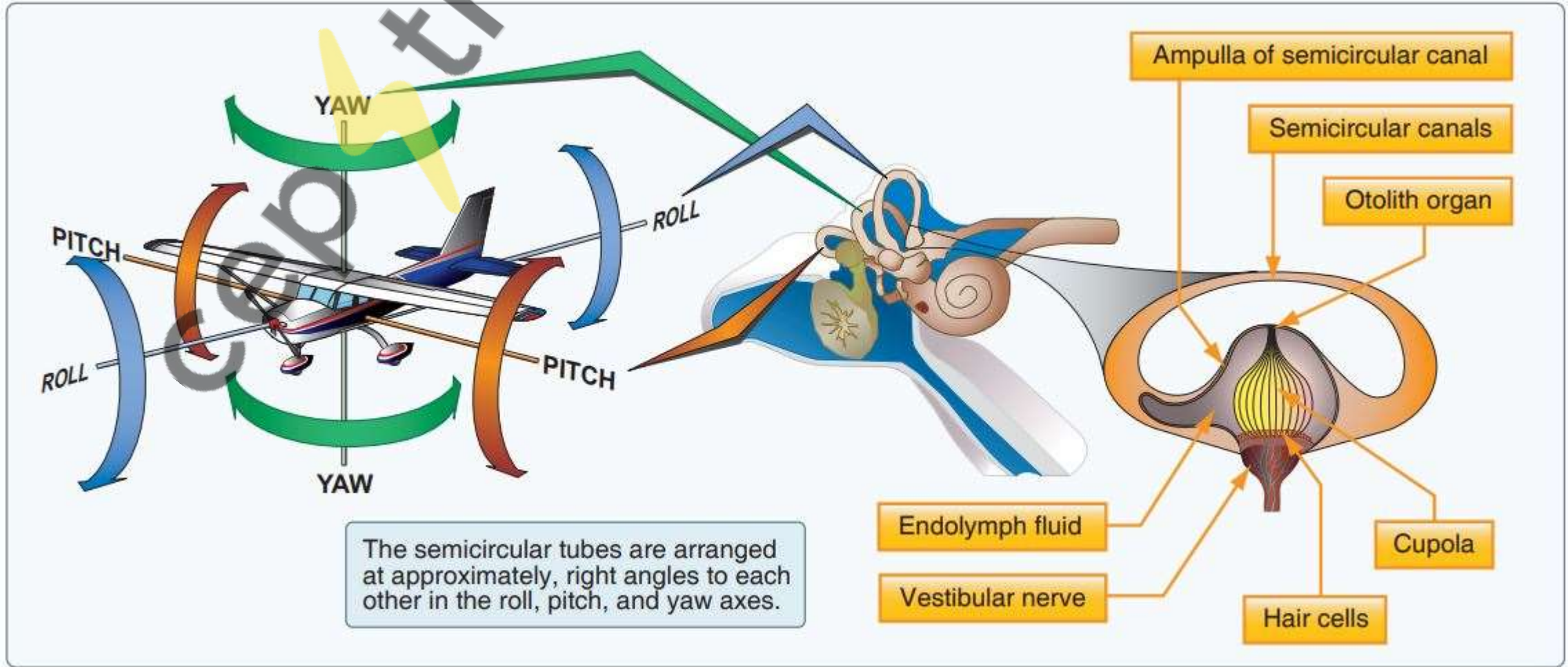
At the top of the otolithic membrane floats calcium carbonate crystals called **otoliths**.

Macula: its role in static equilibrium

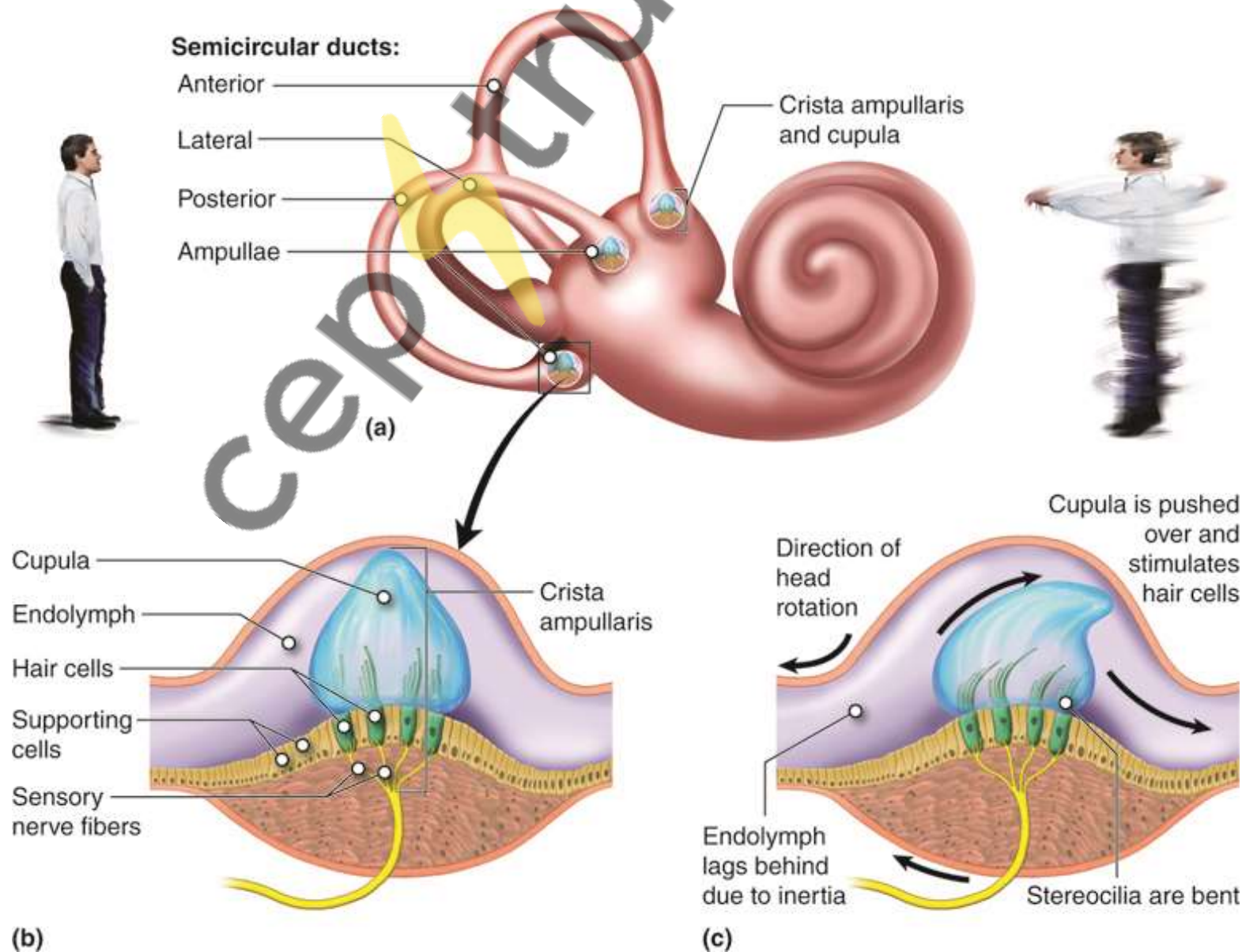


- If the head is tilted forward, the otolithic membrane (along with the otoliths) is pulled by gravity.
- It slides 'downhill' over the hair cells, in the direction of the tilt, bending the hair bundles.
- This causes the hair cells depolarize or repolarize, facilitating release of neurotransmitter at faster or slower rate.
- The hair cells synapse with first order sensory neurons in the vestibulocochlear nerve.
- The also synapse with motor neurons that regulate sensitivity of the hair cells.

The Three Semicircular Ducts



Semicircular ducts: role in dynamic equilibrium



- The three ducts function in dynamic equilibrium.
- The ducts lie at right angles to one another in three mutually perpendicular plane.
- This positioning permits detection of rotational acceleration or deceleration.
- In the ampulla, there is a small elevation called **crista**.
- Covering the crista is a mass of gelatinous material called the **cupula**.
- If the head is rotated, cupula is pushed over thereby stimulating the hair cells.

Quick test on functionality:

Auricle ?

Eardrum ?

Auditory ossicles ?

Eustachian tube ?

Cochlea ?

Vestibular apparatus:

- Utricle and Saccule ?
- Semicircular ducts ?

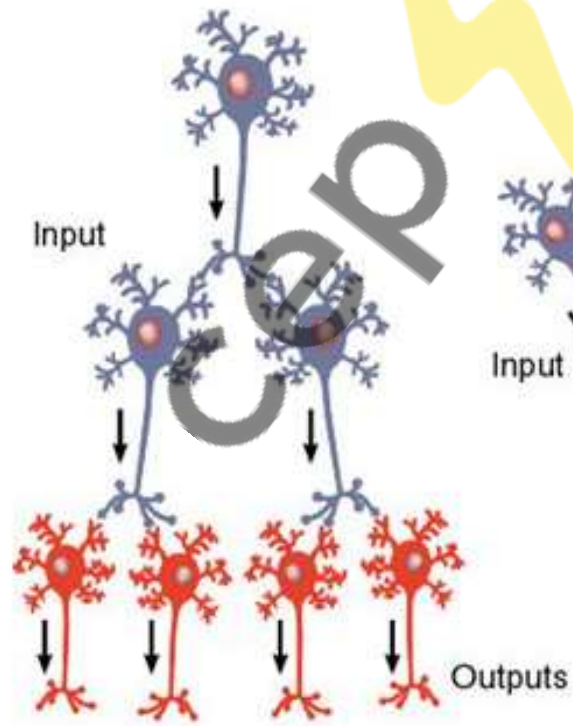
Neural circuits

- CNS contains billions of neurons organized into complicated networks called **neural circuits**.
- These are **functional groups of neurons** that process specific types of information.
- The simplest neural circuit may consist of **one series connection**, but in most cases they are **very complex**.

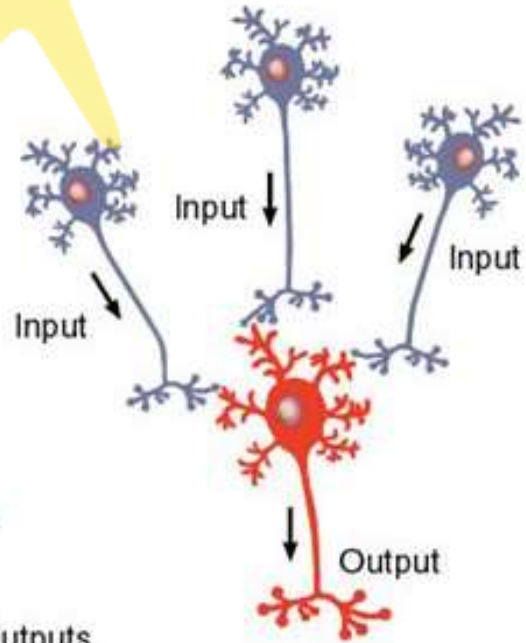
These are not branches of trees, but neural circuits of our brain ...



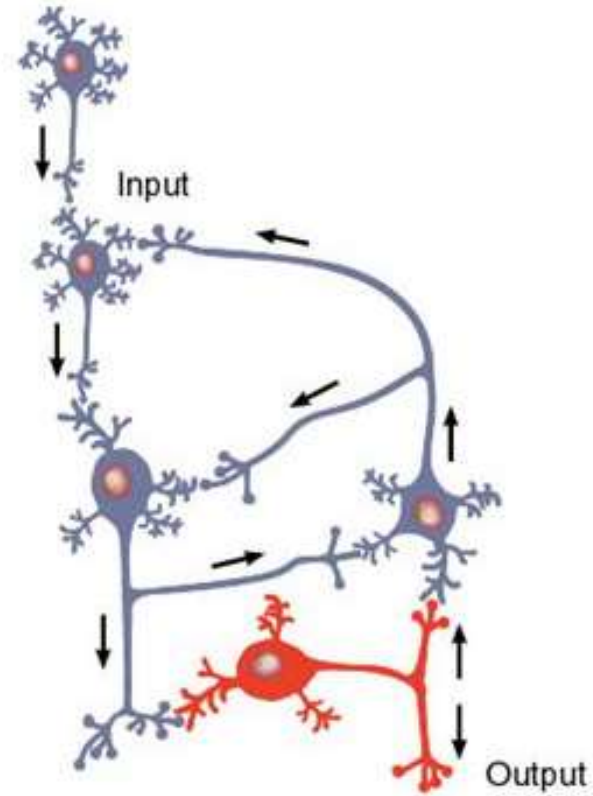
Neural circuits: types



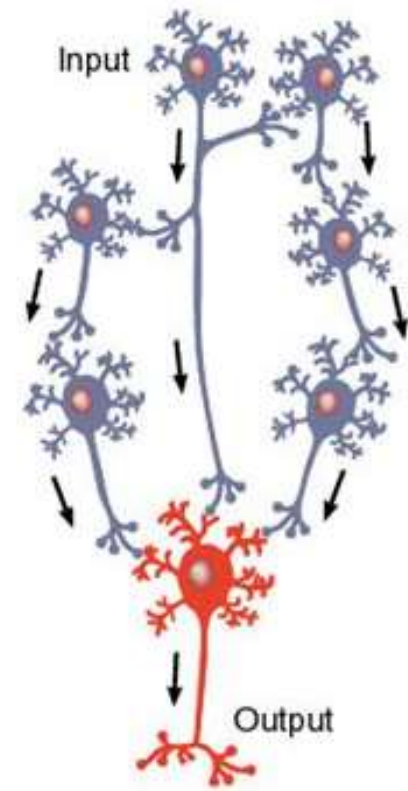
(a) Diverging circuit



(b) Converging circuit



(c) Reverberating circuit



(d) Parallel after-discharge circuit

Definitions

Diverging – A single presynaptic neuron may synapse with several postsynaptic neurons.

Example: Sensory signals are arranged in diverging circuits, allowing a sensory impulse to be relayed to several regions of the brain. This arrangement amplifies the signal.

Converging – Several presynaptic neurons synapse with a single post synaptic neuron. This arrangement permits more effective simulation or inhibition of the postsynaptic neuron.

Example: A single motor neuron that synapses with skeletal muscle fibres at neuromuscular junctions receives input from several pathways that originate in different brain regions.

Reverberating – Branches from later neurons synapse with earlier ones, thereby sending impulses back through the circuit again and again.

Example: The activities of breathing and waking up are considered to be a result of such circuit.

Parallel after-discharge – A single presynaptic cell stimulates a group of neurons, each of which synapses with a common postsynaptic cell.

Example: Mathematical calculations.

Plasticity and Regeneration

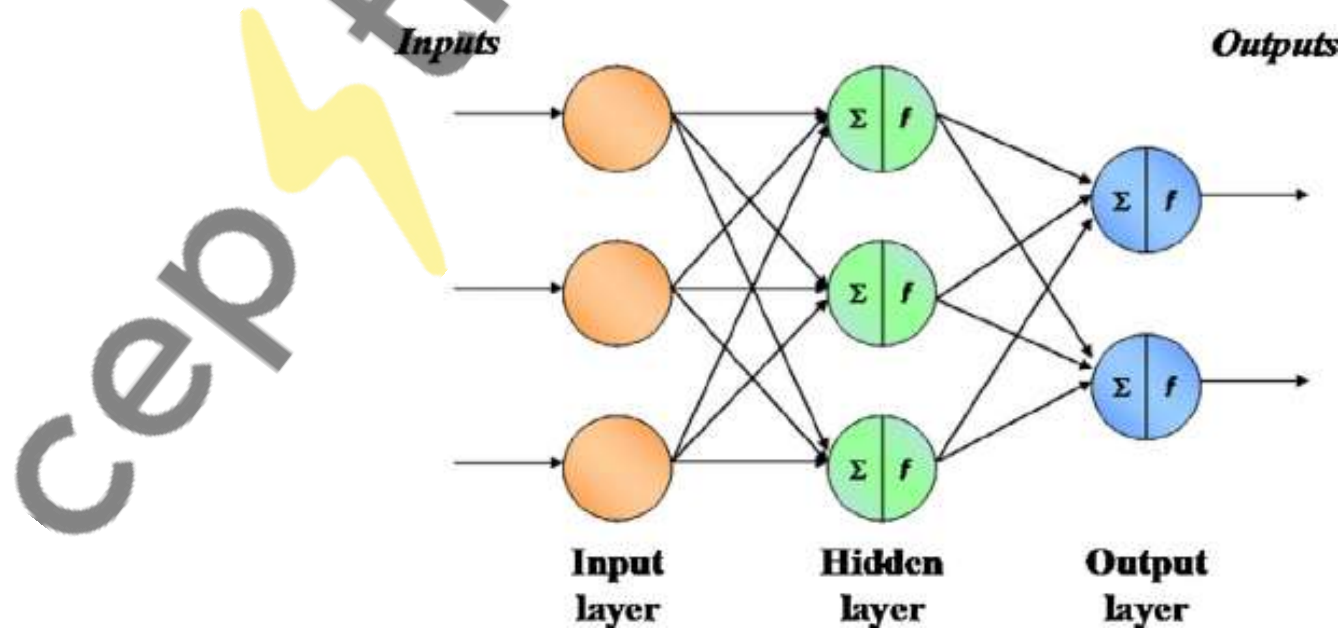
- ❑ The nervous system exhibits capabilities to change based on experience. This property of the neurons is called **plasticity**. This changes may include sprouting of new dendrites, synthesis of new proteins, and changes in synaptic contacts with other neurons. Both chemical and electrical signals drive these changes.
- ❑ The **regeneration** of neurons is the capability to replicate or repair themselves. Mammalian neurons have limited capacity to regenerate. In PNS, damage to dendrites and axons may be repaired to certain extent if the cell body and the Schwann cells remain active. However, CNS exhibits little or no repair to the damage.

Artificial Neural Network (ANN)

An artificial neuron network (ANN) is a computational model based on the structure and functions of biological neural networks. Information that flows through the network affects the structure of the ANN because a neural network changes - or learns, in a sense - based on that input and output.

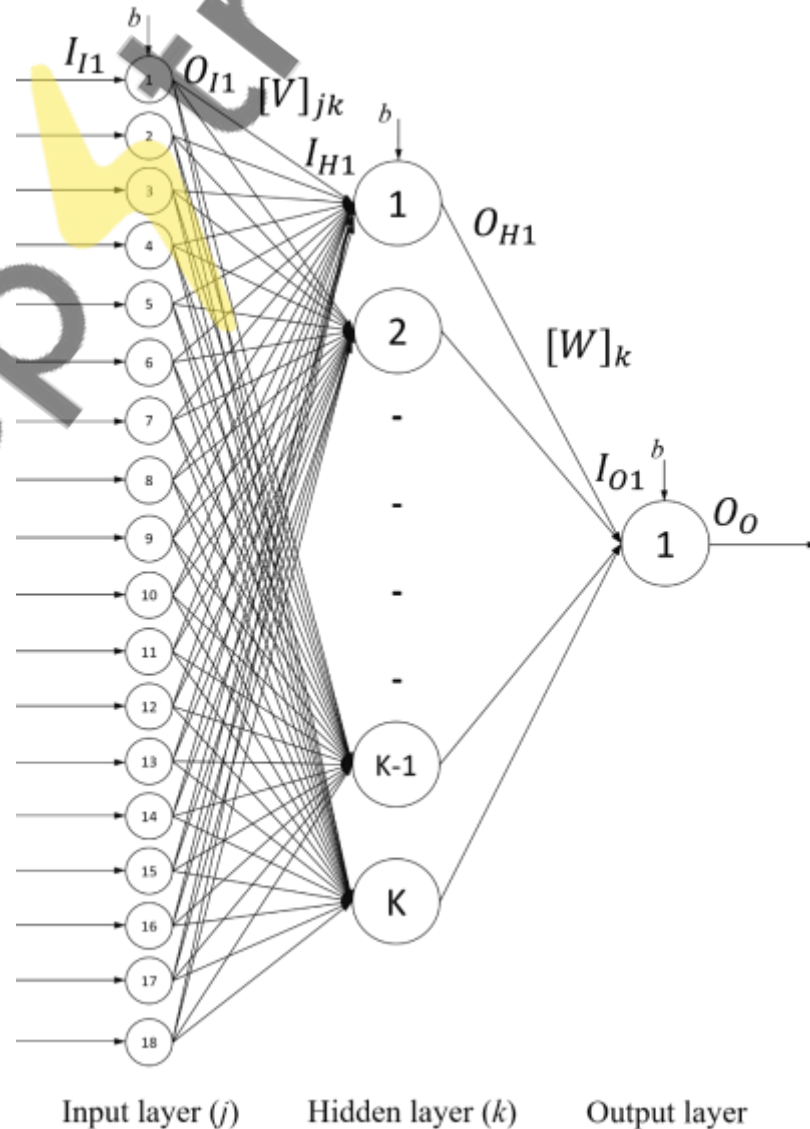
- ✓ ANNs are considered nonlinear statistical data modelling tools where the complex relationships between inputs and outputs are modelled or patterns are found.
- ✓ ANN can actually learn from observing data sets. In this way, it is used as a random function approximation tool.
- ✓ Training an artificial neural network involves choosing from allowed models for which there are several associated algorithms.
- ✓ Usage: Machine Learning, Artificial Intelligence

Structure of ANN



Backpropagation, short for "backward propagation of errors," is an algorithm for supervised learning of artificial neural networks using gradient descent. Given an artificial neural network and an error function, the method calculates the gradient of the error function with respect to the neural network's weights.

Back-propagation Neural Network (BPNN)



$a1$: constant of activation function for input neurons

$a2$: constant of activation function for hidden neurons

$a3$: constant of activation function for output neurons

b : bias value